UNIVERSITÀ COMMERCIALE "LUIGI BOCCONI"

PhD SCHOOL

PhD program in Statistics and Computer Science

Cycle: XXXVIII

Distances Between Laws of Random Probability Measures: Applications to Two-Sample Testing and Merging of Opinions

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Year 2026

Abstract

Distances between probability measures are used across several areas of statistics, including the theory of estimation, hypothesis testing, and computational statistics. They provide natural tools for studying the convergence of estimators, quantifying dissimilarities between observed samples, and measuring discrepancies between empirical and target distributions. While classical applications mainly concern probability measures on Euclidean spaces, there is a need for distances on more complex spaces, such as spaces of laws of random probability measures. This is motivated by Bayesian nonparametrics, where the objects of inference are often random probability measures, and by modern statistical problems in which data objects themselves are naturally represented as probability measures. This thesis focuses on two applications in which such distances play a key role: two-sample testing and merging of opinions.

Two-sample testing assesses whether two populations differ by comparing observed samples, with the Kolmogorov–Smirnov test as a classic example. While numerous extensions address multivariate data, fewer methods have been developed for settings in which the objects of interest are complex, such as probability distributions themselves. In such a case, the random probability measures are typically not observed directly, but only through samples drawn from them. We propose a distance-based two-sample test for testing equality of laws of random probability measures using optimal transport theory, and leverage tools from empirical process theory to establish nonparametric theoretical guarantees. Empirically, we benchmark our method against existing approaches on simulated datasets and apply it to a mortality dataset.

The merging of opinions concerns whether two posterior distributions arising from different prior distributions eventually converge as more data are observed. Distances between posterior distributions provide a quantitative way to study both the occurrence of merging and its rate. We consider a Bayesian nonparametric framework, namely laws induced by completely random measures. Accordingly, the distance is tailored to completely random measures and is defined at the level of their Lévy intensities. Complementing existing results for other Bayesian nonparametric priors, this thesis derives merging rates between posterior distributions induced by the Pitman-Yor process and the Dirichlet process.

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Chapter 1

Introduction

This thesis studies statistical applications of distances on the space of laws of random probability measures. Such objects arise naturally in Bayesian nonparametrics and in modern statistical problems where the observed units are distributions rather than finite-dimensional vectors. In these settings, distances provide a way to compare the laws governing random probability measures, thereby opening the possibility of extending classical statistical questions to this more general framework.

The thesis develops two applications of this distance-based perspective. First, it studies a hypothesis testing problem in which a distance-based approach is used to compare laws of random probability measures. Second, it investigates merging of opinions in Bayesian nonparametric models.

The rest of the introduction is organized as follows. We begin with a historical overview of distances on spaces of probability measures, emphasizing the mathematical, information-theoretic, and statistical motivations that led to their development. We then discuss the motivation for considering distances on laws of random probability measures and review related work in this direction. Finally, we describe the organization of the thesis and specific contributions to the two problems considered: a distance-based approach to hypothesis testing for laws of random probability measures, and the analysis of merging under different prior specifications in Bayesian nonparametric models.

1.1 Metrics on Probability Measures: From Historical Origins to Statistical Applications

Historically, several distinct motivations led to the development of distances between probability measures. The first motivation is purely mathematical, with formal foundations laid following the introduction of metric spaces by Felix Hausdorff and Maurice Fréchet at the beginning of the 20th century. This was followed by work on the space of signed measures that led to the development of the total variation distance. In parallel, Paul

Lévy was working on defining distances between random variables and their laws. He introduced a metric for distribution functions on the real line in Lévy (1925). In addition, in a note written for the book Fréchet (1950), Lévy introduced several notions of distance between random variables and between their induced laws, one of which corresponds to what is now recognized as the Wasserstein distance.

Another motivation for studying distances between probability measures arose in information theory. In Kullback and Leibler (1951), the notion of information for discrimination between two distributions was introduced. The motivation was to quantify how much information observations provide for distinguishing between two hypotheses about the underlying distribution. The information-theoretic point of view was followed by a broader development of divergence measures, such as the chi-square divergence, f-divergences, and the Hellinger distance.

One of the most widely used distances on the space of probability measures is the Wasserstein distance, although it was not originally introduced as a metric on measures. Its origins can be traced back to the development of optimal transport theory. In the late 18th century, Gaspard Monge posed the question of how to move a pile of soil to fill another hole while minimizing the total transportation cost Monge (1781). The problem was formulated in terms of transport maps, which specify where each infinitesimal unit of mass is sent. Around 160 years later, Leonid Kantorovich introduced a relaxation of this formulation in Kantorovich (1942), allowing transport plans rather than deterministic maps. This relaxation led to what is now known as the Monge–Kantorovich problem:

$$\min_{\gamma\in\Gamma(\mu,\nu)} \int_{\mathbb{X}\times\mathbb{Y}} c(x,y)\,\mathrm{d}\gamma(x,y), \tag{1.1}$$

where $\Gamma(\mu,\nu)$ denotes the set of all couplings of μ and ν, that is, probability measures on $\mathbb{X}\times\mathbb{Y}$ with marginals μ and ν. This formulation was later complemented by the duality results of Kantorovich and Rubinshtein (1958).

The Monge–Kantorovich problem includes the Wasserstein distance as a special case. Indeed, when $\mathbb{X}=\mathbb{Y}$ is equipped with a metric d, and the cost is chosen as $c(x,y) = d(x,y)^p$ for $p\geq 1$, the corresponding optimal value defines the p-Wasserstein distance,

$$W_p(\mu,\nu) = \left(\inf_{\gamma\in\Gamma(\mu,\nu)} \int_{\mathbb{X}\times\mathbb{Y}} d(x,y)^p\,\mathrm{d}\gamma(x,y)\right)^{1/p}.$$

However, the terminology itself has a separate historical origin. The name "Vasershtein distance" first appeared in Dobrushin (1970), in the context of studying the existence and uniqueness of random fields, as a reference to earlier work by Leonid Vaserstein. In Vaserstein (1969), Markov processes on denumerable products of spaces were studied, motivated by the analysis of large systems of automata, and that distance was used to study the convergence of Markov processes.

Wasserstein distances have become important tools in statistics and machine learning because they provide a geometrically meaningful discrepancy between probability measures. In particular, they are used in the theory of point estimation, where parameters are estimated by minimizing the Wasserstein distance between the empirical distribution and a model distribution Bassetti et al. (2006); Bernton et al. (2019b). The convergence of empirical estimators based on the Wasserstein distance is studied in Bobkov and Ledoux (2019); Panaretos and Zemel (2019). In hypothesis testing, Wasserstein distances are used to measure departures from a null hypothesis Ramdas et al. (2017); Wang et al. (2022, 2021); Hu and Lin (2025). They are also used in goodness-of-fit testing by comparing the empirical distribution to model-based estimators Hallin et al. (2021). In computational statistics, Wasserstein distances are used to study convergence rates of MCMC algorithms Qin and Hobert (2022); Lavenant and Zanella (2024) and to design likelihood-free inference methods, such as Wasserstein approximate Bayesian computation, by comparing observed and simulated empirical distributions directly Bernton et al. (2019a). In machine learning, they play a central role in generative modeling, especially in Wasserstein generative adversarial networks Arjovsky et al. (2017), and have also been used in domain adaptation Courty et al. (2017). In addition, one can use the Wasserstein distance to define the distance function between labels in classification, which leads to a distance at the level of datasets Alvarez-Melis and Fusi (2020).

Lastly, the Wasserstein distance plays an important role in quantifying the dependence between random probability measures in Bayesian nonparametric models. In many Bayesian nonparametric models, groups of observations may arise from different sources whose underlying distributions are statistically dependent. This dependence can be modelled at the level of a vector of random probability measures, where each such measure is associated with each group. For this type of problem, Catalano et al. (2024) develops a Wasserstein index of dependence for completely random measures to quantify the dependence between random probability measures.

1.2 Motivation and Applications of Distances Between Laws of Random Probability Measures

There are several motivations for studying distances between laws of random probability measures. In this section, we emphasize three of them: the purely mathematical analysis perspective, the Bayesian statistical perspective, and the complex data structure perspective, where probability measures themselves are treated as data objects.

From the purely mathematical analysis perspective, optimal transport provides one important source of examples. The optimal transport problem for laws on spaces of probability measures was studied in Pinzi (2026), where characterization results in terms of Kantorovich potentials are obtained in analogy with classical optimal transport theory, and a connection with the strict Monge problem is established. In addition, the work Pinzi and Savaré (2025a) studies the geometry of probability laws on Wasserstein spaces through a nested superposition principle, giving tools to analyze curves and tangent structures in this setting. Another line of work concerns establishing a Brenier theorem for laws on iterated Wasserstein spaces Beiglböck et al. (2025); Pinzi and Savaré (2025b), and the study of gradient-flow dynamics Bonet et al. (2025).

The second motivation comes from the Bayesian statistical perspective. In Bayesian statistics, priors are probability measures over possible data-generating laws. Therefore, as the framework includes probabilities over probabilities, it is natural to work with distances between such objects.

One line of work studies the merging of opinions. Here, the question is whether posterior distributions arising from different prior distributions become close as more observations are collected. Distances between laws of random probability measures provide a natural way to measure this form of posterior closeness. The paper Catalano and Lavenant (2026) studies this problem in the context of completely random measures. Another important example is the use of the Wasserstein over Wasserstein distance to study posterior concentration and borrowing of information in hierarchical Bayesian nonparametric models Nguyen (2016). In that paper, the goal is to estimate the base probability measure in a Dirichlet process prior from the observations generated through the mixture model arising from latent random probability measures. Such a base measure is endowed with another Dirichlet prior and posterior concentration rates are provided using the Wasserstein over Wasserstein distance.

The third motivation is the complex data structure perspective. In many modern statistical problems, the observations are not scalar or vector-valued points, but probability measures. This occurs in several fields. For example, in macroeconomics, one may observe yearly income distributions or survey distributions of inflation expectations. In demographic studies, one may compare mortality distributions across countries. In biology, single-cell RNA sequencing data can be represented through distributions of cells in a state space. In machine learning, documents, or datasets may also be represented as probability measures. The use of distances on laws of random probability measures is common in machine learning. For example, in generative modelling, images can be represented as probability measures. A generative model then produces random probability measures, and the aim is to choose a model whose generated distributional objects are close to the observed ones. Wasserstein over Wasserstein can be used to define a loss function for this purpose (see Dukler et al. (2019)). Another example is the work of Piening et al. (2026), where flow matching is extended to probability measures over probability measures through a Wasserstein-on-Wasserstein distance.

1.3 Organization and Contributions of the Thesis

The thesis begins with a chapter that collects the background material in Bayesian nonparametrics needed for the subsequent chapters. It starts with the framework for random probability measures and exchangeability, together with de Finetti's representation theorem. It then covers completely random measures, including the role of Lévy intensities and Laplace functionals, identifiability issues under normalization, posterior distributions, and examples. Finally, it reviews the distances on laws of random probability measures, including the Wasserstein over Wasserstein distance and the Hierarchical Integral Probability Metric, as well as the distances on laws of random measures specialized for completely random measures.

The second chapter studies a two-sample testing problem in which the observed units are probability measures. The null hypothesis is that two samples of probability measures are drawn from the same underlying law, against the alternative that the laws differ. Our contribution is to consider this problem when the samples are probability measures themselves and to propose a distance-based approach together with its theoretical properties. Specifically, we derive a threshold based on Rademacher complexity that controls the type-I error, and then introduce a permutation-based threshold that is both practical and theoretically justified, with guarantees on both the type-I error and the power of the resulting test. We note that the theoretical guarantees are established at the level of the raw observations used to estimate the underlying probability measures, which we call hierarchical samples. The method is benchmarked against existing approaches in simulations and applied to a mortality dataset.

The third chapter studies the merging of opinions in Bayesian nonparametric models: whether posterior distributions induced by two different prior distributions become close as the number of observations grows, independently of the data-generating mechanism. The framework for studying this problem is provided by Catalano and Lavenant (2026), which introduces distances between laws of completely random measures and establishes identifiability results that make the comparison of posteriors meaningful. Within this framework, the known results concern merging between two Dirichlet process priors, which occurs at rate $1/n$, and merging between a generalized Gamma CRM and a Gamma CRM, where the rate is $\max\{n^{-1/(1+\sigma)}, k_n/n\}$ and merging may fail when $k_n \sim n$, where k_n

denotes the number of unique values in n observations. The contribution of this thesis is to establish the merging rate between the Pitman–Yor process and the Dirichlet process, showing that it equals k_n/n for any sequence of observations.

Chapter 2

A Bayesian Nonparametric Toolbox

2.1 Random Probability Measures and Exchangeability

In parametric Bayesian statistics, one works with statistical models indexed by finite-dimensional parameters. In many applications, this finite-dimensionality assumption can be restrictive. Bayesian nonparametric statistics therefore provides more flexible modeling approaches. The fundamental object of study is a random probability measure, namely a random variable taking values in the space of probability measures that may have generated the observed data. Before formally defining this object, we introduce the setting for the statistical models.

Let $\mathbb{X}$ be a Polish space equipped with its Borel σ-algebra $\mathcal{B}(\mathbb{X})$. Denote by $\mathcal{P}(\mathbb{X})$ the space of probability measures on $(\mathbb{X}, \mathcal{B}(\mathbb{X}))$. We equip $\mathcal{P}(\mathbb{X})$ with the Borel σ-algebra generated by the topology of weak convergence. This σ-algebra is the smallest one that makes the maps $\phi_A : \mathcal{P}(\mathbb{X}) \to [0, 1]$, defined by

$$\phi_A(P) = P(A),$$

measurable for all $A \in \mathcal{B}(\mathbb{X})$ (see Proposition A.5 in Ghosal and Van der Vaart (2017)).

Definition 2.1.1. Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space. A random probability measure is a measurable map from Ω to $\mathcal{P}(\mathbb{X})$.

In the literature, random probability measures are also often defined in terms of transition probability kernels. Under the measurable structure considered here, these two formulations are equivalent. We will denote the random probability measure by $\widetilde{P}$.

A fundamental idea in Bayesian statistics is that, to make a statistical inference, the order in which the samples are observed should not matter. This symmetry is formalized by the notion of exchangeability, introduced by de Finetti.

Definition 2.1.2. A sequence of random variables $(X_i)_{i\geq 1}$ is exchangeable if

$$\big(X_1, \dots, X_n\big) \overset{d}{=} \big(X_{\sigma(1)}, \dots, X_{\sigma(n)}\big),$$

for all $n \geq 1$ and every permutation σ of $\{1, \dots, n\}$.

By de Finetti's representation theorem, exchangeability is equivalent to the assumption that the law of the observations is the mixture of product probability measures weighted by the prior distribution.

Theorem 2.1.3. *A sequence of random variables $(X_i)_{i\geq 1}$ is exchangeable if and only if there exists a probability measure $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$ such that*

$$\mathbb{P}\left(X_1 \in A_1, \dots, X_n \in A_n\right) = \int_{\mathcal{P}(\mathbb{X})} \prod_{i=1}^{n} P(A_i)\, \mathrm{d}Q(P)$$

for all $n \geq 1$ and $A_1, \dots, A_n \in \mathcal{B}(\mathbb{X})$.

A consequence of the above theorem is the following interpretation of the data generation process:

$$\begin{aligned} &\widetilde{P} \sim Q \\ &X_1, \dots, X_n \mid \widetilde{P} \overset{\text{i.i.d}}{\sim} \widetilde{P}, \end{aligned}$$

where $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$ is referred to as the prior distribution.

In Bayesian nonparametrics, one of the early central challenges was to construct prior distributions on infinite-dimensional parameter spaces whose posterior distributions remained analytically or computationally tractable. A major breakthrough in this direction was the introduction of the Dirichlet process in Ferguson (1973). The Dirichlet process provides a prior distribution over random probability measures and possesses a conjugacy property analogous to that found in finite-dimensional Bayesian models.

The method used by Ferguson to construct the Dirichlet process relies on finite-dimensional distributions. Specifically, one specifies the joint distribution of the probabilities assigned to finite measurable partitions and verifies the consistency conditions required to guarantee the existence of a random probability measure. This approach is analogous to the classical construction of stochastic processes through their finite-dimensional distributions. We now briefly describe this method.

Recall that we aim to define a random variable $\widetilde{P}$ taking values in the space of functions from $\mathcal{B}(\mathbb{X})$ to $[0, 1]$, endowed with the additional properties required of a probability measure. The main tool for this construction is the Kolmogorov extension theorem.

This result allows one to define a stochastic process by specifying a collection of finite-dimensional distributions that satisfy suitable consistency conditions. In our setting, these finite-dimensional distributions are indexed by finite collections of measurable sets $A_1, \ldots, A_n \in \mathcal{B}(\mathbb{X})$. It is sufficient, however, to define these laws on measurable partitions of the space (see page 213 of Ferguson (1973)).

Definition 2.1.4. A measurable partition of a measurable space $(\mathbb{X}, \mathcal{B}(\mathbb{X}))$ is a collection of sets $B_1, \ldots, B_k \in \mathcal{B}(\mathbb{X})$ such that the sets are pairwise disjoint and

$$\mathbb{X} = \bigcup_{i=1}^{k} B_i.$$

The first requirement is that the random measure assigns zero mass to the empty set, namely

$$\mathcal{L}\big(\widetilde{P}(\emptyset)\big) = \delta_0.$$

The second requirement is the consistency condition. This condition states that if a measurable partition is refined into a finer partition, then the induced joint distributions must agree. More precisely, consider two measurable partitions $(B_1, \ldots, B_k)$ and $(C_1, \ldots, C_m)$, where each set B_i can be expressed as a union of consecutive elements of the finer partition: for all $i = 1, \ldots, k$,

$$B_i = \bigcup_{j=r_i}^{r_{i+1}-1} C_j,$$

with indices satisfying $1 = r_1 < r_2 < \cdots < r_{k+1} = m + 1$. Then the consistency condition requires that

$$\mathcal{L}\left(\sum_{j=r_1}^{r_2-1} \widetilde{P}(C_j), \ldots, \sum_{j=r_k}^{r_{k+1}-1} \widetilde{P}(C_j)\right) = \mathcal{L}\big(\widetilde{P}(B_1), \ldots, \widetilde{P}(B_k)\big).$$

Under these conditions, Lemma 1 of Ferguson (1973) guarantees the existence of a random probability measure whose finite-dimensional distributions coincide with those specified in the construction. Using this construction, Ferguson defined the Dirichlet process.

Definition 2.1.5 (Dirichlet process)**.** A random probability measure $\widetilde{P}$ is a Dirichlet process with concentration parameter $\alpha > 0$ and base measure $P_0 \in \mathcal{P}(\mathbb{X})$, written $\widetilde{P} \sim \mathrm{DP}(\alpha, P_0)$, if for every measurable partition $(B_1, \ldots, B_k)$ of $\mathbb{X}$,

$$\big(\widetilde{P}(B_1), \ldots, \widetilde{P}(B_k)\big) \sim \mathrm{Dir}\big(\alpha P_0(B_1), \ldots, \alpha P_0(B_k)\big).$$

2.2 Completely Random Measures

Many Bayesian nonparametric models are constructed by first defining a random measure that need not be a probability measure and then transforming it into a probability measure. Completely random measures (CRM) provide a general framework for this construction.

Introduced in Kingman (1967), completely random measures can be viewed as measure-valued analogues of Lévy processes. In this setting, the index set is a σ-algebra of the underlying probability space, and the stochastic process takes the form of a random measure. The property of independent increments for Lévy processes is extended to this setting as the independence of evaluations of a random measure on disjoint sets. In Bayesian nonparametrics, such random measures are often transformed via normalization into probability measures, which can then be used as priors over distributions. Let us start with the definition of completely random measures.

Although we assume throughout that $\mathbb{X}$ is a Polish space, in Kingman (1967) the space $\mathbb{X}$ is taken to be a general set whose σ-algebra contains all singletons. Since the Borel σ-algebra on a Polish space contains all singletons, this assumption is satisfied in our setting, and we may continue working with $\mathbb{X}$ Polish.

We denote by $\mathcal{M}_B(\mathbb{X})$ the space of finite measures, and by $\mathcal{M}(\mathbb{X})$ the space of locally finite measures. We endow $\mathcal{M}_B(\mathbb{X})$ with the topology of weak convergence. Since $\mathbb{X}$ is Polish, the space $\mathcal{M}_B(\mathbb{X})$ is also Polish. We equip $\mathcal{M}_B(\mathbb{X})$ with its Borel σ-algebra denoted by $\mathcal{B}(\mathcal{M}_B(\mathbb{X}))$.

Definition 2.2.1. Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space. A random measure $\tilde{\mu}$ is a map from Ω to $\mathcal{M}(\mathbb{X})$, such that, for all $A \in \mathcal{B}(\mathbb{X})$, the map

$$\Omega \ni \omega \mapsto \tilde{\mu}(\omega)(A)$$

is measurable.

From now on, we will write $\tilde{\mu}(A)$ instead of $\tilde{\mu}(\omega)(A)$.

Definition 2.2.2. A completely random measure $\tilde{\mu}$ is a random measure such that for all finite collection of mutually disjoint sets $A_1, \dots, A_n \in \mathcal{B}(\mathbb{X})$, the random variables $\tilde{\mu}(A_1), \dots, \tilde{\mu}(A_n)$ are independent.

2.2.1 Decomposition of Completely Random Measures

A key feature of completely random measures is that, despite their general definition, they admit a decomposition into simpler components. In particular, Kingman's representation theorem shows that a completely random measure can be decomposed into a deterministic part, a part supported on fixed atoms, and a purely random atomic part. Such a decomposition is the main result of Kingman (1967) and we describe it below.

Let $\tilde{\mu}$ be a completely random measure satisfying the condition that there exists a countable collection $\{A_n\}$ of sets in $\mathcal{B}(\mathbb{X})$ such that $\mathbb{X} = \cup A_n$ and, for all $n \geq 1$,

$$\mathbb{P}\left(\tilde{\mu}(A_n) < \infty\right) > 0.$$

Then $\tilde{\mu}$ can be represented uniquely as

$$\tilde{\mu} = \mu + \sum_{x \in A} \tilde{\mu}\left(\{x\}\right) \delta_x + \sum_{i=1}^{\infty} J_i \delta_{X_i},$$

where μ is a deterministic non-atomic measure, A is a countable subset of $\mathbb{X}$, and $(J_i)_{i \geq 1}$ and $(X_i)_{i \geq 1}$ are independent sequences of random variables taking values in $(0, \infty)$ and $\mathbb{X}$, respectively.

In most of the literature, attention is restricted to the purely atomic random component of a CRM. This component can also be represented in terms of a Poisson random measure.

Recall that a Poisson random measure is a completely random measure and is characterized by its Lévy intensity ν, which is a σ-finite measure on $(0, \infty) \times \mathbb{X}$, and for every $C \in \mathcal{B}\left((0, \infty) \times \mathbb{X}\right)$ such that $\nu(C) < \infty$, one has

$$\mathcal{N}(C) \sim \text{Poisson}(\nu(C)).$$

To avoid trivial cases, we assume that $\nu \neq 0$. Note that $\mathcal{N}$ is a discrete random measure. The purely atomic random component then admits the representation, for all $A \in \mathcal{B}(\mathbb{X})$,

$$\sum_{i=1}^{\infty} J_i \delta_{X_i}(A) = \int_{(0,\infty) \times A} s \, \mathrm{d}\mathcal{N}(s, x).$$

Moreover, the Lévy intensity ν associated with $\mathcal{N}$ satisfies the integrability condition

$$\int_{(0,\infty) \times \mathbb{X}} \min\{1, s\} \, \mathrm{d}\nu(s, x) < \infty.$$

Since, under this representation, the CRM is completely characterized by the Lévy intensity ν associated with the Poisson random measure $\mathcal{N}$, we write $\tilde{\mu} \sim \text{CRM}(\nu)$. From now on, whenever we refer to CRMs, we mean the component admitting a representation in terms of a Poisson random measure. Note that, by defining the completely random measure as

$$\tilde{\mu}(A) = \int_{(0,\infty)\times A} s \, \mathrm{d}\mathcal{N}(s,x),$$

for $A \in \mathcal{B}(\mathbb{X})$, the fact that it takes values in $\mathcal{M}(\mathbb{X})$ and is measurable is guaranteed by the fact that $\mathcal{N}$ is a random measure.

2.2.2 Lévy Intensity and Laplace Functional

The integrability condition on the Lévy intensity has consequences for the completely random measure. In particular, for $\tilde{\mu} \sim \mathrm{CRM}(\nu)$, we have

$$\tilde{\mu}(\mathbb{X}) < \infty \quad \text{a.s.}$$

Indeed, note that

$$\tilde{\mu}(\mathbb{X}) = \int_{(0,1)\times\mathbb{X}} s \, \mathrm{d}\mathcal{N}(s,x) + \int_{(1,\infty)\times\mathbb{X}} s \, \mathrm{d}\mathcal{N}(s,x).$$

Let us first focus on the first term. By Campbell's theorem,

$$\mathbb{E}\left[\int_{(0,1)\times\mathbb{X}} s \, \mathrm{d}\mathcal{N}(s,x)\right] = \int_{(0,1)\times\mathbb{X}} s \, \mathrm{d}\nu(s,x) = \int_{(0,1)\times\mathbb{X}} \min\{s,1\} \, \mathrm{d}\nu(s,x) < \infty.$$

Hence,

$$\int_{(0,1)\times\mathbb{X}} s \, \mathrm{d}\mathcal{N}(s,x) < \infty \quad \text{a.s.}$$

Next, we consider the second term. Since

$$\int_{(0,\infty)\times\mathbb{X}} \min\{1,s\} \, \mathrm{d}\nu(s,x) < \infty,$$

it follows that

$$\nu\big((1,\infty)\times\mathbb{X}\big) < \infty.$$

Therefore, since $\mathcal{N}\big((1,\infty)\times\mathbb{X}\big) \sim \mathrm{Poisson}\big(\nu((1,\infty)\times\mathbb{X})\big)$, we have that $\mathcal{N}$ is a finite discrete measure. Consequently,

$$\int_{(1,\infty)\times\mathbb{X}} s \, \mathrm{d}\mathcal{N}(s,x) < \infty \quad \text{a.s.}$$

It is also important to note that a sufficient condition for $\tilde{\mu}(\mathbb{X}) > 0$ almost surely is

$$\nu((0,1)\times\mathbb{X}) = \infty.$$

Indeed, since ν is σ-finite, we can find a sequence of sets $(B_i)_{i\geq 1}$ such that

$$B_i \subset B_{i+1}, \quad \nu(B_i) < \infty, \quad \bigcup_{i\geq 1} B_i = (0,\infty) \times \mathbb{X}.$$

Then

$$\mathbb{P}(\mathcal{N}(B_i) = 0) = e^{-\nu(B_i)}.$$

Moreover, it follows that

$$\mathbb{P}\left(\mathcal{N}((0,\infty)\times\mathbb{X}) = 0\right) \leq \inf_{i\geq 1} \mathbb{P}\left(\mathcal{N}(B_i) = 0\right) = \inf_{i\geq 1} e^{-\nu(B_i)} = 0.$$

Thus,

$$\mathcal{N}((0,\infty)\times\mathbb{X}) > 0 \quad \text{a.s.},$$

and we conclude that

$$\tilde{\mu}(\mathbb{X}) > 0 \quad \text{a.s.}$$

Since a Poisson random measure is a discrete random measure of the form

$$\sum_{i\geq 1} \delta_{(J_i, X_i)},$$

the random variables J_i and X_i are called jumps and atoms, respectively.

The Lévy intensity governs the law of the jumps and atoms. By the disintegration theorem, it can be decomposed into a component corresponding to the distribution of atom locations and another component describing the distribution of jump sizes at each location.

In particular, let us denote by $\mathcal{M}_1(\nu)$ the first moment or mean of ν, defined as

$$\mathcal{M}_1(\nu) := \int_{(0,\infty)\times\mathbb{X}} s \,\mathrm{d}\nu(s,x).$$

Then, for $\tilde{\mu} \sim \mathrm{CRM}(\nu)$ with finite mean, it follows from Proposition 3 in Catalano and Lavenant (2026) that

$$\mathrm{d}\nu(s,x) = \mathrm{d}\rho_x(s)\,\mathrm{d}P_0(x),$$

where P_0 is a probability measure on $\mathbb{X}$ and $\rho : \mathbb{X} \times \mathcal{B}((0,\infty)) \to [0,\infty]$ is a transition kernel. Whenever ρ does not depend on x, such a CRM is referred to as a homogeneous CRM.

A completely random measure, similarly to a random probability measure as discussed in Section 2.1, can be uniquely identified by its finite-dimensional distributions, that is,

$$\mathcal{L}\left(\tilde{\mu}(A_1), \ldots, \tilde{\mu}(A_k)\right),$$

for a measurable partition $A_1, \ldots, A_k$ in $\mathcal{B}(\mathbb{X})$. By the independence assumption, these finite-dimensional distributions are identified once we specify the law

$$\mathcal{L}(\tilde{\mu}(A))$$

for $A \in \mathcal{B}(\mathbb{X})$. Equivalently, they are identified by the law of $\tilde{\mu}(f)$ for measurable $f : \mathbb{X} \to [0, \infty]$. For this reason, Laplace functionals play an important role.

For the CRM $\tilde{\mu}$, we have the characterization that, for all measurable functions $f : \mathbb{X} \to [0, \infty]$, or for f integrable with respect to $\tilde{\mu}$,

$$\mathbb{E}\left[e^{-\int_{\mathbb{X}} f(x)\mathrm{d}\tilde{\mu}(x)}\right] = \exp\left\{-\int_{\mathbb{R}_+ \times \mathbb{X}} \left(1 - e^{-sf(x)}\right) \mathrm{d}\nu(s, x)\right\}.$$

This result follows from Lemma J.2 in Ghosal and Van der Vaart (2017).

2.2.3 Normalization and Identifiability

Completely random measures can be transformed into random probability measures. The transformation we consider in this chapter is normalization. In particular, for $\tilde{\mu} \sim \mathrm{CRM}(\nu)$, we consider

$$\tilde{\mu} \mapsto \frac{\tilde{\mu}}{\tilde{\mu}(\mathbb{X})}.$$

As noted in the previous subsection, to guarantee that this transformation is well defined, we need the Lévy intensity to satisfy an integrability condition and $\nu((0, \infty) \times \mathbb{X}) = \infty$. However, for identifiability purposes to be discussed below, in addition to the integrability condition, we will assume a stronger condition, namely that

$$\rho_x((0, \infty)) = \infty \quad \text{for } P_0\text{-almost every } x \in \mathbb{X},$$

where ρ corresponds to the density of jumps in the decomposition of the Lévy intensity. CRMs satisfying this condition are referred to as infinitely active CRMs.

A crucial point to note in normalization is that there may exist two CRMs $\tilde{\mu}^1, \tilde{\mu}^2$ such that

$$\tilde{\mu}^1 \overset{d}{\neq} \tilde{\mu}^2, \qquad \text{but} \qquad \frac{\tilde{\mu}^1}{\tilde{\mu}^1(\mathbb{X})} \overset{d}{=} \frac{\tilde{\mu}^2}{\tilde{\mu}^2(\mathbb{X})}.$$

Therefore, if we work with a distance defined at the level of CRMs, we encounter an identifiability issue: two distinct CRMs may induce the same random probability measure after normalization, yet have strictly positive distance.

To overcome this issue, Catalano and Lavenant (2026) provides conditions under which two CRMs induce the same random probability measure. This allows one to work with a representative element from each equivalence class of CRMs that yield the same normalized measures.

The representative element within each equivalence class of CRMs that induce the same random probability measure through normalization is given by the scaled CRM, defined as follows.

Definition 2.2.3. A completely random measure $\tilde{\mu}$ is called a scaled CRM if

$$\mathbb{E}[\tilde{\mu}(\mathbb{X})] = 1.$$

If $\tilde{\mu}$ is a CRM with finite mean, meaning that $\mathcal{M}_1(\nu) < \infty$, its associated scaled CRM is defined by

$$\tilde{\mu}_S = \frac{\tilde{\mu}}{\mathbb{E}[\tilde{\mu}(\mathbb{X})]}.$$

Then, for two infinitely active completely random measures $\tilde{\mu}^1, \tilde{\mu}^2$ with finite means and associated base measures P_0^1, P_0^2 that are non-Dirac, Corollary 8 in Catalano and Lavenant (2026) states that

$$\frac{\tilde{\mu}^1}{\tilde{\mu}^1(\mathbb{X})} \overset{d}{=} \frac{\tilde{\mu}^2}{\tilde{\mu}^2(\mathbb{X})} \quad \Longleftrightarrow \quad \tilde{\mu}_S^1 \overset{d}{=} \tilde{\mu}_S^2.$$

Note that scaled CRMs are still completely random measures. Lemma 9 in Catalano and Lavenant (2026) provides a way to characterize the Lévy intensity of the scaled CRM.

In particular, let $\tilde{\mu} \sim \mathrm{CRM}(\nu)$ be such that

$$\mathrm{d}\nu(s, x) = \rho_x(s)\,\mathrm{d}s\,\mathrm{d}P_0(x).$$

Then the Lévy intensity ν_S of the scaled CRM $\tilde{\mu}_S$ is given by

$$\mathrm{d}\nu_S(s, x) = \mathbb{E}[\tilde{\mu}(\mathbb{X})]\,\rho_x\big(\mathbb{E}[\tilde{\mu}(\mathbb{X})]\,s\big)\,\mathrm{d}s\,\mathrm{d}P_0(x).$$

2.2.4 Posterior for Completely Random Measures

One of the main reasons for working with completely random measures is their flexibility for modelling and the tractability of the corresponding posterior distributions. We now describe the general form of the posterior distribution for CRMs, as developed in James et al. (2009). Consider the Bayesian model

$$\begin{aligned}\tilde{\mu} &\sim \mathrm{CRM}(\nu),\\ X_1, \dots, X_n \mid \tilde{\mu} &\overset{\text{i.i.d}}{\sim} \frac{\tilde{\mu}}{\tilde{\mu}(\mathbb{X})}.\end{aligned} \tag{2.1}$$

The prior distribution is updated through the posterior distribution. Formally, the posterior is a transition kernel, denoted by $\mathcal{L}(\tilde{\mu} \mid X_{1:n})$, which is a mapping

$$\mathcal{L}(\tilde{\mu} \mid X_{1:n}) : \mathbb{X}^n \times \mathcal{B}(\mathcal{M}_B(\mathbb{X})) \to [0, 1].$$

For a fixed sequence of observations $x_1, \dots, x_n$, we denote by

$$\tilde{\mu}^* \sim \mathcal{L}(\tilde{\mu} \mid X_{1:n} = x_{1:n})$$

a posterior random measure.

Before stating the main theorem, we introduce some additional notation. Given observations $x_1, \dots, x_n$, let $x_1^*, \dots, x_k^*$ denote the $k \leq n$ distinct values, and let, for $i \in \{1, \dots, k\}$,

$$n_i := \#\{j : x_j = x_i^*\}$$

be their corresponding multiplicities.

We also introduce an auxiliary random variable U with density

$$f_U(u) \propto u^{n-1} e^{-\psi(u)} \prod_{i=1}^{k} \tau_{n_i | x_i^*}(u)\, \mathbf{1}_{(0,\infty)}(u),$$

where ψ denotes the Laplace exponent of $\tilde{\mu}$,

$$\psi(u) = \int_{(0,\infty)\times\mathbb{X}} \left(1 - e^{-us}\right) \mathrm{d}\nu(s, x),$$

and

$$\tau_{m|x}(u) = \int_{(0,\infty)} e^{-us} s^m \, \mathrm{d}\rho_x(s).$$

We now state Theorem 1 from James et al. (2009).

Theorem 2.2.4. *Let $\tilde{\mu}$ be a CRM with Lévy intensity $\mathrm{d}\nu(s, x) = \mathrm{d}\rho_x(s)\, \mathrm{d}P_0(x)$, and let $x_1, \dots, x_n$ be observations generated from model (2.1). Define $\tilde{\mu}^*$ such that*

$$\tilde{\mu}^* \mid U \overset{d}{=} \tilde{\mu}_U + \sum_{i=1}^{k} J_i^U \delta_{x_i^*},$$

where:

(i) $\tilde{\mu}_U$ *is a CRM with Lévy intensity*

$$\mathrm{d}\nu_U(s,x) = e^{-Us}\,\mathrm{d}\rho_x(s)\,\mathrm{d}P_0(x);$$

(ii) for each $i = 1, \dots, k$*, the random variable* J_i^U *has density*

$$f_i(s) \propto s^{n_i} e^{-Us}\,\mathrm{d}\rho_{x_i^*}(s);$$

(iii) the random variables $(J_i^U)_{i=1}^k$ *are mutually independent and independent of* $\tilde{\mu}_U$*.*

Then $\mathcal{L}(\tilde{\mu}^*)$ *is a version of the posterior distribution of* $\tilde{\mu}$ *given* $X_{1:n}$ *under model* (2.1)*.*

Note that the posterior random measure is a completely random measure conditional on the random variable U. This observation motivates the notion of a Cox CRM, i.e., a random completely random measure.

Definition 2.2.5. Let $\tilde{\nu}$ be a random Lévy intensity with law F, and let $\mathcal{L}(\tilde{\mu}_\nu)$ denote the law of a CRM with fixed Lévy intensity ν.

A random measure $\tilde{\mu}$ is called a Cox CRM if, for all $A \in \mathcal{B}(\mathcal{M}_B(\mathbb{X}))$,

$$\mathbb{P}\big(\tilde{\mu} \in A\big) = \int_{\mathcal{M}((0,\infty)\times\mathbb{X})} \mathcal{L}(\tilde{\mu}_\nu)(A)\,\mathrm{d}F(\nu).$$

The statistical interpretation is that there exists a random Lévy intensity $\tilde{\nu}$ such that

$$\tilde{\mu} \mid \tilde{\nu} \sim \mathrm{CRM}(\tilde{\nu}).$$

The same identifiability issue arises for Cox CRMs as in the case of CRMs. The notion of scaling extends naturally to this setting and provides a representative element within each equivalence class of Cox CRMs that induce the same random probability measure through normalization.

Definition 2.2.6. Let $\tilde{\mu}$ be a Cox CRM with random Lévy intensity $\tilde{\nu} \sim F$, and suppose that F is concentrated on the set of Lévy intensities defining infinitely active CRMs with finite means. The random measure $\tilde{\mu}_S$ is a scaled Cox CRM if, for all $A \in \mathcal{B}(\mathcal{M}_B(\mathbb{X}))$,

$$\mathbb{P}\big(\tilde{\mu}_S \in A\big) = \int_{\mathcal{M}((0,\infty)\times\mathbb{X})} \mathcal{L}(\tilde{\mu}_{\nu,S})(A)\,\mathrm{d}F(\nu),$$

where $\tilde{\mu}_{\nu,S}$ denotes the scaled CRM associated with ν.

The statistical interpretation of the above definition is that

$$\tilde{\mu}_S \mid \tilde{\nu} = \frac{\tilde{\mu}}{\mathbb{E}[\tilde{\mu}(\mathbb{X}) \mid \tilde{\nu}]}.$$

Under additional assumptions on the Lévy intensities (see Theorem 10 in Catalano and Lavenant (2026)), scaled Cox CRMs serve as representative elements of equivalence classes. In particular, for two Cox CRMs $\tilde{\mu}^1, \tilde{\mu}^2$, we have

$$\frac{\tilde{\mu}^1}{\tilde{\mu}^1(\mathbb{X})} \stackrel{d}{=} \frac{\tilde{\mu}^2}{\tilde{\mu}^2(\mathbb{X})} \quad \Longleftrightarrow \quad \tilde{\mu}^1_S \stackrel{d}{=} \tilde{\mu}^2_S.$$

We denote by $\tilde{\nu}_S$ the random Lévy intensity associated with the scaled Cox CRM. Let us now discuss several examples of CRMs.

2.2.5 The Gamma Process and the Dirichlet Process

We begin this subsection by describing another derivation of the Dirichlet process via completely random measures.

Definition 2.2.7 (Gamma CRM)**.** A completely random measure is gamma if its Lévy intensity is

$$\mathrm{d}\nu(s,x) = \alpha \frac{e^{-bs}}{s} \mathrm{d}s \mathrm{d}P_0(x),$$

where $\alpha > 0$ is a total base measure, $b > 0$ is a scale parameter, and $P_0 \in \mathcal{P}(\mathbb{X})$ is a base probability.

The importance of the gamma CRM stems from the fact that its normalization yields a Dirichlet process. Indeed, define the random probability measure

$$\widetilde{P} = \frac{\tilde{\mu}}{\tilde{\mu}(\mathbb{X})},$$

where $\tilde{\mu}$ is a gamma CRM. We now show that $\widetilde{P}$ is a Dirichlet process with parameters α and P_0.

To this end, it is enough to verify (see Definition 2.1.5) that for any measurable partition $A_1, \ldots, A_n$ of $\mathbb{X}$,

$$\left(\widetilde{P}(A_1), \ldots, \widetilde{P}(A_n)\right) \sim \text{Dirichlet}\left(\alpha P_0(A_1), \ldots, \alpha P_0(A_n)\right).$$

Let $A \in \mathcal{B}(\mathbb{X})$. To obtain the law of $\tilde{\mu}(A)$, we compute its Laplace transform.

Define $f(x) = t\mathbf{1}_A(x)$ for $t > 0$. Then,

$$\begin{aligned}
\mathbb{E}\left[e^{-t\tilde{\mu}(A)}\right] &= \mathbb{E}\left[e^{-\int_{\mathbb{X}} f(x)\mathrm{d}\tilde{\mu}(x)}\right] \\
&= \exp\left\{-\int_{(0,\infty)\times\mathbb{X}} \left(1 - e^{-sf(x)}\right) \mathrm{d}\nu(s,x)\right\} \\
&= \exp\left\{-\alpha P_0(A) \int_0^\infty \left(1 - e^{-st}\right) \frac{e^{-bs}}{s} \mathrm{d}s\right\} \\
&= \exp\left\{-\alpha P_0(A) \int_0^\infty \frac{e^{-bs} - e^{-(b+t)s}}{s} \mathrm{d}s\right\} \\
&= \exp\left\{-\alpha P_0(A) \log\left(\frac{b+t}{b}\right)\right\} \\
&= \left(\frac{b}{b+t}\right)^{\alpha P_0(A)},
\end{aligned}$$

where the last equality follows from the Frullani integral. Hence, this coincides with the Laplace transform of the Gamma$(\alpha P_0(A), b)$ random variable. Therefore, for any measurable partition $A_1, \dots, A_n$,

$$\tilde{\mu}(A_i) \sim \text{Gamma}\left(\alpha P_0(A_i), b\right).$$

Since these random variables are independent and $\tilde{\mu}(\mathbb{X}) = \sum_{i=1}^n \tilde{\mu}(A_i)$, we obtain the desired conclusion.

2.2.6 The Pitman–Yor Process and Equivalences

Another widely used example of a random probability measure is the Pitman–Yor process. It can be defined in several ways: through exchangeable partition probability functions, stick-breaking processes, Poisson–Kingman processes, and CRMs. Here, we provide two definitions using the latter two approaches and show the equivalence between them.

We start by defining the Pitman–Yor process as a special case of a Poisson–Kingman process. Denote by $\Delta^{\downarrow} = \{(s_1, s_2, \dots) : 1 \geq s_1 \geq s_2 \geq \dots \geq 0, \quad \sum_{i=1}^\infty s_i = 1\}$. the space of decreasing sequences in $[0, 1]$.

Definition 2.2.8. Let $\mathcal{N}$ be a Poisson random measure on $(0, \infty)$ with Lévy intensity ρ. Let (J_i) be its random atoms, consider $J_{(1)} \geq J_{(2)} \geq \dots$, and let $T = \sum_{i=1}^\infty J_i$. Then, denote by $\text{PK}(\rho \mid t)$ the conditional distribution of the random vector

$$\left(\frac{J_{(1)}}{T}, \frac{J_{(2)}}{T}, \dots\right) \in \Delta^{\downarrow}$$

given $T = t$, and by $\text{PK}(\rho)$ the distribution of the same vector without conditioning.

For a probability measure η on $(0,\infty)$, the Poisson–Kingman distribution $\mathrm{PK}(\rho,\eta)$ is the law on $\Delta^{\downarrow}$ defined as

$$\int_0^\infty \mathrm{PK}(\rho \mid t)\, \mathrm{d}\eta(t).$$

The corresponding random probability measure on $\mathbb{X}$,

$$\widetilde{P} = \sum_{i\geq 1} \frac{J_i}{T}\delta_{X_i},$$

where $\left(\frac{J_i}{T}\right)_{i\geq 1} \sim \mathrm{PK}(\rho,\eta)$ and $X_1, X_2, \ldots \overset{\text{i.i.d}}{\sim} P_0 \in \mathcal{P}(\mathbb{X})$, is called a Poisson–Kingman process.

Definition 2.2.9. A Pitman–Yor process is a Poisson–Kingman process $\mathrm{PK}(\rho,\eta)$, where

$$\mathrm{d}\rho(s) = \frac{\sigma}{\Gamma(1-\sigma)} s^{-1-\sigma}\mathrm{d}s, \quad \mathrm{d}\eta(t) = \frac{\sigma\Gamma(\theta)}{\Gamma\left(\frac{\theta}{\sigma}\right)} t^{-\theta} f(t)\mathrm{d}t,$$

for $\sigma \in (0,1), \theta > 0$ and f is a probability density with Laplace transform $\lambda \mapsto e^{-\lambda^\sigma}$.

The function ρ with the form given in Definition 2.2.9 is referred to as the intensity function of the σ-stable process. In particular, the σ-stable process is defined as follows.

Definition 2.2.10. $\tilde{\mu}$ is a σ-stable completely random measure if the jump part of the Lévy intensity has the form

$$\mathrm{d}\rho(s) = \frac{\sigma}{\Gamma(1-\sigma)} s^{-1-\sigma}\mathrm{d}s,$$

for some $\sigma \in (0,1)$.

Another way to define the Pitman–Yor process is via normalization of a Cox completely random measure.

Definition 2.2.11. Let $\tilde{\mu}$ be a Cox CRM$(\tilde{\nu})$ with random Lévy intensity

$$\mathrm{d}\tilde{\nu}_\xi(s,x) = \frac{\xi\sigma}{\Gamma(1-\sigma)} \frac{e^{-s}}{s^{1+\sigma}} \mathrm{d}s\, \mathrm{d}P_0(x),$$

where $\xi \sim \Gamma\left(\frac{\theta}{\sigma}, 1\right)$ with density $g(\xi) = \frac{1}{\Gamma\left(\frac{\theta}{\sigma}\right)} \xi^{\frac{\theta}{\sigma}-1} e^{-\xi}$, $\theta \in (0,\infty)$ and $\sigma \in (0,1)$. The corresponding normalized random probability measure,

$$\widetilde{P} = \frac{\tilde{\mu}}{\tilde{\mu}(\mathbb{X})},$$

is called the Pitman–Yor process.

We now show that Definition 2.2.11 and Definition 2.2.9 define the same random probability measures.

Since we work with homogeneous CRMs, our strategy is to show that the distribution of the normalized ranked jumps induced by the Cox completely random measure from Definition 2.2.11 is $\mathrm{PK}(\rho, \eta)$, where ρ and η are as in Definition 2.2.9. To this end, we first reduce the Cox CRM to a simpler form. The motivation is that, for fixed ξ, the associated Lévy intensity resembles the exponential tilting of a σ-stable CRM.

- Step 1: Reduction of the Cox CRM.

Denote by $\tilde{\nu}_\xi^1$ and $\tilde{\nu}_\xi^2$ the random Lévy intensities defined by

$$\mathrm{d}\tilde{\nu}_\xi^1(s,x) = \frac{\xi\sigma}{\Gamma(1-\sigma)}\, s^{-1-\sigma} e^{-s}\, \mathrm{d}s\, \mathrm{d}P_0(x), \qquad \mathrm{d}\tilde{\nu}_\xi^2(s,x) = \frac{\sigma}{\Gamma(1-\sigma)}\, s^{-1-\sigma} e^{-\xi^{\frac{1}{\sigma}} s}\, \mathrm{d}s\, \mathrm{d}P_0(x),$$

and let $\tilde{\mu}^1$ and $\tilde{\mu}^2$ denote the corresponding Cox CRMs. Similarly, define the measures

$$\mathrm{d}\rho_\xi^1(s) = \frac{\xi\sigma}{\Gamma(1-\sigma)}\, s^{-1-\sigma} e^{-s}\, \mathrm{d}s, \qquad \mathrm{d}\rho_\xi^2(s) = \frac{\sigma}{\Gamma(1-\sigma)}\, s^{-1-\sigma} e^{-\xi^{\frac{1}{\sigma}} s}\, \mathrm{d}s.$$

For simplicity of notation, we use ρ^i to denote both the density and the associated measure. Fix $\xi > 0$ and note that $\rho_\xi^2(s) = \xi^{\frac{1}{\sigma}} \rho_\xi^1(\xi^{\frac{1}{\sigma}} s)$. By Lemma SM3 in the Supplementary Material Catalano and Lavenant (2026),

$$\tilde{\mu}^2 \overset{d}{=} \xi^{-\frac{1}{\sigma}} \tilde{\mu}^1.$$

Thus, for fixed ξ, both CRMs induce the same random probability measure, and consequently the two Cox CRMs also induce the same random probability measure.

- Step 2: Matching the distributions of normalized ranked jumps.

Working with the reduced Cox CRM, define

$$\mathrm{d}\tilde{\nu}_\xi(s,x) = \frac{\sigma}{\Gamma(1-\sigma)}\, s^{-1-\sigma} e^{-\xi^{\frac{1}{\sigma}} s}\, \mathrm{d}s\, \mathrm{d}P_0(x), \qquad \mathrm{d}\rho_\xi(s) = \frac{\sigma}{\Gamma(1-\sigma)}\, s^{-1-\sigma} e^{-\xi^{\frac{1}{\sigma}} s}\, \mathrm{d}s,$$

and let $\tilde{\mu}$ denote the associated Cox CRM.

Let $\tilde{\gamma}$ denote the distribution of the normalized ranked jumps from $\tilde{\mu}$. By definition, for every $A \in \mathcal{B}(\Delta^\downarrow)$,

$$\tilde{\gamma}(A) = \int_0^\infty \mathrm{PK}(\rho_\xi)(A)\, g(\xi)\, \mathrm{d}\xi,$$

where g is the density of $\mathrm{Gamma}\big(\frac{\theta}{\sigma}, 1\big)$.

Disintegrating $\mathrm{PK}(\rho_\xi)$ with respect to the distribution of the total jump sum gives

$$\tilde{\gamma}(A) = \int_0^\infty \int_0^\infty \mathrm{PK}(\rho_\xi \mid t)(A)\, f_\xi(t)\, \mathrm{d}t\, g(\xi)\, \mathrm{d}\xi,$$

where f_ξ is the density of the total jump sum of $\mathrm{CRM}(\tilde{\nu}_\xi)$ for fixed ξ, and $\mathrm{PK}(\rho_\xi \mid t)$ denotes the conditional distribution of the normalized ranked jumps given total jump sum t.

Since ρ_ξ is the exponential tilting of the Lévy measure of a σ-stable CRM, we have $\mathrm{PK}(\rho_\xi \mid t) = \mathrm{PK}(\rho \mid t)$ (see Example 14.46 of Ghosal and Van der Vaart (2017)). Therefore,

$$\tilde{\gamma}(A) = \int_0^\infty \int_0^\infty \mathrm{PK}(\rho \mid t)(A)\, f_\xi(t)\, \mathrm{d}t\, g(\xi)\, \mathrm{d}\xi = \int_0^\infty \mathrm{PK}(\rho \mid t)(A) \left[\int_0^\infty f_\xi(t)\, g(\xi)\, \mathrm{d}\xi\right] \mathrm{d}t.$$

It remains to show that

$$\int_0^\infty f_\xi(t)\, g(\xi)\, \mathrm{d}\xi = \frac{\sigma\, \Gamma(\theta)}{\Gamma\big(\frac{\theta}{\sigma}\big)}\, t^{-\theta}\, f(t),$$

where f is the density of the total jump sum of a σ-stable CRM. The density f is characterized by its Laplace transform $\lambda \mapsto e^{-\psi(\lambda)}$ with $\psi(\lambda) = \lambda^\sigma$ (see Example 14.43 in Ghosal and Van der Vaart (2017)). By Example 14.46 in Ghosal and Van der Vaart (2017),

$$f_\xi(t) = f(t)\, e^{\psi\left(\xi^{\frac{1}{\sigma}}\right) - \xi^{\frac{1}{\sigma}} t} = f(t)\, e^{\xi - \xi^{\frac{1}{\sigma}} t}.$$

Hence,

$$\begin{aligned}
\int_0^\infty f_\xi(t)\, g(\xi)\, \mathrm{d}\xi &= \int_0^\infty f(t)\, e^{\xi - \xi^{\frac{1}{\sigma}} t}\, \frac{1}{\Gamma\big(\frac{\theta}{\sigma}\big)}\, \xi^{\frac{\theta}{\sigma} - 1} e^{-\xi}\, \mathrm{d}\xi \\
&= \frac{f(t)}{\Gamma\big(\frac{\theta}{\sigma}\big)} \int_0^\infty \xi^{\frac{\theta}{\sigma} - 1} e^{-\xi^{\frac{1}{\sigma}} t}\, \mathrm{d}\xi \\
&= \frac{\sigma\, \Gamma(\theta)}{\Gamma\big(\frac{\theta}{\sigma}\big)}\, t^{-\theta}\, f(t),
\end{aligned}$$

where the last equality follows from the substitution $\tau = \xi^{\frac{1}{\sigma}}$.

2.3 Distances Between Laws of Random Measures and RPMs

In this section, we describe several ways to define distances on the laws of random probability measures and, more generally, on random measures. The distances on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$

that are mainly used for two-sample testing are Wasserstein over Wasserstein (WoW) and the Hierarchical Integral Probability Metric (HIPM). The other distances we consider for random measures are used to study the merging of opinions.

2.3.1 Distances on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$

We consider two distance functions on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$. The first, referred to as Wasserstein over Wasserstein, is the natural lifting of the Wasserstein distance on $\mathcal{P}(\mathbb{X})$. As we mentioned in Section 1.2, it has been widely used in Bayesian nonparametrics Nguyen (2016); Nguyen and Wei (2026) and machine learning Ho et al. (2017); Dukler et al. (2019); Alvarez-Melis and Fusi (2020); Piening et al. (2026), with recent works also studying its geometric structure Pinzi and Savaré (2025a,b); Beiglböck et al. (2025) and the gradient flows it generates Bonet et al. (2025). The second, termed the Hierarchical Integral Probability Metric, is a very recent development defined in Catalano and Lavenant (2024). Let us start with the definition of Wasserstein over Wasserstein.

Only in this subsection and in the chapter on two-sample testing, we assume that the sample space $\mathbb{X}$ is Polish with bounded diameter. Using the distance function $d_{\mathbb{X}}$ on $\mathbb{X}$, one can define the Wasserstein distance on $\mathcal{P}(\mathbb{X})$ by

$$\mathcal{W}_1(P^1, P^2) = \inf_{\pi \in \Gamma(P^1,P^2)} \int_{\mathbb{X}\times\mathbb{X}} d_{\mathbb{X}}(x,y)\, \pi(dx,dy), \qquad P^1, P^2 \in \mathcal{P}(\mathbb{X}),$$

where

$$\Gamma(P^1,P^2) = \left\{\pi \in \mathcal{P}(\mathbb{X}^2) : \pi(A\times\mathbb{X}) = P^1(A),\ \pi(\mathbb{X}\times A) = P^2(A) \quad \forall A \in \mathcal{B}(\mathbb{X})\right\}.$$

Note that this is the Wasserstein distance of order 1. Since we do not consider other Wasserstein distances, we denote it simply by $\mathcal{W}$ instead of $\mathcal{W}_1$.

In general, $\mathcal{W}$ is a distance function on the Wasserstein space of order 1, defined as

$$\mathcal{P}_1(\mathbb{X}) = \left\{P \in \mathcal{P}(\mathbb{X}) : \int_{\mathbb{X}} d_{\mathbb{X}}(x,x_0) P(dx) < \infty, \quad x_0 \in \mathbb{X}\right\}.$$

Since we assume that $\mathbb{X}$ has bounded diameter, all $P \in \mathcal{P}(\mathbb{X})$ belong to $\mathcal{P}_1(\mathbb{X})$. The assumption that $\mathbb{X}$ is Polish and $d_{\mathbb{X}}$ is bounded implies that the metric space $(\mathcal{P}(\mathbb{X}), \mathcal{W})$ is also Polish; see Remark 7.1.7 in Ambrosio et al. (2008). Note also that $(\mathcal{P}(\mathbb{X}), \mathcal{W})$ has bounded diameter, since $d_{\mathbb{X}}$ is bounded. Therefore, instead of defining a distance function on $\mathcal{P}_1(\mathcal{P}(\mathbb{X}))$, we can define a distance function on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ using $\mathcal{W}$ as the ground metric.

Definition 2.3.1 (Wasserstein over Wasserstein)**.** Given $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, the Wasserstein over Wasserstein (WoW) distance between Q^1 and Q^2 is

$$\mathcal{W}_{\mathcal{W}}(Q^1, Q^2) = \inf_{\pi \in \Gamma(Q^1, Q^2)} \int_{\mathcal{P}(\mathbb{X}) \times \mathcal{P}(\mathbb{X})} \mathcal{W}(P^1, P^2)\, \pi(dP^1, dP^2).$$

Here, $\Gamma(Q^1, Q^2)$ denotes the set of probability measures on $\mathcal{P}(\mathbb{X}) \times \mathcal{P}(\mathbb{X})$ with marginals Q^1 and Q^2. Again, Remark 7.1.7 in Ambrosio et al. (2008), together with the fact that $\mathcal{W}$ has bounded diameter, implies that the metric space $(\mathcal{P}(\mathcal{P}(\mathbb{X})), \mathcal{W}_{\mathcal{W}})$ is Polish with bounded diameter.

The second distance function we consider is HIPM, introduced in Catalano and Lavenant (2024). The primary intuition behind its definition is as follows. Let Q^1, Q^2 be probability measures on $\mathcal{P}(\mathbb{X})$, and let $\widetilde{P}^1 \sim Q^1$ and $\widetilde{P}^2 \sim Q^2$. We compute the distance between Q^1 and Q^2 by considering the distances between the random integrals $\widetilde{P}^1(f)$ and $\widetilde{P}^2(f)$, where f ranges over a class of functions from $\mathbb{X}$ to $\mathbb{R}$. By considering these projections, we shift the focus from $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ to $\mathcal{P}(\mathbb{R})$, the latter being significantly easier to handle. Such a projection is justified by the fact that, for a random probability measure $\widetilde{P}$, the laws of the random integrals $\widetilde{P}(f)$, where f belongs to the set of continuous and bounded functions, are sufficient to characterize the law of $\widetilde{P}$ (see Theorem 4.11 in Kallenberg (2017)). This leads to the following definition.

Definition 2.3.2 (Hierarchical integral probability metric)**.** Given $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, the hierarchical integral probability metric between Q^1 and Q^2 is defined as

$$d_{\mathcal{F}}(Q^1, Q^2) = \sup_{f \in \mathcal{F}} \mathcal{W}(\mathcal{L}(\widetilde{P}^1(f)), \mathcal{L}(\widetilde{P}^2(f))),$$

where $\widetilde{P}^k \sim Q^k$ for $k = 1, 2$ and $\mathcal{F}$ is a class of bounded and measurable functions from $\mathbb{X}$ to $\mathbb{R}$.

Alternatively, the HIPM can be expressed as

$$d_{\mathcal{F}}(Q^1, Q^2) = \sup_{f \in \mathcal{F}} \mathcal{W}(Q^1 \circ \hat{f}^{-1}, Q^2 \circ \hat{f}^{-1}),$$

where $\hat{f}$ denotes the lifting of the function f, i.e., $\hat{f}(P) = \int_{\mathbb{X}} f \, \mathrm{d}P$, and $Q^k \circ \hat{f}^{-1}$ is the corresponding push-forward measure.

As long as the function class $\mathcal{F}$ is sufficiently rich, this construction defines a valid distance on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$. Hereafter, we set $\mathcal{F} = \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})$ and denote the corresponding metric $d_{\mathcal{F}}$ by d_{Lip}.

Both distance functions possess desirable topological properties. In particular, by Theorem 3.1 in Catalano and Lavenant (2024), WoW metrizes weak convergence on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$, and HIPM does the same when $\mathbb{X}$ is compact.

There is a relation between the two distance functions; specifically, they are special cases of an Integral Probability Metric defined on a generic Polish space.

Definition 2.3.3 (Integral Probability Metric)**.** Let $(\mathbb{Y}, d_{\mathbb{Y}})$ be a Polish space and $\mathcal{H}$ a class of bounded and measurable functions from $\mathbb{Y}$ to $\mathbb{R}$. The Integral Probability Metric (IPM) between $P^1, P^2 \in \mathcal{P}(\mathbb{Y})$ is

$$\mathcal{I}_{\mathcal{H}}(P^1, P^2) = \sup_{f \in \mathcal{H}} \left| \int_{\mathbb{Y}} f(y) P^1(dy) - \int_{\mathbb{Y}} f(y) P^2(dy) \right|.$$

First, let us show that WoW can be written as an integral probability metric. By Remark 6.5 in Villani (2009), if P^1, P^2 belong to $\mathcal{P}_1(\mathbb{X})$, the Wasserstein space of order 1, then the Wasserstein distance can be written as

$$\mathcal{W}(P^1, P^2) = \sup_{f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})} \left| \int_{\mathbb{X}} f(x) P^1(dx) - \int_{\mathbb{X}} f(x) P^2(dx) \right|, \tag{2.2}$$

where $\mathrm{Lip}_1(\mathbb{X}, \mathbb{R})$ denotes the set of 1-Lipschitz continuous functions from $\mathbb{X}$ to $\mathbb{R}$. Since $(\mathcal{P}(\mathcal{P}(\mathbb{X})), \mathcal{W}_{\mathcal{W}})$ is a Polish space with bounded diameter, we can use the above result to obtain, for all $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$,

$$\mathcal{W}_{\mathcal{W}}(Q^1, Q^2) = \sup_{f \in \mathrm{Lip}_1(\mathcal{P}(\mathbb{X}), \mathbb{R}, \mathcal{W})} \left| \int_{\mathcal{P}(\mathbb{X})} f(P) Q^1(dP) - \int_{\mathcal{P}(\mathbb{X})} f(P) Q^2(dP) \right|,$$

where $\mathrm{Lip}_1(\mathcal{P}(\mathbb{X}), \mathbb{R}, \mathcal{W})$ denotes the set of 1-Lipschitz continuous functions from $\mathcal{P}(\mathbb{X})$ to $\mathbb{R}$ with respect to $\mathcal{W}$.

Similarly, we show that HIPM can be written as an integral probability metric. For $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$,

$$\begin{aligned}
d_{\mathrm{Lip}}(Q^1, Q^2) &= \sup_{f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})} \mathcal{W}(Q^1 \circ \hat{f}^{-1}, Q^2 \circ \hat{f}^{-1}) \\
&= \sup_{f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})} \sup_{g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R})} \left| \int_{\mathbb{R}} g(x)(Q^1 \circ \hat{f}^{-1})(dx) - \int_{\mathbb{R}} g(x)(Q^2 \circ \hat{f}^{-1})(dx) \right| \\
&= \sup_{f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})} \sup_{g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R})} \left| \int_{\mathcal{P}(\mathbb{X})} g(\widehat{f}(P)) Q^1(dP) - \int_{\mathcal{P}(\mathbb{X})} g(\widehat{f}(P)) Q^2(dP) \right| \\
&= \sup_{f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})} \sup_{g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R})} \left| \int_{\mathcal{P}(\mathbb{X})} g(P(f)) Q^1(dP) - \int_{\mathcal{P}(\mathbb{X})} g(P(f)) Q^2(dP) \right| \\
&= \sup_{h \in \mathcal{D}} \left| \int_{\mathcal{P}(\mathbb{X})} h(P) Q^1(dP) - \int_{\mathcal{P}(\mathbb{X})} h(P) Q^2(dP) \right|,
\end{aligned}$$

where $\mathcal{D} = \{h : \mathcal{P}(\mathbb{X}) \to \mathbb{R} : h(P) = g(P(f)),\ g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R}), f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})\}$.

By representing both distance functions as Integral Probability Metrics, we see that HIPM is upper bounded by the WoW distance. Indeed, we can show that $\mathcal{D}$ is a subset of $\mathrm{Lip}_1(\mathcal{P}(\mathbb{X}), \mathbb{R}, \mathcal{W})$: let $h \in \mathcal{D}$ and $P^1, P^2 \in \mathcal{P}(\mathbb{X})$; then

$$|h(P^1) - h(P^2)| = |g(P^1(f)) - g(P^2(f))| \leq |P^1(f) - P^2(f)| \leq \sup_{f \in \mathrm{Lip}_1(\mathbb{X},\mathbb{R})} |P^1(f) - P^2(f)|$$
$$= \mathcal{W}(P^1, P^2).$$

2.3.2 Distances on $\mathcal{M}_B(\mathbb{X})$

Unlike in the previous subsection, we only assume that $\mathbb{X}$ is a Polish space equipped with its Borel σ-algebra $\mathcal{B}(\mathbb{X})$. Our motivation for defining a distance between laws of random measures stems from the study of completely random measures. Since we will be working with CRMs taking values in the space of finite measures on $\mathbb{X}$, we begin by introducing a distance on the space $\mathcal{M}_B(\mathbb{X})$.

Recall that the space $\mathcal{M}_B(\mathbb{X})$ is endowed with the Borel σ-algebra generated by the topology of weak convergence. Analogously to the case of $\mathcal{P}(\mathbb{X})$, this σ-algebra coincides with the smallest σ-algebra that makes all maps of the form

$$\mu \mapsto \mu(f), \quad f \in C_b(\mathbb{X}),$$

measurable.

Since two measures in $\mathcal{M}_B(\mathbb{X})$ may have different total masses, the standard Wasserstein distance is not directly applicable. Instead, we consider the bounded Lipschitz distance.

Definition 2.3.4. Given $\mu^1, \mu^2 \in \mathcal{M}_B(\mathbb{X})$, the bounded Lipschitz distance BL is defined as

$$\mathrm{BL}(\mu^1, \mu^2) = \sup \left\{ \int_{\mathbb{X}} f \mathrm{d}\mu^1 - \int_{\mathbb{X}} f \mathrm{d}\mu^2 \ : \ f \text{ is 1-Lipschitz and } \sup_{x \in \mathbb{X}} |f(x)| \leq 1 \right\}.$$

By lifting the distance BL to the space of laws of random measures, denoted by $\mathcal{P}(\mathcal{M}_B(\mathbb{X}))$, we obtain a distance on $\mathcal{P}(\mathcal{M}_B(\mathbb{X}))$.

Definition 2.3.5. Given $Q^1, Q^2 \in \mathcal{P}(\mathcal{M}_B(\mathbb{X}))$, the distance $\mathcal{W}_{\mathrm{BL}}$ is defined as

$$\mathcal{W}_{\mathrm{BL}}(Q^1, Q^2) = \min_{\pi \in \Gamma(Q^1, Q^2)} \int_{\mathcal{M}_B(\mathbb{X}) \times \mathcal{M}_B(\mathbb{X})} \mathrm{BL}(\mu^1, \mu^2) \, \mathrm{d}\pi(\mu^1, \mu^2),$$

where $\Gamma(Q^1, Q^2)$ denotes the set of probability measures on $\mathcal{M}_B(\mathbb{X}) \times \mathcal{M}_B(\mathbb{X})$ with marginals Q^1 and Q^2.

Remark 2.3.6. As mentioned in the previous subsection, the Wasserstein distance is, in general, a metric on the space of probability measures with finite first moments. In the present setting, since BL is bounded, $\mathcal{W}_{\mathrm{BL}}$ is a well-defined distance on $\mathcal{P}(\mathcal{M}_B(\mathbb{X}))$.

If $\tilde{\mu}^1$ and $\tilde{\mu}^2$ are random measures with laws Q^1 and Q^2, respectively, we refer to $\mathcal{W}_{\mathrm{BL}}(\tilde{\mu}^1, \tilde{\mu}^2)$ as the distance between the random measures, with the understanding that this denotes the distance between their laws, i.e., $\mathcal{W}_{\mathrm{BL}}(Q^1, Q^2)$. For a more detailed overview of the properties of $\mathcal{W}_{\mathrm{BL}}$, see Catalano and Lavenant (2026).

2.3.3 Distance for Completely Random Measures

In the previous subsection, we defined a distance between laws of random measures taking values in the space of finite measures. Although this framework also includes completely random measures, the resulting distance is generally not tractable in this setting. Since our goal is to work with distances that can be explicitly computed or bounded, we now introduce a distance tailored specifically to CRMs.

As discussed earlier, a completely random measure is uniquely characterized by its Lévy intensity, which is a σ-finite measure on $(0, \infty) \times \mathbb{X}$. For this reason, we define the distance at the level of Lévy intensities. Note that, since Lévy intensities have infinite total mass, we cannot use the bounded Lipschitz distance. Recall that, given a Lévy intensity ν, its disintegration can be written as

$$\mathrm{d}\nu(s, x) = \mathrm{d}\rho_x(s)\mathrm{d}P_0(x),$$

where P_0 is a probability measure on $\mathbb{X}$ and, for each $x \in \mathbb{X}$, ρ_x is a measure on $(0, \infty)$. This decomposition suggests defining a discrepancy between two Lévy intensities by coupling atoms in $\mathbb{X}$ and comparing the corresponding measures on $(0, \infty)$ along such a coupling. To do so, we first need a distance on the space of measures on $(0, \infty)$.

Since the measures ρ_x may have infinite total mass, neither the standard Wasserstein distance nor the bounded Lipschitz distance is directly applicable. We therefore use the notion introduced in Figalli and Gigli (2010), which extends the Wasserstein distance to positive measures that may have infinite total mass.

Definition 2.3.7. Let ρ^1, ρ^2 be two positive Borel measures on $(0, \infty)$ with finite first moments about 0. The extended Wasserstein distance between ρ^1 and ρ^2 is defined by

$$\mathcal{W}_*(\rho^1, \rho^2) = \int_0^\infty |U_1(s) - U_2(s)| \mathrm{d}s,$$

where

$$U_i(s) = \rho^i((s, \infty))$$

is the tail integral of ρ^i.

Having introduced distances on measures on $(0,\infty)$ and on the space $\mathbb{X}$, we can now define a discrepancy between Lévy intensities.

Definition 2.3.8. Let ν^1, ν^2 be two Lévy intensities such that $\mathcal{M}_1(\nu^i) < \infty$, and let

$$\mathrm{d}\nu^i(s,x) = \mathrm{d}\rho^i_x(s)\mathrm{d}P^i_0(x),$$

for $i = 1,2$, be their canonical decompositions. The adapted extended Wasserstein discrepancy between ν^1 and ν^2 is defined as

$$\mathrm{AD}(\nu^1,\nu^2) = \inf_{\pi\in\Gamma(P^1_0,P^2_0)} \int_{\mathbb{X}\times\mathbb{X}} \left[\frac{\mathcal{M}_1(\nu^1)+\mathcal{M}_1(\nu^2)}{2} d_{\mathbb{X},t}(x,y) + \mathcal{W}_*(\rho^1_x,\rho^2_y)\right] \mathrm{d}\pi(x,y),$$

where

$$d_{\mathbb{X},t}(x,y) = \min\{d_{\mathbb{X}}(x,y), 2\}$$

denotes the truncated metric on $\mathbb{X}$.

The use of the truncated metric $d_{\mathbb{X},t}$ instead of the original metric $d_{\mathbb{X}}$ is motivated by the bounds developed later. More precisely, our goal is to upper bound Wasserstein over bounded Lipschitz by the discrepancy AD. One of the key ingredients in that argument is the fact that the bounded Lipschitz distance is upper bounded by the Wasserstein distance on $\mathcal{P}(\mathbb{X})$. Working with the truncated metric leads to sharper bounds.

It is also worth noting that AD is not a distance. Indeed, the factor

$$\frac{\mathcal{M}_1(\nu^1)+\mathcal{M}_1(\nu^2)}{2}$$

makes the triangle inequality fail. Thus, AD should be regarded as a discrepancy rather than a distance. The following remark is stated in Catalano and Lavenant (2026) (see Remark 2). Due to its importance for the computations involved in merging, we restate it here.

Remark 2.3.9. Consider two completely random measures with Lévy intensities ν^1, ν^2 of the form

$$\mathrm{d}\nu^1(s,x) = \mathrm{d}\rho^1_x(s)\mathrm{d}P^1_0(x), \qquad \mathrm{d}\nu^2(s,x) = \mathrm{d}\rho^2(s)\mathrm{d}P^2_0(x),$$

that is, one of them is a homogeneous CRM. Then the adapted extended Wasserstein discrepancy between ν^1 and ν^2 decomposes as the sum of a scaled distance between the laws of the atoms and the expected distance between the laws of the jumps, i.e.,

$$\mathrm{AD}(\nu^1,\nu^2) = \frac{\mathcal{M}_1(\nu^1)+\mathcal{M}_1(\nu^2)}{2}\mathcal{W}(P^1_0,P^2_0) + \mathbb{E}_{X\sim P^1_0}\left[\mathcal{W}_*(\rho^1_X,\rho^2)\right].$$

We now discuss the distance between Cox CRMs. The motivation stems from the fact that posterior distributions are random completely random measures, namely Cox CRMs. Since Cox CRMs are characterized by random Lévy intensities, it is natural to define a distance at the level of their laws by lifting the adapted extended Wasserstein discrepancy.

More precisely, let $\tilde{\nu}^1, \tilde{\nu}^2$ be random Lévy intensities. We define

$$\mathcal{W}_{\mathrm{AD}}(\mathcal{L}(\tilde{\nu}^1), \mathcal{L}(\tilde{\nu}^2)) = \inf_{\pi \in \Gamma(\mathcal{L}(\tilde{\nu}^1), \mathcal{L}(\tilde{\nu}^2))} \mathbb{E}_{(\nu^1, \nu^2) \sim \pi} \left[\mathrm{AD}(\nu^1, \nu^2) \right].$$

Finally, since in the case of merging rates our goal is to consider CRMs (or Cox CRMs) that uniquely determine a random probability measure through normalization, we restrict attention to scaled CRMs and Cox CRMs. In this case, the first moment of the Lévy intensity $\mathcal{M}_1(\nu)$ is equal to 1. As a consequence, by Proposition 5 in Catalano and Lavenant (2026), the quantity $\mathcal{W}_{\mathrm{AD}}$ defines a distance on the corresponding laws.

Our main tool for controlling the Wasserstein over bounded Lipschitz distance is provided by Theorem 6 in Catalano and Lavenant (2026), which states that, in the case of scaled Cox CRMs $\tilde{\mu}^1, \tilde{\mu}^2$ with random Lévy intensities $\tilde{\nu}^1, \tilde{\nu}^2$,

$$\mathcal{W}_{\mathrm{BL}}\big(\mathcal{L}\left(\tilde{\mu}^1\right), \mathcal{L}\left(\tilde{\mu}^2\right)\big) \leq \mathcal{W}_{\mathrm{AD}}(\mathcal{L}(\tilde{\nu}^1), \mathcal{L}(\tilde{\nu}^2)).$$

Chapter 3

Two-sample test for laws of random probabilities

3.1 Introduction

The two-sample test is a fundamental statistical procedure used to test the null hypothesis that two datasets are generated from the same underlying distribution. Specifically, given two sets of samples, the goal is to assess whether their associated distributions are statistically indistinguishable. The hypotheses are formulated as follows:

- Null hypothesis (H_0): The two distributions are identical.
- Alternative hypothesis (H_1): The two distributions are distinct.

Classic examples include the Kolmogorov–Smirnov test Kolmogorov (1933); Smirnov (1939), which compares empirical cumulative distribution functions derived from independent and identically distributed samples. Other well-known univariate methods include the classical t-test and the Wilcoxon rank-sum test. As modern applications frequently involve multivariate observations in $\mathbb{R}^d$, several extensions have been developed, including kernel-based methods Gretton et al. (2012) and approaches based on the Wasserstein distance Ramdas et al. (2017); Wang et al. (2022, 2021); Hu and Lin (2025).

Increasingly, however, data objects have become more complex, often appearing as probability measures themselves. Settings where observed samples are modeled as probability measures are now prevalent (see for an overview Petersen et al. (2022)); examples include mortality distributions across different countries, income distributions within various regions, and distributional summaries of functional connectivity in the brain. Two-sample testing, in the case of probability measures as samples, naturally motivates the study of the laws of random probability measures, that is, probability measures defined on the space of probability measures.

This problem was previously addressed in Dubey and Müller (2019) using Fréchet means and variances. However, that approach has limitations: specifically, if the laws are not uniquely characterized by their Fréchet means and variances, the test may fail to provide a high rejection rate when the null hypothesis is false. In addition, the energy test introduced in Székely and Rizzo (2004) can be generalized to general metric spaces; however, it may be challenging to verify that the space we consider satisfies the conditions required for such a method to apply (see Subsection 3.6.2). We adopt a fully nonparametric perspective, making no distributional assumptions on the underlying laws of the random probability measures. It is important to mention that, in the majority of cases, these probability measures are latent and are instead estimated from the raw observations they generate. Therefore, theoretical guarantees are provided directly in terms of the raw observations used to estimate the underlying probability measures.

In this work, we adopt a distance-based approach. This framework first requires a metric to quantify the distance between two datasets. To decide whether to reject the null hypothesis, the core idea is that if the observed distance is sufficiently “large,” it is “unlikely” that the underlying distributions are the same. To formalize the terms ”large” and ”unlikely,” one must determine a threshold based on a pre-specified significance level θ. This threshold is defined such that the probability of the observed distance exceeding it under the null hypothesis is at most θ. Consequently, the testing procedure is to reject the null hypothesis if the observed distance exceeds this threshold.

Our objective is to establish theoretical guarantees for both the false positive rate (the probability of incorrectly rejecting the null hypothesis when it is true) and the power (the probability of correctly rejecting the null hypothesis when it is false).

The rest of this chapter is organized as follows. In Section 3.2, we formalize the sampling scheme and discuss its connections to exchangeability. Section 3.3 reviews existing distance functions on laws of random probability measures. In Section 3.4, we develop a theoretical threshold based on Rademacher complexity and illustrate its practical limitations through simulations. As a remedy, in Section 3.5, we introduce a practical threshold based on a permutation approach and establish its theoretical guarantees. Finally, in Section 3.6, we benchmark our method against existing approaches in the literature on simulated datasets and apply it to a mortality dataset.

3.2 Hierarchical Sampling and Exchangeability

We assume that the space $\mathbb{X}$ on which probability measures are defined is a Polish space (i.e., a complete, separable metric space) with a bounded diameter, meaning $\mathrm{diam}(\mathbb{X}) = \sup_{x,y\in\mathbb{X}} d_{\mathbb{X}}(x, y) < \infty$. Recall that $\mathcal{P}(\mathbb{X})$ denotes the space of probability measures on $\mathbb{X}$, and $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ denotes the space of probability measures on $\mathcal{P}(\mathbb{X})$. Unless stated otherwise, the σ-algebra on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ is the Borel σ-algebra generated by the topology of weak convergence, where the continuous functions on $\mathcal{P}(\mathbb{X})$ are defined through the weak convergence on $\mathcal{P}(\mathbb{X})$. We refer to Kallenberg (2017) for a detailed study of the topological and measurable properties of $\mathcal{P}(\mathcal{P}(\mathbb{X}))$.

In the standard two-sample testing framework, for given laws of random probability measures $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, one observes two samples

$$\widetilde{P}_1^1, \ldots, \widetilde{P}_n^1 \overset{\text{i.i.d}}{\sim} Q^1, \quad \widetilde{P}_1^2, \ldots, \widetilde{P}_n^2 \overset{\text{i.i.d}}{\sim} Q^2.$$

In practice, it is rare to observe the probability measures themselves directly, especially when dealing with continuous probability distributions. Rather, one obtains observations from each latent probability measure. Therefore, we assume that given a law of a random probability measure $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, instead of observing $\widetilde{P}_1, \ldots, \widetilde{P}_n$ from Q, we observe samples from each $\widetilde{P}_i$. Such a collection of samples is referred to as a hierarchical sample:

Definition 3.2.1 (Hierarchical sample)**.** Given $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, the hierarchical sample refers to the collection of random variables $\{X_{i,j} : i = 1, \ldots, n,\ j = 1, \ldots, m\}$, sampled as

$$\begin{pmatrix} X_{1,1} & X_{1,2} & \cdots & X_{1,m} \mid \widetilde{P}_1 \overset{\text{i.i.d}}{\sim} \widetilde{P}_1 \\ X_{2,1} & X_{2,2} & \cdots & X_{2,m} \mid \widetilde{P}_2 \overset{\text{i.i.d}}{\sim} \widetilde{P}_2 \\ \vdots & \vdots & \ddots & \vdots \\ X_{n,1} & X_{n,2} & \cdots & X_{n,m} \mid \widetilde{P}_n \overset{\text{i.i.d}}{\sim} \widetilde{P}_n \end{pmatrix},$$

where $\widetilde{P}_1, \ldots, \widetilde{P}_n \overset{\text{i.i.d}}{\sim} Q$.

Throughout this chapter, n and m will refer to the number of rows and columns of the hierarchical sample, respectively. We denote the hierarchical sample by $\mathbf{X}_{n,m} = (X_{i,j})$, where boldface letters refer to matrices and non-bold letters refer to their individual elements. With a slight abuse of notation, we will also sometimes denote $\mathbf{X}_{n,m} \sim Q$. Hierarchical samples are what we observe to decide whether two laws of random probability measures are the same.

Hierarchical samples are not new and appear across several literatures under different names and motivations. In Baxter (2000), hierarchical samples arise in the context of task-specific data: the latent probability measures correspond to task-specific data-generating distributions, while the law of the random probability measure describes the distribution over tasks. In Szabó et al. (2016), hierarchical samples are introduced as a two-stage sampled setting and are used to study consistency for distribution regression problems. They also appear in Haviv et al. (2024) in the context of generative modeling over families of distributions, and in Warren et al. (2025), where samples are used to consistently recover principal curves whose points are probability measures in Wasserstein space. Let us now discuss the limitations of the hierarchical sampling framework in the context of the two-sample testing problem.

Let $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Suppose that we observe two hierarchical samples $\mathbf{X}_{n,m} \sim Q^1$ and $\mathbf{Y}_{n,m} \sim Q^2$. Note that the n sequences of length m from the first hierarchical sample are i.i.d. draws from the following law on $\mathbb{X}^m$: for $i = 1, \ldots, n$,

$$P^{1,(m)}(A_1 \times \cdots \times A_m) := \mathbb{P}(X_{i,1} \in A_1, \ldots, X_{i,m} \in A_m) = \int_{\mathcal{P}(\mathbb{X})} \prod_{j=1}^{m} P(A_j) Q^1(dP).$$

Similarly, for the second hierarchical sample, we have

$$P^{2,(m)}(A_1 \times \cdots \times A_m) := \mathbb{P}(Y_{i,1} \in A_1, \ldots, Y_{i,m} \in A_m) = \int_{\mathcal{P}(\mathbb{X})} \prod_{j=1}^{m} P(A_j) Q^2(dP).$$

Such a characterization implies that the sequences $X_{i,1}, \ldots, X_{i,m}$ and $Y_{i,1}, \ldots, Y_{i,m}$ are finitely exchangeable. This means that each of the hierarchical samples, $\mathbf{X}_{n,m}$ and $\mathbf{Y}_{n,m}$, contains n finitely exchangeable sequences that are independent of one another. It is a well-known fact in Bayesian nonparametrics that a finite exchangeable sequence is not always in one-to-one correspondence with the law of a random probability measure. An implication of this is that it might happen that the laws of the random probability measures are different, yet the hierarchical samples we observe are statistically indistinguishable; that is, all the finite exchangeable sequences we observe have the same probability distribution.

Indeed, consider two distinct Dirichlet processes

$$Q^1 = \mathrm{DP}(\alpha_1, P_0), \quad Q^2 = \mathrm{DP}(\alpha_2, P_0),$$

with $\alpha_1 \neq \alpha_2$. Then $Q^1 \neq Q^2$. Suppose that the sizes of the hierarchical samples from Q^1 and Q^2 are $n \times 1$. Then, the laws characterizing the two hierarchical samples are the same; i.e., for all $A \in \mathcal{B}(\mathbb{X})$,

$$P^{1,(1)}(A) = \mathbb{P}(X_{i,1} \in A) = \int_{\mathcal{P}(\mathbb{X})} P(A)\, Q^1(dP) = P_0(A),$$
$$P^{2,(1)}(A) = \mathbb{P}(Y_{i,1} \in A) = \int_{\mathcal{P}(\mathbb{X})} P(A)\, Q^2(dP) = P_0(A).$$

In this case, no testing procedure can achieve a low Type II error, which is the probability of failing to reject the null hypothesis when it is false. To overcome this limitation, we must ensure that

$$Q^1 = Q^2 \quad \text{if and only if} \quad P^{1,(m)} = P^{2,(m)}.$$

This is indeed the case for any $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, provided m is sufficiently large, as established by the following proposition.

Proposition 3.2.2. *Let $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Then there exists $m_0 \geq 1$ such that for all $m \geq m_0$,*

$$Q^1 = Q^2 \quad \textit{if and only if} \quad P^{1,(m)} = P^{2,(m)}.$$

Proof. If $Q^1 = Q^2$, the result follows immediately from the definition of $P^{1,(m)}$ and $P^{2,(m)}$.

We now prove the converse. For any $m \geq 1$,

$$P^{1,(m)} = P^{2,(m)} \quad \text{if and only if} \quad \mathcal{L}\left(\frac{1}{m}\sum_{i=1}^{m}\delta_{X_i}\right) = \mathcal{L}\left(\frac{1}{m}\sum_{i=1}^{m}\delta_{Y_i}\right),$$

where $(X_1, \ldots, X_m) \sim P^{1,(m)}$ and $(Y_1, \ldots, Y_m) \sim P^{2,(m)}$. By de Finetti's theorem (Theorem 2.1.3),

$$\frac{1}{m}\sum_{i=1}^{m}\delta_{X_i} \xrightarrow{w} \widetilde{P}^1, \qquad \frac{1}{m}\sum_{i=1}^{m}\delta_{Y_i} \xrightarrow{w} \widetilde{P}^2 \qquad \text{a.s.},$$

where $\widetilde{P}^1 \sim Q^1$ and $\widetilde{P}^2 \sim Q^2$. Since almost sure convergence implies convergence in distribution, it follows that

$$\mathcal{L}\left(\frac{1}{m}\sum_{i=1}^{m}\delta_{X_i}\right) \xrightarrow{w} Q^1, \qquad \mathcal{L}\left(\frac{1}{m}\sum_{i=1}^{m}\delta_{Y_i}\right) \xrightarrow{w} Q^2.$$

Now assume that $Q^1 \neq Q^2$. Since $\mathcal{P}(\mathbb{X})$ equipped with the topology of weak convergence is a Polish space, bounded continuous functions separate probability measures on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$; hence there exists $f \in C_b(\mathcal{P}(\mathbb{X}), \mathbb{R})$ such that $Q^1(f) \neq Q^2(f)$. Set $\epsilon = \frac{1}{2}\left|Q^1(f) - Q^2(f)\right| > 0$. By the weak convergence established above, there exists $m_0 \geq 1$ such that for all $m \geq m_0$,

$$\left|\mathbb{E}\,\frac{1}{m}\sum_{i=1}^{m} f(X_i) - Q^1(f)\right| < \epsilon, \qquad \left|\mathbb{E}\,\frac{1}{m}\sum_{i=1}^{m} f(Y_i) - Q^2(f)\right| < \epsilon.$$

By the triangle inequality,

$$\begin{aligned}\left|\mathbb{E}\,\frac{1}{m}\sum_{i=1}^{m} f(X_i) - \mathbb{E}\,\frac{1}{m}\sum_{i=1}^{m} f(Y_i)\right| &\geq \left|Q^1(f) - Q^2(f)\right| \\ &\quad - \left|\mathbb{E}\,\frac{1}{m}\sum_{i=1}^{m} f(X_i) - Q^1(f)\right| \\ &\quad - \left|\mathbb{E}\,\frac{1}{m}\sum_{i=1}^{m} f(Y_i) - Q^2(f)\right| \\ &> 2\epsilon - \epsilon - \epsilon = 0,\end{aligned}$$

from which we conclude that $P^{1,(m)} \neq P^{2,(m)}$. □

As noted above, hierarchical samples $\mathbf{X}_{n,m} \sim Q^1$ and $\mathbf{Y}_{n,m} \sim Q^2$ are used to decide whether $Q^1 = Q^2$. We aim to develop a testing scheme inspired by the Kolmogorov–Smirnov (KS) test. Recall that in the KS test, one rejects the null hypothesis if the distance between the empirical measures of the samples exceeds a given threshold. In Section 3.3, we will equip the space $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ with a distance function; for the moment, assume that d defines a distance on that space. Then, we could naturally extend the idea of the KS test to our setting if we had access to the samples $\widetilde{P}_1^1, \ldots, \widetilde{P}_n^1 \overset{\text{i.i.d}}{\sim} Q^1$ and $\widetilde{P}_1^2, \ldots, \widetilde{P}_n^2 \overset{\text{i.i.d}}{\sim} Q^2$, by rejecting H_0 if

$$d\left(\frac{1}{n}\sum_{i=1}^{n} \delta_{\widetilde{P}_i^1}, \frac{1}{n}\sum_{i=1}^{n} \delta_{\widetilde{P}_i^2}\right)$$

exceeds some threshold. However, recall that we do not observe $\widetilde{P}_i^1$ and $\widetilde{P}_i^2$. Nevertheless, we can estimate them from the observations that they generated; that is, we estimate each of them using the corresponding sequences $X_{i,1}, \ldots, X_{i,m}$ and $Y_{i,1}, \ldots, Y_{i,m}$ as

$$\widehat{P}_{i,m}^1 = \frac{1}{m}\sum_{j=1}^{m} \delta_{X_{i,j}}, \quad \widehat{P}_{i,m}^2 = \frac{1}{m}\sum_{j=1}^{m} \delta_{Y_{i,j}},$$

for $i \in \{1, \cdots, n\}$. Thus, the idea of the test is to reject H_0 if

$$d\left(\frac{1}{n}\sum_{i=1}^{n} \delta_{\widehat{P}_{i,m}^1}, \frac{1}{n}\sum_{i=1}^{n} \delta_{\widehat{P}_{i,m}^2}\right)$$

exceeds some threshold. As the estimators in the distance function are the fundamental objects of this chapter, we give specific names and notation for them.

Definition 3.2.3 (Hierarchical empirical estimator)**.** Given $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, and a hierarchical sample $\mathbf{X}_{n,m}$, the hierarchical empirical estimator of Q is defined as

$$\widehat{Q}_{n,m} = \frac{1}{n}\sum_{i=1}^{n} \delta_{\widehat{P}_{i,m}},$$

where $\widehat{P}_{i,m} = \frac{1}{m}\sum_{j=1}^{m} \delta_{X_{i,j}}$ for all $i = 1, \ldots, n$.

Note that the law of each $\widehat{P}^k_{i,m}$ is different from Q^k for $k = 1, 2$. Since these random probability measures are the main objects of study in this chapter, we introduce a specific notation for their laws.

Definition 3.2.4. Let $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$ and $m \geq 1$. We define $\mathcal{T}_m[Q] \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$ as the law of $\frac{1}{m}\sum_{i=1}^{m} \delta_{X_i}$ for $\widetilde{P} \sim Q$ and $X_1, \ldots, X_m | \widetilde{P} \overset{\text{i.i.d}}{\sim} \widetilde{P}$.

As noted above, a direct consequence of de Finetti's theorem is that, for any $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$,

$$\mathcal{T}_m[Q] \overset{w}{\to} Q \quad \text{as } m \to \infty.$$

To summarize, given Q^1 and Q^2, since we estimate the random probability measures $\widehat{P}^k_{i,m}$ from the hierarchical samples $\mathbf{X}_{n,m}$ and $\mathbf{Y}_{n,m}$, we effectively observe

$$\widehat{P}^1_{1,m}, \ldots, \widehat{P}^1_{n,m} \overset{\text{i.i.d}}{\sim} \mathcal{T}_m[Q^1], \quad \widehat{P}^2_{1,m}, \ldots, \widehat{P}^2_{n,m} \overset{\text{i.i.d}}{\sim} \mathcal{T}_m[Q^2]. \tag{3.1}$$

We aim to determine whether Q^1 and Q^2 are identical by comparing the empirical measures supported on these two samples. For such a comparison, we require a distance function on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$.

3.3 Distances and Rademacher Complexity

In this chapter, we work with two distance functions on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ based on optimal transport: the Wasserstein over Wasserstein distance (WoW; see Definition 2.3.1) and the Hierarchical Integral Probability Metric (HIPM; see Definition 2.3.2). For their topological properties and their relation to the Integral Probability Metric (IPM; see Definition 2.3.3), we refer the reader to Subsection 2.3.1. In this section, we discuss the sample complexity of these distances, that is, the rate at which the estimation error of the empirical estimator decays with sample size, and its relevance to the threshold function developed in Section 3.4.

Let $\mathbb{Y}$ be a general Polish space and $\mathcal{H}$ be a class of measurable and bounded functions from $\mathbb{Y}$ to $\mathbb{R}$. Let also $\mathcal{I}_{\mathcal{H}}$ denote the Integral Probability Metric associated to class $\mathcal{H}$ (see Definition 2.3.3).

Given a class of functions $\mathcal{H}$, we define its Rademacher complexity

$$\mathcal{R}_n(\mathcal{H}) = \sup_P \mathbb{E}\left(\sup_{h\in\mathcal{H}} \left| \frac{1}{n} \sum_{j=1}^{n} \epsilon_j h(Y_j) \right| \right), \tag{3.2}$$

where the supremum is taken over all probability measures P on $\mathbb{Y}$ with $Y_1, \ldots, Y_n \overset{\text{i.i.d}}{\sim} P$, and $\epsilon_1, \ldots, \epsilon_n$ are i.i.d. Rademacher random variables independent of the Y_i's. For any probability distribution P on $\mathbb{Y}$, letting $Y_1, \ldots, Y_n \overset{\text{i.i.d}}{\sim} P$ and $\hat{P}_n = \frac{1}{n}\sum_{i=1}^n \delta_{Y_i}$ denote the empirical estimator, we can control the random variable $\mathcal{I}_{\mathcal{H}}(P, \hat{P}_n)$. Following, for instance, the proof of Theorem 4.10 in Wainwright (2019), we have the following bias bound:

$$\mathbb{E}\, \mathcal{I}_{\mathcal{H}}(P, \hat{P}_n) \le 2\, \mathcal{R}_n(\mathcal{H}^*), \tag{3.3}$$

where $\mathcal{H}^* = \{h - h(y_0) : h \in \mathcal{H}, y_0 \in \mathbb{Y}\}$, together with the following deviation bound: if all functions in $\mathcal{H}$ are bounded by M, then with probability $1 - \theta$,

$$|\mathcal{I}_{\mathcal{H}}(P, \hat{P}_n) - \mathbb{E}\, \mathcal{I}_{\mathcal{H}}(P, \hat{P}_n)| \le M \sqrt{\frac{2}{n} \log \frac{2}{\theta}}. \tag{3.4}$$

As a consequence, for $P^1, P^2 \in \mathcal{P}(\mathbb{Y})$, with probability $1 - \theta$,

$$|\mathcal{I}_{\mathcal{H}}(\hat{P}_n^1, \hat{P}_n^2) - \mathcal{I}_{\mathcal{H}}(P^1, P^2)| \le 4\, \mathcal{R}_n(\mathcal{H}^*) + 2M \sqrt{\frac{2}{n} \log \frac{4}{\theta}}. \tag{3.5}$$

Indeed, by the triangle inequality for IPMs,

$$|\mathcal{I}_{\mathcal{H}}(\hat{P}_n^1, \hat{P}_n^2) - \mathcal{I}_{\mathcal{H}}(P^1, P^2)| \le \mathcal{I}_{\mathcal{H}}(P^1, \hat{P}_n^1) + \mathcal{I}_{\mathcal{H}}(P^2, \hat{P}_n^2).$$

Applying (3.4) to each term with parameter $\theta/2$ and combining via a union bound, with probability $1 - \theta$:

$$\mathcal{I}_{\mathcal{H}}(P^i, \hat{P}_n^i) \le \mathbb{E}\, \mathcal{I}_{\mathcal{H}}(P^i, \hat{P}_n^i) + M \sqrt{\frac{2}{n} \log \frac{4}{\theta}},$$

for $i = 1, 2$. Applying the bias bound (3.3) and summing yields (3.5).

Since concentration inequalities hold for any integral probability metric, and the distance functions we consider are special cases of it, we will show in Section 3.4 that such inequalities can be obtained in our setting.

Since the growth of the threshold is primarily governed by the Rademacher complexity of the function class, we now recall the known bounds on the Rademacher complexity for the WoW and HIPM.

The work Catalano and Lavenant (2024) establishes an upper bound on the Rademacher complexity of the class $\mathrm{Lip}_1^*(\mathcal{P}(\mathbb{X}), \mathbb{R})$. Namely, there exists $C > 0$

such that, for n large enough,

$$\mathcal{R}_n(\mathrm{Lip}_1^*(\mathcal{P}(\mathbb{X}), \mathbb{R})) \leq C \frac{\log \log(n)}{\log(n)}, \tag{3.6}$$

which immediately entails a quantitative convergence result for empirical estimators in $\mathcal{W}_{\mathcal{W}}$ distance via (3.3). Though it is unclear whether this bound is tight, the work Catalano and Lavenant (2024) also provides a lower bound. Specifically, let $\mathbb{X}$ be a bounded subset of $\mathbb{R}^d$ and $\widetilde{P}_1, \ldots, \widetilde{P}_m \overset{\text{i.i.d}}{\sim} Q$. Then Catalano and Lavenant (2024, Theorem 4.1) shows that for $Q = \mathrm{DP}(\alpha, P_0)$, a Dirichlet process with absolutely continuous base measure P_0 and $\alpha > 0$, for every $\gamma > 0$ there exists $c_\gamma > 0$ such that, for n large enough,

$$\mathbb{E}\mathcal{W}_{\mathcal{W}}\left(\frac{1}{n}\sum_{i=1}^{n} \delta_{\widetilde{P}_i}, Q\right) \geq c_\gamma n^{-\gamma}.$$

That is, the convergence rate of the WoW is slower than any polynomial rate in n.

In addition, the work Catalano and Lavenant (2024) provides bounds on the Rademacher complexity associated with the HIPM. In Appendix 3.A, we show that the HIPM is obtained as a special case of an integral probability metric with the function class

$$\mathcal{D}^* = \{h : \mathcal{P}(\mathbb{X}) \to \mathbb{R} : h(P) = g(P(f)),\ g \in \mathrm{Lip}_1^0([-K, K], \mathbb{R}), f \in \mathcal{F}^*\},$$

where $K = \mathrm{diam}(\mathbb{X})$. Then, Catalano and Lavenant (2024, Theorem 4.3) shows that for the HIPM there exist constants C_1, C_2, and, for $d > 2$, $C_d > 0$ such that

$$\mathcal{R}_n(\mathcal{D}^*) \leq \begin{cases} C_1 n^{-1/2} & \text{if } d = 1, \\ C_2 n^{-1/2}\log(n) & \text{if } d = 2, \\ C_d n^{-1/d} & \text{if } d > 2. \end{cases} \tag{3.7}$$

By (3.3), it follows that the sample complexity of the HIPM is polynomial in n and is unaffected by the infinite dimensionality of $\mathcal{P}(\mathbb{X})$. These results extend to the hierarchical empirical estimator $\widehat{Q}_{n,m}$ defined in Definition 3.2.3; see Catalano and Lavenant (2024, Theorem 4.4).

With a distance function on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$ in hand, the hypothesis testing problem takes the following form. Let d denote either the WoW or the HIPM distance, and let $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. We test the null hypothesis

$$H_0 : d(Q^1, Q^2) = 0$$

against the alternative

$$H_1 : d(Q^1, Q^2) > 0.$$

The test is based on independent samples

$$\widehat{P}^1_{1,m}, \ldots, \widehat{P}^1_{n,m} \overset{\text{i.i.d}}{\sim} \mathcal{T}_m[Q^1], \qquad \widehat{P}^2_{1,m}, \ldots, \widehat{P}^2_{n,m} \overset{\text{i.i.d}}{\sim} \mathcal{T}_m[Q^2].$$

The testing scheme we develop will reject H_0 if the distance between the hierarchical empirical estimators exceeds some threshold. In the following sections, we discuss methods for determining this threshold.

3.4 Threshold via Rademacher Complexity

This section develops a uniform, nonparametric threshold for hypothesis testing derived from Rademacher complexity. The primary objective is to establish a concentration inequality for the distance between two hierarchical empirical estimators. Intuitively, given two laws of random probability measures Q^1, Q^2, the distance $d(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m})$, whether d is the WoW or the HIPM distance, should be close to $d(Q^1, Q^2)$. A concentration inequality quantifies how close these quantities are with high probability. In particular, for $\theta \in (0, 1)$, it provides a radius $r(n, m, \theta)$ such that the probability that $d(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m})$ lies within $r(n, m, \theta)$ of $d(Q^1, Q^2)$ is at least $1-\theta$. This implies that under the null hypothesis, for every $\theta \in (0, 1)$, $d(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m})$ will be at most $r(n, m, \theta)$ with probability at least $1-\theta$. Therefore, if we use $r(n, m, \theta)$ as a threshold function and reject H_0 whenever

$$d(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) > r(n, m, \theta),$$

we are guaranteed that the probability of a Type I error is at most θ. We now discuss how to obtain the threshold function $r(n, m, \theta)$ and its relationship to the Rademacher complexity of the function class defining the distance.

Since both the WoW and the HIPM distances are IPMs (see Subsection 2.3.1) on the infinite-dimensional space $\mathcal{P}(\mathcal{P}(\mathbb{X}))$, our first strategy is to adapt the techniques developed in Sriperumbudur et al. (2012) for standard IPMs. These build thresholds through concentration inequalities for empirical processes, by analyzing the Rademacher complexity of the function class generating the IPM. The main difference is that we must account for a hierarchical empirical estimator, rather than a standard one. Moreover, instead of Talagrand's inequality as in Sriperumbudur et al. (2012, Proposition B.1), we use McDiarmid's inequality, which yields the bound (3.4) already established above. The reason is that Talagrand's inequality yields sharper variance-adaptive concentration rates for the empirical process around its mean, but these better rates do not transfer to the threshold itself, which is dominated by the uniform Rademacher complexity of the function class.

Theorem 3.4.1 (Threshold for IPM)**.** *Let $\mathcal{G}$ and $\mathcal{H}$ be classes of functions generating IPMs on $\mathcal{P}(\mathbb{X})$ and $\mathcal{P}(\mathcal{P}(\mathbb{X}))$, uniformly bounded by $M_{\mathcal{G}}$ and $M_{\mathcal{H}}$, respectively. Assume also that functions in $\mathcal{H}$ are 1-Lipschitz with respect to $\mathcal{I}_{\mathcal{G}}$. Define for any $n, m \geq 1$ and $\theta \in (0,1)$:*

$$r(n,m,\theta) = 4\mathcal{R}_n(\mathcal{H}) + 4\mathcal{R}_m(\mathcal{G}) + 2M_{\mathcal{G}}\sqrt{\frac{\log\frac{2}{\theta}}{2n}} + 2M_{\mathcal{H}}\sqrt{\frac{2\log\frac{4}{\theta}}{n}}.$$

Then for any $\theta \in (0,1)$ $n, m \geq 1$,

$$\mathbb{P}\left(\left|\mathcal{I}_{\mathcal{H}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) - \mathcal{I}_{\mathcal{H}}(Q^1, Q^2)\right| > r(n,m,\theta)\right) \leq \theta.$$

Proof. Note that to bound the gap

$$\left|\mathcal{I}_{\mathcal{H}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) - \mathcal{I}_{\mathcal{H}}(Q^1, Q^2)\right|,$$

we cannot directly apply (3.5), since $\widehat{Q}^k_{n,m}$ are not empirical measures of i.i.d. random variables from Q^k, $k = 1, 2$. Instead, we decompose the gap as

$$\begin{aligned}\left|\mathcal{I}_{\mathcal{H}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) - \mathcal{I}_{\mathcal{H}}(Q^1, Q^2)\right| \leq{}& \left|\mathcal{I}_{\mathcal{H}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) - \mathcal{I}_{\mathcal{H}}(\widehat{Q}^1_{n}, \widehat{Q}^2_{n})\right| \\ &+ \left|\mathcal{I}_{\mathcal{H}}(\widehat{Q}^1_{n}, \widehat{Q}^2_{n}) - \mathcal{I}_{\mathcal{H}}(Q^1, Q^2)\right|,\end{aligned}$$

where

$$\widehat{Q}^1_n = \frac{1}{n}\sum_{i=1}^n \delta_{\widetilde{P}^1_i}, \quad \widehat{Q}^2_n = \frac{1}{n}\sum_{i=1}^n \delta_{\widetilde{P}^2_i}$$

are empirical measures on $\widetilde{P}^1_1, \ldots, \widetilde{P}^1_n \overset{\text{i.i.d}}{\sim} Q^1$ and $\widetilde{P}^2_1, \ldots, \widetilde{P}^2_n \overset{\text{i.i.d}}{\sim} Q^2$.

The strategy of the proof is to obtain separate concentration inequalities for each term in the decomposition and then combine them via a union bound. We begin by bounding the first term.

Let $d_{n,m} = \left|\mathcal{I}_{\mathcal{H}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) - \mathcal{I}_{\mathcal{H}}(\widehat{Q}^1_{n}, \widehat{Q}^2_{n})\right|$. By the triangle inequality,

$$d_{n,m} \leq \mathcal{I}_{\mathcal{H}}(\widehat{Q}^1_{n,m}, \widehat{Q}^1_{n}) + \mathcal{I}_{\mathcal{H}}(\widehat{Q}^2_{n,m}, \widehat{Q}^2_{n}).$$

For $k = 1, 2$, we have

$$\begin{aligned}
\mathcal{I}_{\mathcal{H}}(\widehat{Q}^k_{n,m}, \widehat{Q}^k_n) &= \sup_{h\in\mathcal{H}} \left| \widehat{Q}^k_{n,m}(h) - \widehat{Q}^k_n(h) \right| \\
&= \sup_{h\in\mathcal{H}} \left| \frac{1}{n}\sum_{i=1}^n h\left(\widehat{P}^k_{i,m}\right) - \frac{1}{n}\sum_{i=1}^n h\left(\widetilde{P}^k_i\right) \right| \\
&\le \frac{1}{n}\sum_{i=1}^n \sup_{h\in\mathcal{H}} \left| h\left(\widehat{P}^k_{i,m}\right) - h\left(\widetilde{P}^k_i\right) \right| \\
&\le \frac{1}{n}\sum_{i=1}^n \sup_{h\in\mathcal{H}} \mathcal{I}_{\mathcal{G}}(\widehat{P}^k_{i,m}, \widetilde{P}^k_i) \\
&= \frac{1}{n}\sum_{i=1}^n \mathcal{I}_{\mathcal{G}}(\widehat{P}^k_{i,m}, \widetilde{P}^k_i).
\end{aligned}$$

We use Hoeffding's inequality to bound $\frac{1}{n}\sum_{i=1}^n \mathcal{I}_{\mathcal{G}}(\widehat{P}^k_{i,m}, \widetilde{P}^k_i)$. The reason for preferring Hoeffding's inequality over a union bound is that in the latter case, one would need to impose stronger conditions on the relative growth rates of n and m to guarantee that the threshold converges to 0 as $n, m \to \infty$. Note that the random variables $\frac{1}{n}\mathcal{I}_{\mathcal{G}}(\widehat{P}^k_{1,m}, \widetilde{P}^k_1), \dots, \frac{1}{n}\mathcal{I}_{\mathcal{G}}(\widehat{P}^k_{n,m}, \widetilde{P}^k_n)$ are independent and take values in $\left[0, \frac{M_{\mathcal{G}}}{n}\right]$. Applying Hoeffding's inequality, we find that for all $t > 0$,

$$\mathbb{P}\left(\frac{1}{n}\sum_{i=1}^n \mathcal{I}_{\mathcal{G}}(\widehat{P}^k_{i,m}, \widetilde{P}^k_i) > \mathbb{E}\frac{1}{n}\sum_{i=1}^n \mathcal{I}_{\mathcal{G}}(\widehat{P}^k_{i,m}, \widetilde{P}^k_i) + t \right) \le \exp\left(-\frac{2nt^2}{M_{\mathcal{G}}^2}\right).$$

By setting $\exp\left(-\frac{2nt^2}{M_{\mathcal{G}}^2}\right) = \theta$ and inverting, we get that for all $\theta \in (0, 1)$,

$$\mathbb{P}\left(\frac{1}{n}\sum_{i=1}^n \mathcal{I}_{\mathcal{G}}(\widehat{P}^k_{i,m}, \widetilde{P}^k_i) > \mathbb{E}\frac{1}{n}\sum_{i=1}^n \mathcal{I}_{\mathcal{G}}(\widehat{P}^k_{i,m}, \widetilde{P}^k_i) + M_{\mathcal{G}}\sqrt{\frac{\log\frac{1}{\theta}}{2n}} \right) \le \theta.$$

Using the symmetrization argument (see Lemma 2.3.1 in Van der Vaart and Wellner (1996)), we obtain

$$\mathbb{E}\left[\mathcal{I}_{\mathcal{G}}\left(\widehat{P}^k_{i,m}, \widetilde{P}^k_i\right) \mid \widetilde{P}^k_i\right] \le 2\,\mathcal{R}_m(\mathcal{G}).$$

Using the tower property of conditional expectation,

$$\mathbb{E}\frac{1}{n}\sum_{i=1}^n \mathcal{I}_{\mathcal{G}}\left(\widehat{P}^k_{i,m}, \widetilde{P}^k_i\right) = \mathbb{E}\,\mathcal{I}_{\mathcal{G}}\left(\widehat{P}^k_{1,m}, \widetilde{P}^k_1\right) = \mathbb{E}\left[\mathbb{E}\left[\mathcal{I}_{\mathcal{G}}\left(\widehat{P}^k_{1,m}, \widetilde{P}^k_1\right) \mid \widetilde{P}^k_1\right]\right] \le 2\,\mathcal{R}_m(\mathcal{G}).$$

Thus, for all $\theta \in (0,1)$ and $n, m \geq 1$,

$$\mathbb{P}\left(\frac{1}{n}\sum_{i=1}^{n}\mathcal{I}_{\mathcal{G}}(\widehat{P}^k_{i,m}, \widetilde{P}^k_i) > 2\,\mathcal{R}_m(\mathcal{G}) + M_{\mathcal{G}}\sqrt{\frac{\log\frac{1}{\theta}}{2n}}\right) \leq \theta.$$

Since

$$\mathcal{I}_{\mathcal{H}}(\widehat{Q}^k_{n,m}, \widehat{Q}^k_n) \leq \frac{1}{n}\sum_{i=1}^{n}\mathcal{I}_{\mathcal{G}}(\widehat{P}^k_{i,m}, \widetilde{P}^k_i),$$

we obtain

$$\mathbb{P}\left(\mathcal{I}_{\mathcal{H}}(\widehat{Q}^k_{n,m}, \widehat{Q}^k_n) > 2\mathcal{R}_m(\mathcal{G}) + M_{\mathcal{G}}\sqrt{\frac{\log\frac{1}{\theta}}{2n}}\right) \leq \theta.$$

Applying a union bound,

$$\begin{aligned}
\mathbb{P}\left(d_{n,m} > 2\left(2\mathcal{R}_m(\mathcal{G}) + M_{\mathcal{G}}\sqrt{\frac{\log\frac{2}{\theta}}{2n}}\right)\right) &\leq \mathbb{P}\left(\mathcal{I}_{\mathcal{H}}(\widehat{Q}^1_{n,m}, \widehat{Q}^1_n) > 2\mathcal{R}_m(\mathcal{G}) + M_{\mathcal{G}}\sqrt{\frac{\log\frac{2}{\theta}}{2n}}\right) \\
&\quad + \mathbb{P}\left(\mathcal{I}_{\mathcal{H}}(\widehat{Q}^2_{n,m}, \widehat{Q}^2_n) > 2\mathcal{R}_m(\mathcal{G}) + M_{\mathcal{G}}\sqrt{\frac{\log\frac{2}{\theta}}{2n}}\right) \\
&\leq \frac{\theta}{2} + \frac{\theta}{2} = \theta.
\end{aligned}$$

The next step is to bound

$$\left|\mathcal{I}_{\mathcal{H}}(\widehat{Q}^1_n, \widehat{Q}^2_n) - \mathcal{I}_{\mathcal{H}}(Q^1, Q^2)\right|.$$

Since $\widehat{Q}^k_n$ are empirical measures of i.i.d. random variables drawn from Q^k, we can apply (3.5). The result follows by combining the bound for $d_{n,m}$ with (3.5) via a union bound. $\square$

Theorem 3.4.1 can be specialized to both the WoW and HIPM distances. Indeed, for the WoW distance we set $\mathcal{H} = \mathrm{Lip}^*_1(\mathcal{P}(\mathbb{X}), \mathbb{R})$ and $\mathcal{G} = \mathrm{Lip}^*_1(\mathbb{X}, \mathbb{R})$. On the other hand, for the HIPM we set $\mathcal{H} = \mathcal{D}^*$ (see (3.13) in Appendix 3.A), with $\mathcal{G} = \mathrm{Lip}^*_1(\mathbb{X}, \mathbb{R})$. In both cases, in addition to the Rademacher complexity terms, the threshold contains a term arising from McDiarmid's inequality that vanishes at the parametric rate $n^{-1/2}$ and is typically dominated by the Rademacher complexity terms. The two distances differ in the first term $\mathcal{R}_n(\mathcal{H})$, which highlights the benefit of using a smaller function class $\mathcal{H}$, such as the one used in the HIPM.

Remark 3.4.2. Theorem 3.4.1 admits an alternative proof in the cases of the HIPM and

WoW. Indeed, consider the class of functions $\mathcal{H} \subset \mathrm{Lip}_1^*(\mathcal{P}(\mathbb{X}), \mathbb{R})$ and decompose the gap

$$\begin{aligned}\left|\mathcal{I}_{\mathcal{H}}\left(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}\right) - \mathcal{I}_{\mathcal{H}}\left(Q^1, Q^2\right)\right| \leq \left|\mathcal{I}_{\mathcal{H}}\left(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}\right) - \mathcal{I}_{\mathcal{H}}\left(\mathcal{T}_m[Q^1], \mathcal{T}_m[Q^2]\right)\right| \\ + \left|\mathcal{I}_{\mathcal{H}}\left(\mathcal{T}_m[Q^1], \mathcal{T}_m[Q^2]\right) - \mathcal{I}_{\mathcal{H}}\left(Q^1, Q^2\right)\right|.\end{aligned}$$

For the first term, we apply (3.5). For the second term, note that for $k = 1, 2$,

$$\mathcal{I}_{\mathcal{H}}\left(\mathcal{T}_m[Q^k], Q^k\right) \leq \mathcal{W}_{\mathcal{W}}\left(\mathcal{T}_m[Q^k], Q^k\right).$$

This is bounded uniformly in Q by $2\mathcal{R}_m(\mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R}))$. Indeed, let $\widetilde{P} \sim Q^k$ and $X_1, \ldots, X_m \overset{\text{i.i.d}}{\sim} \widetilde{P}$. Using the coupling between $\widetilde{P}$ and $m^{-1}\sum_{i=1}^m \delta_{X_i}$, we get

$$\mathcal{W}_{\mathcal{W}}(\mathcal{T}_m[Q^k], Q^k) \leq \mathbb{E}\mathcal{W}\left(\widetilde{P}, \frac{1}{m}\sum_{i=1}^m \delta_{X_i}\right).$$

Then, the result follows from (2.2) and bound (3.3), which is uniform in $\widetilde{P}$.

For the particular case when $\mathbb{X}$ is a compact subset of $\mathbb{R}^d$, the threshold takes the following form. For the WoW, using the bound (3.6) on the Rademacher complexity and the standard bound on the Rademacher complexity of centered 1-Lipschitz functions from $\mathbb{X}$ to $\mathbb{R}$, we get that there exists $N \geq 1$ such that for all $n \geq N$ and fixed $\theta \in (0, 1)$,

$$r_d(n, m, \theta) = \begin{cases} \mathcal{O}\left(\frac{\log(\log(n))}{\log(n)} + \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}}\right), & \text{if } d = 1 \\ \mathcal{O}\left(\frac{\log(\log(n))}{\log(n)} + \frac{1}{\sqrt{n}}\log(n) + \frac{1}{\sqrt{m}}\log(m)\right), & \text{if } d = 2 \\ \mathcal{O}\left(\frac{\log(\log(n))}{\log(n)} + \frac{1}{n^{\frac{1}{d}}} + \frac{1}{m^{\frac{1}{d}}}\right), & \text{if } d > 2. \end{cases} \tag{3.8}$$

On the other hand, for the HIPM,

$$r_d(n, m, \theta) = \begin{cases} \mathcal{O}\left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}}\right), & \text{if } d = 1 \\ \mathcal{O}\left(\frac{1}{\sqrt{n}}\log(n) + \frac{1}{\sqrt{m}}\log(m)\right), & \text{if } d = 2 \\ \mathcal{O}\left(\frac{1}{n^{\frac{1}{d}}} + \frac{1}{m^{\frac{1}{d}}}\right), & \text{if } d > 2. \end{cases} \tag{3.9}$$

Remark 3.4.3. If $\mathbb{X} = [a, b]$, the exact form of the threshold function in Theorem 3.4.1 can be derived, since the bound for the Rademacher complexity is known in closed form. In this case, we obtain a finite-sample guarantee for the Type I error. Note that for the WoW distance, even if the exact threshold were known, the finite-sample guarantee would hold only for $n \geq N$ for some integer $N \geq 1$.

A consequence of Theorem 3.4.1 is a testing scheme that, given a significance level

θ, rejects the null hypothesis whenever the observed distance between the hierarchical empirical estimators exceeds the threshold. In addition to controlling the Type I error, the theorems below also ensure consistency. Intuitively, under H_1, the observed distance $d(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m})$, whether d is the WoW or the HIPM distance, will be close to $d(Q^1, Q^2)$. Hence, as long as the threshold $r(n, m, \theta)$ converges to 0 as $n, m \to \infty$, we will eventually have $d(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) > r(n, m, \theta)$. The theorems below formalize this intuition. Note that in the limits below, n and m tend to infinity independently.

Theorem 3.4.4 (Consistency for HIPM)**.** *Let $\mathbb{X}$ be a compact subset of $\mathbb{R}^d$, and let $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Let θ denote the significance level, and let $r_d(n, m, \theta)$ be the threshold function* (3.9)*. Then:*

- *Type I error: If $Q^1 = Q^2$, then for all $n, m \geq 1$,*

$$\mathbb{P}\left(d_{\text{Lip}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) > r_d(n, m, \theta)\right) \leq \theta.$$

- *Consistency: If $Q^1 \neq Q^2$, then*

$$\lim_{n,m\to\infty} \mathbb{P}\left(d_{\text{Lip}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) > r_d(n, m, \theta)\right) = 1.$$

Proof. The first point follows directly from Theorem 3.4.1, because $d_{\text{Lip}}(Q^1, Q^2) = 0$ under the assumption that $Q^1 = Q^2$.

For the second point, fix $\delta > 0$ and choose $\theta' \in (0, \theta)$ such that $1 - \theta' > \delta$. Note that since the threshold function from Theorem 3.4.1 is a decreasing function of θ, we have $r_d(n, m, \theta') \geq r_d(n, m, \theta)$ for all $n, m \geq 1$. By Theorem 3.4.1, for all $n \geq 1$ and $m \geq 1$,

$$\mathbb{P}\left(\left|d_{\text{Lip}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) - d_{\text{Lip}}(Q^1, Q^2)\right| < r_d(n, m, \theta')\right) \geq 1 - \theta' > \delta.$$

We also have that

$$\begin{aligned}\mathbb{P}\Big(\big|d_{\text{Lip}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) - d_{\text{Lip}}(Q^1, Q^2)\big| < r_d(n, m, \theta')\Big)& \\ \leq \mathbb{P}\left(d_{\text{Lip}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) > d_{\text{Lip}}(Q^1, Q^2) - r_d(n, m, \theta')\right)&.\end{aligned}$$

Note that for a fixed θ', the threshold function $r_d(n, m, \theta') \to 0$ as $n, m \to \infty$, which implies that there exist n_0, m_0 such that for all $n \geq n_0$ and $m \geq m_0$,

$$r_d(n, m, \theta) < \frac{d_{\text{Lip}}(Q^1, Q^2)}{2}.$$

This inequality implies that

$$d_{\mathrm{Lip}}(Q^1, Q^2) - r_d(n, m, \theta') > r_d(n, m, \theta').$$

Therefore, we have that for every $n \geq n_0$ and $m \geq m_0$,

$$\begin{aligned}
\mathbb{P}\left(d_{\mathrm{Lip}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) > r_d(n, m, \theta)\right) &\geq \mathbb{P}\left(d_{\mathrm{Lip}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) > r_d(n, m, \theta')\right) \\
&\geq \mathbb{P}\left(d_{\mathrm{Lip}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) > d_{\mathrm{Lip}}(Q^1, Q^2) - r_d(n, m, \theta')\right) \\
&> 1 - \theta' > \delta.
\end{aligned}$$

□

Theorem 3.4.5 (Consistency for WoW)**.** *Let $\mathbb{X}$ be a compact subset of $\mathbb{R}^d$, and let $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Let θ denote the significance level, and let $r_d(n, m, \theta)$ be the threshold function* (3.8)*. Then, there exists $N \geq 1$ such that:*

- *Type I error: If $Q^1 = Q^2$, then for all $n \geq N, m \geq 1$,*

$$\mathbb{P}\left(\mathcal{W}_{\mathcal{W}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) > r_d(n, m, \theta)\right) \leq \theta.$$

- *Consistency: If $Q^1 \neq Q^2$, then*

$$\lim_{n,m\to\infty} \mathbb{P}\left(\mathcal{W}_{\mathcal{W}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) > r_d(n, m, \theta)\right) = 1.$$

Proof. The first point follows directly from Theorem 3.4.1, because $\mathcal{W}_{\mathcal{W}}(Q^1, Q^2) = 0$ under the assumption that $Q^1 = Q^2$.

For the second point, fix $\delta > 0$ and choose $\theta' \in (0, \theta)$ such that $1 - \theta' > \delta$. Note that since the threshold function from Theorem 3.4.1 is a decreasing function of θ, we have $r_d(n, m, \theta') \geq r_d(n, m, \theta)$ for all $n, m \geq 1$. By Theorem 3.4.1, there exists $N' \geq 1$ such that for all $n \geq N'$ and $m \geq 1$,

$$\mathbb{P}\left(\left|\mathcal{W}_{\mathcal{W}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) - \mathcal{W}_{\mathcal{W}}(Q^1, Q^2)\right| < r_d(n, m, \theta')\right) \geq 1 - \theta' > \delta.$$

We also have that

$$\begin{aligned}
&\mathbb{P}\Big(\left|\mathcal{W}_{\mathcal{W}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) - \mathcal{W}_{\mathcal{W}}(Q^1, Q^2)\right| < r_d(n, m, \theta')\Big) \\
&\qquad \leq \mathbb{P}\left(\mathcal{W}_{\mathcal{W}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) > \mathcal{W}_{\mathcal{W}}(Q^1, Q^2) - r_d(n, m, \theta')\right).
\end{aligned}$$

Note that for a fixed θ', the threshold function $r_d(n, m, \theta') \to 0$ as $n, m \to \infty$, which implies that there exist $n_0, m_0 \geq 1$ such that for all $n \geq n_0$ and $m \geq m_0$,

$$r_d(n, m, \theta) < \frac{\mathcal{W}_{\mathcal{W}}(Q^1, Q^2)}{2}.$$

This inequality implies that

$$\mathcal{W}_{\mathcal{W}}(Q^1, Q^2) - r_d(n, m, \theta') > r_d(n, m, \theta').$$

Therefore, for every $n \geq \max\{n_0, N'\}$ and $m \geq m_0$,

$$\begin{aligned}
\mathbb{P}\left(\mathcal{W}_{\mathcal{W}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) > r_d(n, m, \theta)\right) &\geq \mathbb{P}\left(\mathcal{W}_{\mathcal{W}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) > r_d(n, m, \theta')\right) \\
&\geq \mathbb{P}\left(\mathcal{W}_{\mathcal{W}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) > \mathcal{W}_{\mathcal{W}}(Q^1, Q^2) - r_d(n, m, \theta')\right) \\
&> 1 - \theta' > \delta.
\end{aligned}$$

□

While theoretically appealing, Theorems 3.4.4 and 3.4.5 have practical limitations. Achieving high power requires the threshold to decay to zero at a sufficiently fast rate. However, the logarithmic terms present in the threshold for the WoW distance result in a threshold function that converges to zero at a suboptimal rate. Furthermore, although using the HIPM provides a parametric rate, the magnitude of the constant in the threshold often makes the testing scheme impractical for finite samples.

These limitations are demonstrated by our numerical simulations. We restrict our simulations to the HIPM, as the threshold associated with the WoW distance is even more conservative than that of the HIPM.

Suppose that we have two different laws of random probability measures, $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. For a fixed significance level $\theta \in (0, 1)$, to achieve high power the theoretical threshold must be smaller than the observed distance. To demonstrate that the observed distance is far smaller than the theoretical threshold, we compare it against the empirical threshold, defined as the $(1-\theta)$-quantile of the random variable $d_{\text{Lip}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m})$. Figure 3.1 shows how the theoretical and empirical thresholds differ for a fixed θ as n and m increase.

Fix $\theta = 0.05$ and consider two Dirichlet processes $Q^1 = \text{DP}(1.0, P_0)$ and $Q^2 = \text{DP}(1.5, P_1)$, where $P_0 = \text{Uniform}[0, 1]$ and $P_1 = \text{Beta}(2, 2)$.

As shown in Figure 3.1, the theoretical threshold is significantly larger than the empirical threshold, even for large values of n and m. Consequently, the null hypothesis is never rejected when it is false. The reason for such a large theoretical threshold is that it is designed to be uniform over the entire space of laws of random probability measures.

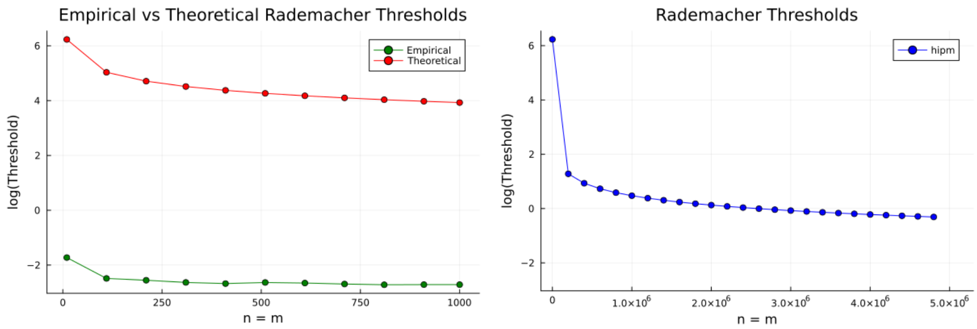


Figure 3.1: Comparison of Empirical and Rademacher Thresholds

3.5 Threshold via a Permutation Approach

In this section, we introduce a permutation-based framework for two-sample testing that yields improved performance in practice. As discussed earlier, the threshold derived from Rademacher complexity is overly conservative, resulting in low power for finite sample sizes. The permutation approach alleviates this issue by providing a more accurate approximation of the quantile function of the test statistic under the null hypothesis H_0, while typically yielding a much smaller threshold under the alternative.

Recall that, given $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, we observe hierarchical samples $\mathbf{X}_{n,m} \sim Q^1$ and $\mathbf{Y}_{n,m} \sim Q^2$ as in Definition 3.2.1 and we construct $\widehat{Q}^1_{n,m}$ and $\widehat{Q}^2_{n,m}$, the hierarchical empirical estimators, as in Definition 3.2.3. Equivalently, given the notation $\mathcal{T}_m$ introduced in Definition 3.2.4 and the equivalent formulation (3.1), we observe

$$\widehat{P}^1_{1,m}, \ldots, \widehat{P}^1_{n,m} \overset{\text{i.i.d}}{\sim} \mathcal{T}_m[Q^1], \qquad \widehat{P}^2_{1,m}, \ldots, \widehat{P}^2_{n,m} \overset{\text{i.i.d}}{\sim} \mathcal{T}_m[Q^2].$$

Using these samples, we compute the distance $d(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m})$, where d can be either WoW or HIPM. Our goal is to estimate the quantile function of the random variable $d(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m})$ assuming that $Q^1 = Q^2$, and use it as a testing threshold. Ideally, such a quantile could be estimated empirically from multiple independent realizations of this random variable. In practice, however, only a single realization is available.

We circumvent this limitation by noting that under the null hypothesis $Q^1 = Q^2$, all random probability measures $\widehat{P}^k_{i,m}$ share the same distribution. Consequently, if we randomly shuffle these objects and recompute the distance, the resulting value constitutes another realization from the distribution of $d(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m})$. By repeating this process, we can obtain several samples of the distance between hierarchical empirical estimators. Although these samples are not independent, this dependence is typically negligible in practice when n is sufficiently large.

To implement the permutation test, we first form the pooled collection of random probability measures

$$\left\{\widehat{P}_{1,m},\cdots,\widehat{P}_{2n,m}\right\} := \left\{\widehat{P}^1_{1,m},\ldots,\widehat{P}^1_{n,m},\widehat{P}^2_{1,m},\ldots,\widehat{P}^2_{n,m}\right\}.$$

A random permutation of this collection is determined by a random vector $R = (R_1,\ldots,R_{2n})$ drawn uniformly from all permutations of $\{1,\ldots,2n\}$. For notational brevity, we omit the explicit dependence of R on n.

We then define the permuted hierarchical empirical estimators as

$$\widetilde{Q}^1_{n,m} = \frac{1}{n}\sum_{i=1}^{n}\delta_{\widehat{P}_{R_i,m}} \quad \text{and} \quad \widetilde{Q}^2_{n,m} = \frac{1}{n}\sum_{i=n+1}^{2n}\delta_{\widehat{P}_{R_i,m}}. \tag{3.10}$$

Finally, we compute the distance $d(\widetilde{Q}^1_{n,m},\widetilde{Q}^2_{n,m})$.

Repeating this process yields approximate samples from the null distribution of $d(\widehat{Q}^1_{n,m},\widehat{Q}^2_{n,m})$. Then, for a given significance level $\theta\in(0,1)$, we set the testing threshold equal to the empirical $(1-\theta)$-quantile of these values.

In practice, since we work with the hierarchical samples $\mathbf{X}_{n,m}$ and $\mathbf{Y}_{n,m}$, permutation is implemented by shuffling rows across the two matrices. The resulting procedure is summarized in Algorithm 1.

Algorithm 1 Permutation test with $d\in\{d_{\mathrm{Lip}},\mathcal{W}_{\mathcal{W}}\}$

Require: Hierarchical samples $\mathbf{X}_{n,m},\mathbf{Y}_{n,m}$; significance level $\theta\in(0,1)$; number of permutations n_{perm}

Ensure: Decision: $1=$ reject H_0, $0=$ do not reject H_0

```
1: d_obs ← d(X_{n,m}, Y_{n,m})                      ▷ Observed test statistic
2: Z ← vcat(X_{n,m}, Y_{n,m})                       ▷ Pool the 2n rows
3: for i = 1, ..., n_perm do
4:     R ← permute({1, ..., 2n})
5:     d_i ← d(Z_{R_{1:n}}, Z_{R_{n+1:2n}})
6: end for
7: r_θ ← q̂_{1−θ}(d_1, ..., d_{n_perm})              ▷ Empirical (1 − θ)-quantile
8: return 1{d_obs > r_θ}
```

We study the theoretical properties of the proposed procedure for the distance d_{Lip} by analyzing the asymptotic distributions of the observed test statistic and the corresponding threshold under the null and alternative hypotheses. To this end, we appropriately rescale the observed d_{Lip} distance and define the test statistic

$$T_{n,m} = \sqrt{\frac{n}{2}}\,d_{\mathrm{Lip}}\left(\widehat{Q}^1_{n,m},\widehat{Q}^2_{n,m}\right).$$

The corresponding threshold is defined as

$$r(n,m,\theta) = \inf\left\{t \in \mathbb{R} : \mathbb{P}\left(\widetilde{T}_{n,m} > t \mid \mathbf{X}_{n,m}, \mathbf{Y}_{n,m}\right) \leq \theta\right\}, \tag{3.11}$$

where

$$\widetilde{T}_{n,m} = \sqrt{\frac{n}{2}}\, d_{\text{Lip}}(\widetilde{Q}^1_{n,m}, \widetilde{Q}^2_{n,m}).$$

Note that this scaling is introduced solely for theoretical analysis and is not required for the practical implementation of the algorithm.

For the distance d_{Lip} in dimension 1, we are able to establish theoretical guarantees for the permutation test described above. The reason is that the class of measurable functions $\mathcal{D}^*$ used to express the distance as an IPM in this case is a Donsker class. This implies that the asymptotic distributions of both the test statistic and the permutation threshold can be expressed in terms of norms of Brownian bridges. The permutation test can be implemented for both metrics, but the asymptotic guarantees are only available for the HIPM metric in dimension 1. The reason is that the function class generating WoW as an IPM, namely $\text{Lip}_1(\mathcal{P}(\mathbb{X}), \mathbb{R}, \mathcal{W})$, is not a Donsker class in general, so the CLT-based argument used for d_{Lip} does not extend to that setting. We emphasize, however, that this does not imply that the consistency result fails for WoW; it only means that the current proof technique does not apply.

Theorem 3.5.1 (Consistency of permutation approach)**.** *Given $\mathbb{X} = [a,b]$ and $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Let θ denote the significance level and $r(n,m,\theta)$ be the threshold from the permutation approach. Then*

- *if $Q^1 = Q^2$, then*
$$\lim_{n,m\to\infty} \mathbb{P}\left(T_{n,m} > r(n,m,\theta)\right) = \theta.$$
- *if $Q^1 \neq Q^2$, then*
$$\lim_{n,m\to\infty} \mathbb{P}\left(T_{n,m} > r(n,m,\theta)\right) = 1.$$

The main strategy of the proof of Theorem 3.5.1 is to show that the test statistic and the permutation threshold converge in distribution to the same random variable under the null hypothesis; under the alternative, the test statistic diverges to infinity while the permutation threshold converges to an almost surely bounded random variable, so that the test statistic eventually exceeds the threshold. Since the proof relies on several auxiliary results, we defer it to Appendix 3.C.

3.6 Simulation Studies

In this section, we examine the finite-sample performance of the proposed HIPM and WoW testing procedures. Our numerical analysis is organized into three parts. First, we investigate the validity of the permutation-based test by evaluating the Type I error rate and statistical power of both HIPM and WoW using a synthetic dataset. Second, we compare our methods with existing methods from the literature. Finally, we demonstrate the practical utility of the proposed framework through an application to a mortality dataset.

3.6.1 Empirical Comparison Between HIPM and WoW

We compare the use of HIPM and WoW within our permutation-based test by analysing the Type I error and the power in several simulated settings, considering both parametric and nonparametric data-generating processes, in settings that satisfy the assumptions of Theorem 3.5.1 and in settings that do not, notably $\mathbb{X} = \mathbb{R}$, but are still of practical interest. We consider the following baselines for parametric and nonparametric models.

1. Parametric model that fulfills the assumptions:

$$Q^1 = \mathcal{L}(\tilde{P}), \qquad \tilde{P}|\tilde{\mu} = \text{Beta}(\tilde{\mu}, 1-\tilde{\mu}), \qquad \tilde{\mu} \sim \text{Beta}(a_0 = 1, b_0 = 1);$$

2. Parametric model that does not fulfill the assumptions:

$$Q^1 = \mathcal{L}(\tilde{P}), \qquad \tilde{P}|\tilde{\mu} = \mathcal{N}(\tilde{\mu}, \sigma^2 = 1), \qquad \tilde{\mu} \sim \mathcal{N}(\mu_0 = 0, \sigma_0^2 = 1);$$

3. Nonparametric model that fulfills the assumptions:

$$Q^1 = \text{DP}(\alpha = 1, P_0 = \text{Unif}[0,1]);$$

4. Nonparametric model that does not fulfill the assumptions:

$$Q^1 = \text{DP}(\alpha = 1, P_0 = N(0,1)),$$

where $\text{Beta}(a, b)$ is the Beta distribution with shape parameters a, b, $\mathcal{N}(\mu, \sigma^2)$ is the Gaussian distribution with mean μ and variance σ^2, $\text{Unif}[0, 1]$ is the Uniform distribution on the interval $[0, 1]$, and $\text{DP}(\alpha, P_0)$ denotes the Dirichlet process with concentration $\alpha > 0$ and base measure $P_0 \in \mathcal{P}(\mathbb{X})$. Note that if P_0 has full support on $\mathbb{X}$, $\text{DP}(\alpha, P_0)$ has full weak support on $\mathcal{P}(\mathbb{X})$.

To investigate the behaviour of the Type I error, we simulate data $S = 1000$ times with $Q^1 = Q^2$ set equal to each of the four models above, perform our permutation-based test for both WoW and HIPM, and report the percentage of times it rejects the (correct) $H_0 : Q^1 = Q^2$ for each significance level θ. In each simulation, we set $n = m = n_{\text{perm}} = 100$.

The results are shown in Figure 3.2, where we see that both WoW and HIPM control the Type I error at approximately the significance level θ across all simulation settings. We report only one of the four graphs, as the remaining graphs show no qualitative differences. Interestingly, WoW also appears to control the Type I error even though our theory does not guarantee it, and both methods behave well in unbounded settings.

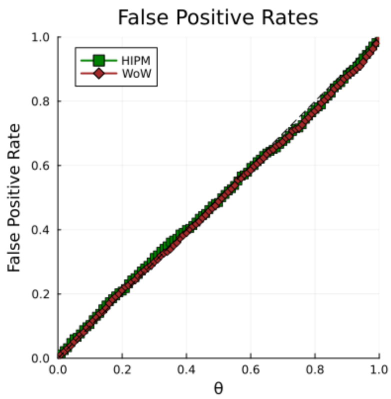


Figure 3.2: False Positive Rates for HIPM and WoW

To investigate the power of the test, we simulate the data under the alternative hypothesis $Q^1 \neq Q^2$. For each of the four laws considered above, we associate an alternative law and estimate the power, resulting in the following four cases:

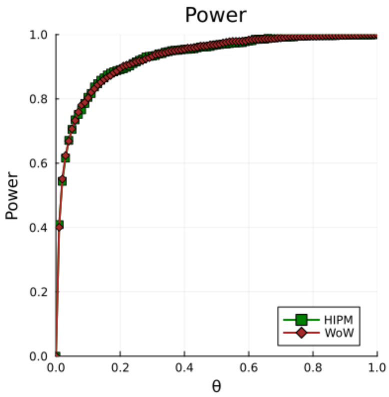
A. $Q^1 = \mathcal{L}(\tilde{P}), \quad \tilde{P}|\tilde{\mu} = \text{Beta}(\tilde{\mu}, 1-\tilde{\mu}), \quad \tilde{\mu} \sim \text{Beta}(a_0 = 1, b_0 = 1)$
$Q^2 = \mathcal{L}(\tilde{P}), \quad \tilde{P}|\tilde{\mu} = \text{Beta}(\tilde{\mu}, 1-\tilde{\mu}), \quad \tilde{\mu} \sim \text{Beta}(a_0 = 1, b_0 = 1.5)$

B. $Q^1 = \mathcal{L}(\tilde{P}), \quad \tilde{P}|\tilde{\mu} = \mathcal{N}(\tilde{\mu}, \sigma^2 = 1), \quad \tilde{\mu} \sim \mathcal{N}(\mu_0 = 0, \sigma_0^2 = 1)$
$Q^2 = \mathcal{L}(\tilde{P}), \quad \tilde{P}|\tilde{\mu} = \mathcal{N}(\tilde{\mu}, \sigma^2 = 1), \quad \tilde{\mu} \sim \mathcal{N}(\mu_0 = 0, \sigma_0^2 = 1.5)$

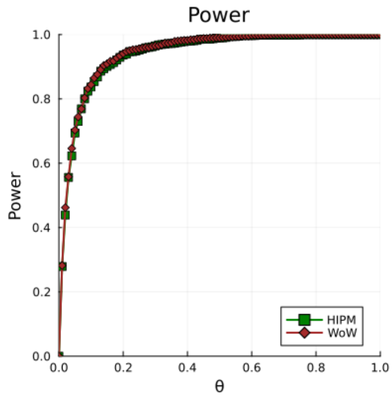
C. $Q^1 = \text{DP}(\alpha = 1, P_0 = \text{Unif}[0,1])$
$Q^2 = \text{DP}(\alpha = 1, P_0 = \text{Beta}(a_0 = 1, b_0 = 1.4))$

D. $Q^1 = \text{DP}(\alpha = 1, P_0 = \mathcal{N}(0,1))$
$Q^2 = \text{DP}(\alpha = 2, P_0 = \mathcal{N}(0,1))$

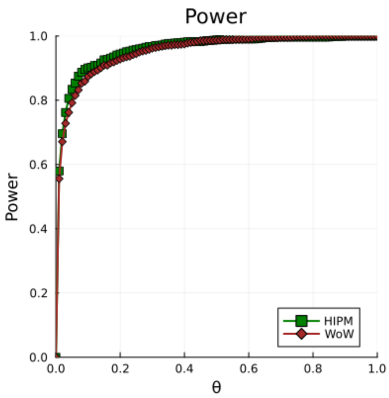
As before, we simulate data $S = 1000$ times, perform our permutation-based test for both WoW and HIPM, and report the empirical power, that is, the percentage of times it rejects the (incorrect) null $H_0 : Q^1 = Q^2$, for each significance level θ, setting $n = m = n_{\text{perm}} = 100$.



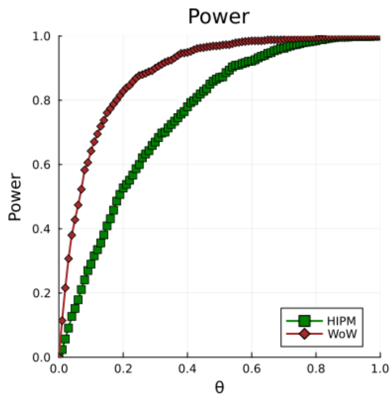
(a) Case A



(b) Case B



(c) Case C



(d) Case D

Figure 3.3: Power for WoW and HIPM

The results presented in Figure 3.3 indicate that HIPM and WoW achieve comparable power in Cases A and B. In Case C, HIPM exhibits slightly higher power than WoW. A more pronounced difference emerges in Case D, where WoW achieves substantially higher power than HIPM. Understanding the conditions under which one method dominates the other remains an open question. One possible direction is to examine the signal-to-noise ratio of the underlying testing problem.

3.6.2 Empirical comparison with existing methods

Several frameworks for two-sample testing in general metric spaces have been developed in the literature. As $(\mathcal{P}(\mathbb{X}), \mathcal{W})$ is a metric space, those methods can be specialized to that setting. We benchmark our proposed scheme against two existing methods. The first is that of Dubey and Müller (2019), which we refer to as the Fréchet test. The second is the Energy test introduced in Székely and Rizzo (2004).

The Fréchet test uses the concepts of Fréchet mean and variance for random probability measures. Recall that the mean of a real-valued random variable X is the number $a \in \mathbb{R}$ such that

$$a = \arg\min_{a' \in \mathbb{R}} \mathbb{E}(X - a')^2.$$

This characterization generalizes to a general metric space. In the space $(\mathcal{P}(\mathbb{X}), \mathcal{W}_2)$, the analogous notion is referred to as the Fréchet mean, defined as

Definition 3.6.1 (Fréchet Mean)**.** Let $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. The Fréchet mean is the probability measure $\mu_F \in \mathcal{P}_2(\mathbb{X})$ defined as

$$\mu_F = \arg\min_{P \in \mathcal{P}_2(\mathbb{X})} \mathbb{E}_{\widetilde{P} \sim Q} \left[\mathcal{W}_2^2(P, \widetilde{P}) \right],$$

where $\mathcal{P}_2(\mathbb{X})$ denotes the space of probability measures on $\mathbb{X}$ with finite second moments.

Accordingly, we describe the dispersion around that point, which is referred to as the Fréchet variance.

Definition 3.6.2 (Fréchet Variance)**.** Let $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. The Fréchet variance V_F is defined as

$$V_F = \min_{P \in \mathcal{P}_2(\mathbb{X})} \mathbb{E}_{\widetilde{P} \sim Q} \left[\mathcal{W}_2^2(P, \widetilde{P}) \right] = \mathbb{E}_{\widetilde{P} \sim Q} \left[\mathcal{W}_2^2(\mu_F, \widetilde{P}) \right]$$

The asymptotic theory of the Fréchet test is developed for bounded metric spaces under additional regularity assumptions. In the case of $\mathcal{P}(\mathbb{X})$, these assumptions are known to hold when the Wasserstein distance is of order 2 and $\mathbb{X}$ is a compact interval of the real line. It is important to note that the Fréchet test targets the surrogate null hypothesis of equality of Fréchet means and Fréchet variances associated to laws of random probability measures.

The Energy test was first developed as a testing procedure for samples taking values in Euclidean spaces. It can be extended to a general metric space $(\mathbb{Y}, d)$, provided that the underlying space is of strongly negative type (see Lyons (2013) for a comprehensive overview).

We compare the methods in two different scenarios using the same baseline as Dubey and Müller (2019). In each scenario, we examine the rejection rates of each testing scheme for several pairs of laws of random probability measures. The random probability measures we consider are

$$\tilde{P}^i | \tilde{\mu}^{(i)} = \mathcal{N}(\tilde{\mu}^{(i)}, 1), \qquad \tilde{\mu}^{(i)} \sim \mathcal{N}_T(\mu_{0i}, \sigma_{0i}^2, a, b), \tag{3.12}$$

for $i = 1, 2$, where $\mathcal{N}_T(\cdot, \cdot, a, b)$ denotes the truncated normal distribution on $[a, b]$. To obtain the set of pairs, we vary a parameter governing the random means. In particular, we consider the following settings:

1. The difference between two distributions is dictated by changing the mean parameter of the random mean, reproducing the setting of Figure 1 in Dubey and Müller (2019), where Q^i are as in (3.12) with $\sigma_{01} = \sigma_{02} = 0.5$, $a = -10$, $b = 10$ and

$$\mu_{01} = 0, \qquad \mu_{02} = \delta, \qquad \delta \in [-1, 1].$$

2. The difference between two distributions is dictated by changing the variance parameter of the random mean, reproducing the setting of Figure 1 in Dubey and Müller (2019), where Q^i are as in (3.12) with $\mu_{01} = \mu_{02} = 0$, $a = -10$, $b = 10$ and

$$\sigma_{01} = 0.2, \qquad \sigma_{02} = 0.2\tau, \qquad \tau \in [0, 2.5].$$

A low type I error is reflected by a low rejection rate under H_0, corresponding respectively to $\delta = 0$ and $\tau = 1$. High power is reflected by a steep increase in the rejection curve away from H_0.

For each setting we simulate the data $S = 1000$ times, perform all tests on the same data, and report the percentage of times they reject the corresponding null hypothesis. We set the hierarchical sample sizes to $n = 100$, $m = 200$. Our tests use a permutation approach with $n_{\text{perm}} = 100$ re-samples, while the Fréchet test uses a bootstrap approach with $n_{\text{boot}} = 100$ re-samples. For the Fréchet test, we work directly with samples from the Gaussian distributions rather than hierarchical samples. Results are reported in Figure 3.4.

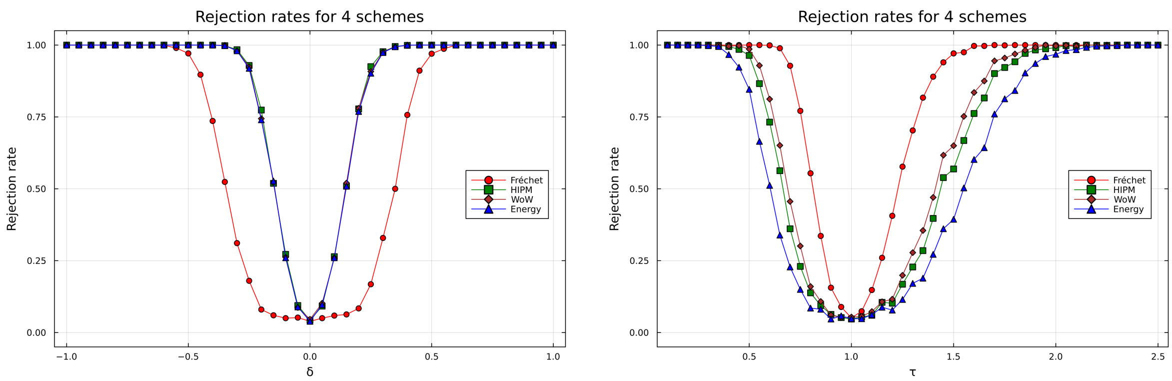


Figure 3.4: Rejection rates for WoW, HIPM, Fréchet and Energy tests

As seen in Figure 3.4, all testing schemes maintain low Type I error across both settings. When the distributions differ in mean (left plot), the Energy, WoW, and HIPM tests achieve higher power than the Fréchet test. In contrast, when the distributions differ in variance (right plot), the Fréchet test attains the highest power among all methods. Furthermore, in the variance setting, both HIPM and WoW outperform the Energy test.

The final simulation on the synthetic dataset is designed to demonstrate the necessity of distance-based testing. Note that the underlying assumption behind the Fréchet test is that the Fréchet mean and variance are sufficient to distinguish between two different laws of random probability measures. However, laws of RPMs are not always uniquely characterized by their Fréchet means and variances.

Indeed, we can construct a simple counterexample. Consider

$$\begin{aligned} Q^1 &= \mathcal{L}\widetilde{P}^1 \quad \text{where} \quad \widetilde{P}^1|\tilde{\mu} = \mathcal{N}\left(\widetilde{\mu}^1, 1\right), \quad \widetilde{\mu}^1 \sim \tfrac{1}{2}\delta_{-1} + \tfrac{1}{2}\delta_1, \\ Q^2 &= \mathcal{L}\widetilde{P}^2 \quad \text{where} \quad \widetilde{P}^2|\tilde{\mu} = \mathcal{N}\left(\widetilde{\mu}^2, 1\right), \quad \widetilde{\mu}^2 \sim \tfrac{1}{8}\delta_{-2} + \tfrac{3}{4}\delta_0 + \tfrac{1}{8}\delta_2, \end{aligned}$$

where $Q^1 \neq Q^2$, yet their associated Fréchet means and variances coincide (see Remark 3.D.2).

Moreover, if we compare Q^1 with a mixture of Q^1 and Q^2, the resulting laws remain different while still inducing the same Fréchet means and variances (see Remark 3.D.3). Specifically, letting

$$M_\lambda = \lambda Q^1 + (1-\lambda)Q^2,$$

we consider the family of pairs

$$\{(Q^1, M_\lambda) : \lambda \in [0,1]\}.$$

Note that $\lambda = 1$ corresponds to estimating the Type I error, while other values of λ correspond to estimating the power of the testing procedures. We consider mixtures of the two laws because, as the parameter varies, it becomes more challenging for the testing schemes to detect the presence of different laws. We plot the rejection rates in Figure 3.5, where the simulation parameters are the same as those used in Figure 3.4.

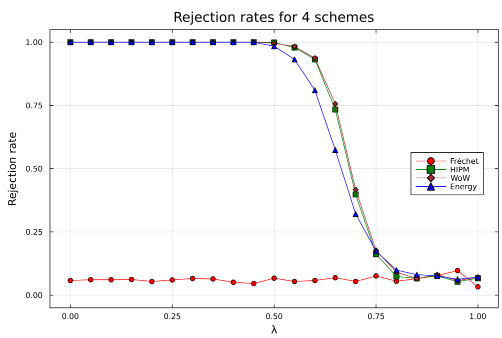


Figure 3.5: Rejection rates for WoW, HIPM, Fréchet and Energy tests

As observed in Figure 3.5, since the two distributions induce the same Fréchet means and variances, the method proposed in Dubey and Müller (2019) fails to detect the presence of different laws. In contrast, the HIPM and WoW tests perform slightly better than the Energy test in identifying the discrepancy.

To conclude, although the Fréchet test can exhibit higher power than the other testing schemes in certain settings, its main limitation is that it fails to detect the difference when the laws of random probability measures are not fully characterized by the Fréchet mean and variance. The Energy test, as well as permutation tests based on HIPM or WoW, mitigate this limitation. Moreover, HIPM and WoW may be preferable to the Energy test in practice, since the underlying metric space, in our case, the space of probability measures, may fail to be of strongly negative type, or this property may be difficult to verify.

3.6.3 Mortality Data

We apply our proposed testing scheme to human mortality data obtained from the Human Mortality Database (https://www.mortality.org/). The dataset contains period life tables for various countries and calendar years. A period life table summarizes mortality conditions in a given calendar year by describing how a hypothetical cohort would die if it faced the age-specific mortality proportions observed in that year. The resulting mortality proportions are then used to derive the age-at-death distribution of the hypothetical cohort.

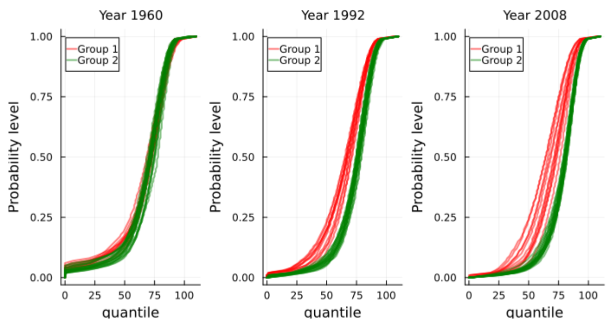
Our analysis compares mortality proportions from 1960 to 2010 between two historically motivated groups of countries. Following the classification in Dubey and Müller (2019), the first group consists of countries that were either part of the Soviet Union or belonged to the Soviet-aligned Eastern Bloc during much of the 20th century. The second group consists mainly of Western or non-Soviet high-income countries, including several countries that were politically neutral during the Cold War.

- Group 1 : Belarus, Bulgaria, Czechia, Estonia, Hungary, Latvia, Lithuania, Poland, Russia, Slovakia, and Ukraine.

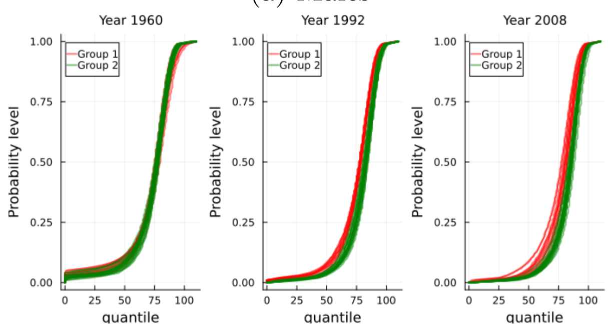
- Group 2 : Australia, Austria, Belgium, Canada, Denmark, Finland, France, Iceland, Ireland, Italy, Japan, Luxembourg, Netherlands, New Zealand, Norway, Spain, Sweden, Switzerland, United Kingdom, and United States of America.

The age range is $\{0, 1, \cdots, 110\}$, where 110 denotes that death occurred at that age or later. We choose not to truncate the age range for the following reasons. First, visual inspection of the probability mass functions and quantiles does not reveal any abnormalities, particularly at higher ages (see Figure 3.6). Second, truncating the age range leads to significant data loss. For example, among females, the maximum percentage of data lost across countries is at least 20 percent for all time periods when the age interval is truncated to $\{0, 1, \ldots, 85\}$.

In Figure 3.6, we present the quantile plots of the probability mass functions corresponding to age at death for both males and females. We consider the years 1960, 1992, and 2008 to capture different historical periods: the Cold War era closer to the post-war period, the years shortly after the collapse of the Soviet Union, and a more recent period following the breakup. The plots are shown separately for females and males. As observed, noticeable differences can be visually identified, particularly in the more recent time periods. To assess whether there is a difference in mortality proportions between the two



(a) Males



(b) Females

Figure 3.6: Quantile plots of the age-at-death probability mass functions for the two country groups

groups of countries in each time period, we apply tests using HIPM and WoW separately for males and females. We set the significance level to $\theta = 0.05$ and use $n_{\text{perm}} = 100$ permutations to compute a p-value for each year.

We also compare the p-values obtained from our approach with those derived from the following alternative methods:

- Averaging: Given $\widehat{P}_1^1, \cdots, \widehat{P}_{n_1}^1$ from group 1 and $\widehat{P}_1^2, \cdots, \widehat{P}_{n_2}^2$ from group 2, we obtain average probability measures inside each group, i.e.

$$\overline{P}^1 = \frac{1}{n_1} \sum_{i=1}^{n_1} \widehat{P}_i^1, \quad \overline{P}^2 = \frac{1}{n_2} \sum_{i=1}^{n_2} \widehat{P}_i^2.$$

The test statistic is obtained by computing Wasserstein-1 distance between $\overline{P}^1$ and $\overline{P}^2$. The threshold is obtained via a permutation approach, in which the observed probability measures are permuted across the two groups.

- Pooling: This approach is included to emphasize the importance of preserving the probability-measure structure within each group. Since the goal is to compare mortality proportions across the two groups, one might instead ignore the country-level distributional structure and simply aggregate deaths within each group. This aggregation produces a single probability measure for each group, and the test statistic is then computed as the Wasserstein-1 distance between these two measures. As before, the significance threshold is obtained using a permutation procedure. Because the data for each country consist of death counts by age, we first convert these counts into observations of age at death. Specifically, if there are k deaths at age a, we generate k observations equal to a. This is done separately for each group. The resulting age-at-death observations are then permuted across the two groups, which allows the permutation procedure to operate directly on individual age-at-death observations rather than on aggregated counts.

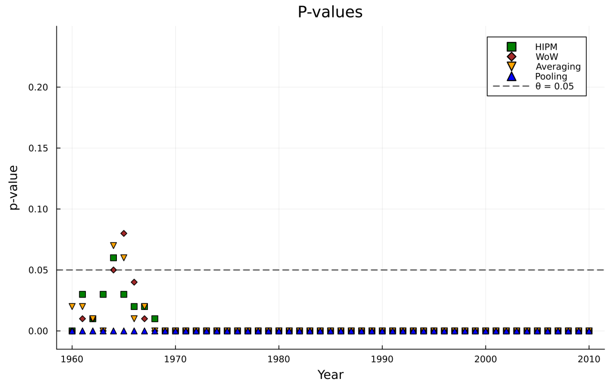
Note that the test statistic is identical for both the pooling and averaging approaches; the difference between the two testing schemes lies in the permutation procedure. In particular, the averaging approach preserves the probability measure structure within each group, whereas the pooling approach does not.

Examining the p-values from all methods in Figure 3.7, we observe that for females, the difference between the groups is statistically significant in 1960. During the following decade, this difference becomes less significant under the methods based on HIPM, WoW, and Averaging, which may be explained by economic growth. After the 1970s, the disparity in mortality proportions between the groups increases again, potentially reflecting the economic decline in the countries of Group 1. Following the breakup of the Soviet Union, the observed differences may be associated with countries that were unable to integrate well into the Western bloc.

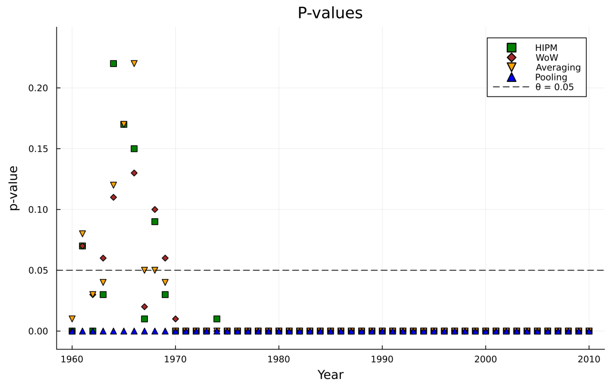
A similar trend is observed for males; however, the key difference is that during the period from 1960 to 1970, although the differences in mortality proportions were decreasing, this was statistically significant in some of those years.

Note also that the p-values for Averaging, HIPM, and WoW follow a similar trend. This is because the average measure is in this case sufficient to detect the difference between the two groups.

We also observe an important issue concerning the pooling approach, which is of broader relevance to hierarchical sampling settings. In particular, the pooling procedure



(a) Males



(b) Females

Figure 3.7: P-values for mortality proportions

may fail to control the Type I error in hierarchical sampling settings. This is because pooling ignores the fact that observations within each country are generated from a latent random probability measure. Under hierarchical sampling, where we observe n sequences of length m, each generated from a different latent random probability measure, the empirical distribution obtained by pooling all observations converges to the mean measure of the law of the random probability measure. Hence, the pooled test statistic converges to the Wasserstein distance between the two group-level mean measures. In contrast, the permutation samples behave as if they were drawn directly from this mean measure, with an effective sample size of nm. As a result, the permutation threshold converges to zero faster than the observed distance, leading to systematically small p-values. This issue disappears when the latent probability measures are identical.

Appendix 3.A Function class for HIPM

Recall that $(\mathbb{X}, d_{\mathbb{X}})$ is a Polish space with bounded diameter $\mathrm{diam}(\mathbb{X}) = K$. Recall also the definition of the function class $\mathcal{D}$ over which the HIPM is defined.

$$\mathcal{D} = \{h : \mathcal{P}(\mathbb{X}) \to \mathbb{R} : h(P) = g(P(f)),\ g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R}),\ f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})\}$$

Note that, as g can be any constant function, $\mathcal{D}$ is not uniformly bounded. However, as we are working with Lipschitz functions and the sample space has bounded diameter, it is natural to expect that a uniformly bounded function class can be extracted from $\mathcal{D}$ without changing the distance function.

We show that after reducing the function class $\mathcal{D}$ to another function class $\mathcal{D}^*$, where

$$\mathcal{D}^* = \{h : \mathcal{P}(\mathbb{X}) \to \mathbb{R} : h(P) = g(P(f)),\ g \in \mathrm{Lip}_1^0([-K, K], \mathbb{R}),\ f \in \mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R})\}, \quad (3.13)$$

where, for a fixed $x_0 \in \mathbb{X}$,

$$\mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R}) = \{f^* = f - f(x_0) : f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})\},$$

and

$$\mathrm{Lip}_1^0([-K, K], \mathbb{R}) = \{g^* = g - g(0) : g \in \mathrm{Lip}_1([-K, K], \mathbb{R})\},$$

the HIPM distance is unchanged and $\mathcal{D}^*$ is uniformly bounded by K.

Recall that

$$d_{\mathrm{Lip}}(Q^1, Q^2) = \sup_{h \in \mathcal{D}} \left| \int_{\mathcal{P}(\mathbb{X})} h(P)\, \mathrm{d}Q^1(P) - \int_{\mathcal{P}(\mathbb{X})} h(P)\, \mathrm{d}Q^2(P) \right|.$$

Step 1. Reduce $\mathcal{D}$ to

$$\mathcal{D}' = \{h : \mathcal{P}(\mathbb{X}) \to \mathbb{R} : h(P) = g(P(f)) \text{ s.t. } g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R}), f \in \mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R})\}.$$

Since $\mathrm{Lip}_1(\mathbb{X}, \mathbb{R})$ contains all constant functions, it is not uniformly bounded. For this reason, for a fixed $x_0 \in \mathbb{X}$, we consider

$$\mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R}) = \{f^* = f - f(x_0) : f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R})\}.$$

Note that the functions in $\mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R})$ are 1-Lipschitz, and they are uniformly bounded by K, because for all $f^* \in \mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R})$,

$$\sup_{x \in \mathbb{X}} |f^*(x)| = \sup_{x \in \mathbb{X}} |f(x) - f(x_0)| \leq \sup_{x \in \mathbb{X}} d_{\mathbb{X}}(x, x_0) \leq K.$$

We verify that replacing $\mathrm{Lip}_1(\mathbb{X}, \mathbb{R})$ by $\mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R})$ does not change the distance; it suffices to show that $\mathcal{D} \subset \mathcal{D}'$, since the inclusion $\mathcal{D}' \subset \mathcal{D}$ is immediate.

Let $h(P) = g(P(f))$ where $f \in \mathrm{Lip}_1(\mathbb{X}, \mathbb{R}), g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R})$. Define $\widetilde{g}(x) = g(x + f(x_0))$ and $f^*(x) = f(x) - f(x_0)$. Note that $\widetilde{g} \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R})$, because for every $x, y \in \mathbb{R}$,

$$|\widetilde{g}(x) - \widetilde{g}(y)| = |g(x + f(x_0)) - g(y + f(x_0))| \leq |(x + f(x_0)) - (y + f(x_0))| = |x - y|.$$

Hence,

$$\widetilde{g}(P(f^*)) = \widetilde{g}(P(f) - f(x_0)) = g(P(f)) = h(P).$$

Since $f^* \in \mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R})$ and $\widetilde{g} \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R})$, it follows that $h \in \mathcal{D}'$.

Step 2. Reduce the function class $\mathcal{D}'$ to the function class

$$\mathcal{D}'' = \{h : \mathcal{P}(\mathbb{X}) \to \mathbb{R} : h(P) = g(P(f)) \text{ s.t. } g \in \mathrm{Lip}_1([-K, K], \mathbb{R}), f \in \mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R})\}.$$

Since $\mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R})$ is uniformly bounded by K, every function $g \in \mathrm{Lip}_1(\mathbb{R}, \mathbb{R})$ is evaluated only at arguments in $[-K, K]$. Indeed, for every $f^* \in \mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R})$,

$$\sup_{P \in \mathcal{P}(\mathbb{X})} |P(f^*)| \leq \sup_{P \in \mathcal{P}(\mathbb{X})} P|f^*| \leq K.$$

Consequently, we restrict attention to $\mathrm{Lip}_1([-K, K], \mathbb{R})$ in place of $\mathrm{Lip}_1(\mathbb{R}, \mathbb{R})$.

Step 3. Reduce $\mathcal{D}''$ to

$$\mathcal{D}^* = \{h : \mathcal{P}(\mathbb{X}) \to \mathbb{R} : h(P) = g(P(f)) \text{ s.t. } g \in \mathrm{Lip}_1^0([-K, K], \mathbb{R}), f \in \mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R})\}.$$

Note that $\mathrm{Lip}_1([-K, K], \mathbb{R})$ is not uniformly bounded, as it contains all constant functions. For this reason, we consider

$$\mathrm{Lip}_1^0([-K, K], \mathbb{R}) = \{g^* = g - g(0) : g \in \mathrm{Lip}_1([-K, K], \mathbb{R})\}.$$

This function class is uniformly bounded, because for all $g^* \in \mathrm{Lip}_1^0([-K, K], \mathbb{R})$,

$$\sup_{x \in [-K,K]} |g^*(x)| = \sup_{x \in [-K,K]} |g(x) - g(0)| \leq \sup_{x \in [-K,K]} |x| \leq K.$$

It remains to verify that the supremum is unchanged when $\mathrm{Lip}_1([-K, K], \mathbb{R})$ is replaced by $\mathrm{Lip}_1^0([-K, K], \mathbb{R})$. To this end, we show that

$$\sup_{h' \in \mathcal{D}''} \left|Q^1(h') - Q^2(h')\right| = \sup_{h \in \mathcal{D}^*} \left|Q^1(h) - Q^2(h)\right|.$$

Note that the right-hand side is at most the left-hand side, since $\mathcal{D}^* \subset \mathcal{D}''$.

Let $h' \in \mathcal{D}''$, i.e. $h'(P) = g(P(f^*))$ for $f^* \in \mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R})$, $g \in \mathrm{Lip}_1([-K, K], \mathbb{R})$. Define

$g^*(x) := g(x) - g(0)$. We have,

$$
\begin{aligned}
\Big| \int_{\mathcal{P}(\mathbb{X})} h'(P)\,\mathrm{d}Q^1(P) - \int_{\mathcal{P}(\mathbb{X})} h'(P)\,\mathrm{d}Q^2(P) \Big| \\
&= \Big| \int_{\mathcal{P}(\mathbb{X})} g(P(f^*))\,\mathrm{d}Q^1(P) - \int_{\mathcal{P}(\mathbb{X})} g(P(f^*))\,\mathrm{d}Q^2(P) \Big| \\
&= \Big| \int_{\mathcal{P}(\mathbb{X})} g^*(P(f^*))\,\mathrm{d}Q^1(P) - \int_{\mathcal{P}(\mathbb{X})} g^*(P(f^*))\,\mathrm{d}Q^2(P) \Big|,
\end{aligned}
$$

since Q^1 and Q^2 are probability measures. Consequently, for every $h' \in \mathcal{D}''$,

$$
\left|Q^1(h') - Q^2(h')\right| \leq \sup_{h \in \mathcal{D}^*} \left|Q^1(h) - Q^2(h)\right|.
$$

Taking the supremum over $h' \in \mathcal{D}''$ yields the conclusion.

In summary, the function class over which the HIPM is defined is

$$
\mathcal{D}^* = \{h : \mathcal{P}(\mathbb{X}) \to \mathbb{R} : h(P) = g(P(f)),\ g \in \mathrm{Lip}_1^0([-K, K], \mathbb{R}),\ f \in \mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R})\}.
$$

Since $\mathrm{Lip}_1^0([-K, K], \mathbb{R})$ is uniformly bounded by K, it follows that $\mathcal{D}^*$ is also uniformly bounded by K. When it is clear from the context, we write g and f in place of g^* and f^*.

Appendix 3.B Weak Convergence of Empirical Processes

In this section, we describe the general strategy for establishing weak convergence (also known as convergence in distribution) of empirical processes and discuss conditions under which this convergence can be metrized. Finally, we introduce the notion of uniform Donsker classes of functions.

3.B.1 Weak convergence on $\ell^\infty(T)$

Let T be a set and let $\ell^\infty(T)$ denote the space of bounded functions from T to $\mathbb{R}$ equipped with the supnorm. The space $(\ell^\infty(T), \|\cdot\|_\infty)$ is equipped with the Borel sigma-algebra. We work with $\ell^\infty(T)$ since empirical processes take values in this space. Weak convergence on $\ell^\infty(T)$ is defined as follows.

Definition 3.B.1 (Weak convergence on $\ell^\infty(T)$)**.** A sequence of probability measures $(\mu_n)_{n\geq 1}$ on $\ell^\infty(T)$ converges weakly to a probability measure μ on $\ell^\infty(T)$ if for all $f \in$

$C_b(\ell^\infty(T), \mathbb{R})$,

$$\lim_{n\to\infty} \int_{\ell^\infty(T)} f \,\mathrm{d}\mu_n = \int_{\ell^\infty(T)} f \,\mathrm{d}\mu.$$

A sequence of random variables $(\mathbb{Z}_n)_{n\geq 1}$ converges weakly to a random variable $\mathbb{Z}$ if their associated laws converge weakly. We use the notation

$$\mathbb{Z}_n \xrightarrow{w} \mathbb{Z}, \qquad \mathcal{L}(\mathbb{Z}_n) \xrightarrow{w} \mathcal{L}(\mathbb{Z}).$$

In the case of a separable probability measure, weak convergence can be metrized by the bounded Lipschitz metric (see Definition 2.3.4). We recall the definition of a separable probability measure.

Definition 3.B.2 (Separable probability measure)**.** A probability measure μ on a measurable space $(\ell^\infty(T), \mathcal{B}(\ell^\infty(T)))$ is separable if there exists a separable subset $A \subset \ell^\infty(T)$ such that $\mu(A) = 1$.

For a sequence of probability measures $(\mu_n)_{n\geq 1}$ on $\ell^\infty(T)$ and a separable probability measure $\mu \in \mathcal{P}(\ell^\infty(T))$, we have (see page 73 in Van der Vaart and Wellner (1996))

$$\mu_n \xrightarrow{w} \mu \quad \iff \quad \lim_{n\to\infty} \mathrm{BL}(\mu_n, \mu) = 0. \tag{3.14}$$

Separability is closely related to the tightness of a probability measure. We recall the definition of tightness.

Definition 3.B.3 (Tightness)**.** A probability measure μ on a metric space $(\ell^\infty(T), \|\cdot\|_\infty)$ is tight if, for every $\epsilon > 0$, there exists a compact subset $A \subset \ell^\infty(T)$ such that

$$\mu(A) \geq 1 - \epsilon.$$

On complete metric spaces, separability of probability measures is equivalent to tightness (see Lemma 1.3.2 in Van der Vaart and Wellner (1996)). This equivalence applies in our setting, since the space $(\ell^\infty(T), \|\cdot\|_\infty)$ is complete.

The general strategy for showing the weak convergence of random variables taking values in $\ell^\infty(T)$ is based on the following theorem.

Theorem 3.B.4. *The sequence $\mathbb{Z}_n : \Omega_n \to \ell^\infty(T)$ converges weakly to the tight limit $\mathbb{Z}$ if and only if the following conditions hold:*

- *i) (Finite-dimensional convergence) The marginals*

$$(\mathbb{Z}_n(t_1), \ldots, \mathbb{Z}_n(t_k)) \xrightarrow{w} (\mathbb{Z}(t_1), \ldots, \mathbb{Z}(t_k))$$

  for every $k \geq 1$ and $t_1, \ldots, t_k \in T$.

- *ii) (Asymptotic equicontinuity) There exists a semimetric ρ on T such that (T,ρ) is totally bounded and the sequence $\mathbb{Z}_n$ is asymptotically uniformly ρ-equicontinuous in probability, i.e.*

$$\lim_{\delta\to 0}\limsup_{n}\mathbb{P}\left(\sup_{s,t\in T,\rho(s,t)<\delta}|\mathbb{Z}_n(s)-\mathbb{Z}_n(t)|>\epsilon\right)=0$$

Proof. The proof follows by combining Theorem 1.5.7, Theorem 1.5.4, and point (ii) of Lemma 1.3.8 from Van der Vaart and Wellner (1996). □

3.B.2 Empirical processes

Empirical processes generalize the central limit theorem to functions of random variables by establishing uniform convergence over classes of functions. The main object is the empirical measure constructed from observed random variables, from which one defines a stochastic process indexed by a function class; the central interest is a uniform central limit theorem over this class. In this subsection, we take $\mathbb{Y}$ to be a Polish space.

Definition 3.B.5 (Empirical process)**.** Let $\mathcal{H}$ be a class of measurable functions from $\mathbb{Y}$ to $\mathbb{R}$, let $P\in\mathcal{P}(\mathbb{Y})$ be a probability measure, and let $Y_1,Y_2,\ldots\overset{\text{i.i.d}}{\sim}P$. The empirical process is the sequence of random variables $(G_{n,P})_{n\geq 1}$, each taking values in $\mathbb{R}^{\mathcal{H}}$, defined by

$$G_{n,P}(f)=\sqrt{n}(P_n(f)-P(f)),$$

where $f\in\mathcal{H}$ and $P_n=\frac{1}{n}\sum_{i=1}^{n}\delta_{Y_i}$.

Even though the empirical process is a sequence of random variables, we denote it by $G_{n,P}$. Note that, in this setting, $T=\mathcal{H}$. To guarantee that such random variables take values in $\ell^\infty(\mathcal{H})$, we assume that the function class $\mathcal{H}$ satisfies:

$$\sup_{f\in\mathcal{H}}|f(y)-P(f)|<\infty \tag{3.15}$$

for all $y\in\mathbb{Y}$. In what follows, we assume that condition (3.15) is satisfied.

Provided that the second moment exists for some fixed $f\in\mathcal{H}$, $G_{n,P}(f)$ converges in distribution to a normally distributed random variable with mean 0 and variance $P(f^2)-(P(f))^2$. If such convergence holds uniformly over $f\in\mathcal{H}$, then the function class $\mathcal{H}$ has a special name:

Definition 3.B.6 (Donsker class)**.** Let $G_{n,P}$ be an empirical process associated with $P\in\mathcal{P}(\mathbb{Y})$ and let $\mathcal{H}$ be a class of measurable functions from $\mathbb{Y}$ to $\mathbb{R}$. The class $\mathcal{H}$ is

P-Donsker if the empirical process $G_{n,P}$ converges in distribution in the space $\ell^\infty(\mathcal{H})$ to a tight random variable.

If such convergence holds, the limiting random variable also has a special name.

Definition 3.B.7 (Brownian bridge)**.** Let $P \in \mathcal{P}(\mathbb{Y})$ and let $\mathcal{H}$ be a class of measurable functions from $\mathbb{Y}$ to $\mathbb{R}$. The P-Brownian bridge process indexed by $\mathcal{H}$ is the zero-mean Gaussian process $(G_P(f))_{f\in\mathcal{H}}$ such that, for every $k \geq 1$ and $f_1, \dots, f_k \in \mathcal{H}$,

$$(G_P(f_1), \dots, G_P(f_k)) \sim \mathcal{N}\left(0, \Sigma^k\right),$$

where the covariance matrix Σ^k is given by

$$\Sigma^k_{ij} = P(f_i f_j) - P(f_i)P(f_j)$$

for $i, j = 1, \dots, k$.

The main tool for establishing the weak convergence of empirical processes is Theorem 3.B.4. For marginal convergence, one typically uses the classical central limit theorem. For the asymptotic equicontinuity, the main tool we will use is the following maximal inequality (Lemma 19.34 in Van der Vaart (1998)).

Lemma 3.B.8. *Let $\mathcal{H}$ be a class of measurable functions $f : \mathbb{Y} \to \mathbb{R}$ such that $Pf^2 < \delta^2$ for all $f \in \mathcal{H}$, and let $G_{n,P}$ be the empirical process. Let F be an envelope function of $\mathcal{H}$ and define*

$$\alpha(\delta) = \frac{\delta}{\sqrt{\log \mathcal{N}_{[]}\left(\delta, \mathcal{H}, \mathrm{L}_2(P)\right)}}.$$

Then

$$\mathbb{E}\left\|G_{n,P}\right\|_{\mathcal{H}} \lesssim J_{[]}(\delta, \mathcal{H}, \mathrm{L}_2(P)) + \sqrt{n}\, P\left\{F\mathbf{1}\big(F > \sqrt{n}\alpha(\delta)\big)\right\},$$

where $\|G_{n,P}\|_{\mathcal{H}} = \sup_{f\in\mathcal{H}} |G_{n,P}(f)|$,

$$J_{[]}(\delta, \mathcal{H}, \mathrm{L}_2(P)) = \int_0^\delta \sqrt{\log \mathcal{N}_{[]}\left(\epsilon, \mathcal{H} \cup \{0\}, \mathrm{L}_2(P)\right)}\, \mathrm{d}\epsilon,$$

and $\mathcal{N}_{[]}$ denotes the bracketing number.

3.B.3 Uniform Donsker Classes

In the previous subsection, we introduced the notion of a Donsker class, in which the property is defined relative to a fixed probability measure. The concept of a uniform Donsker class extends this idea by requiring that weak convergence hold uniformly over a family of probability measures. This notion is useful in the context of hierarchical sampling because, as the number of columns in a hierarchical sample increases, the distributions of the random probability measures constructed from exchangeable sequences also vary.

Definition 3.B.9. Let $\mathcal{H}$ be a class of real-valued measurable functions on $\mathbb{Y}$, and let $(G_{n,P})_{P\in\mathcal{P}}$ be a family of empirical processes associated with a set of probability measures $\mathcal{P} \subset \mathcal{P}(\mathbb{Y})$. The class $\mathcal{H}$ is Donsker uniformly over $\mathcal{P}$ if

$$\lim_{n\to\infty} \sup_{P\in\mathcal{P}} \mathrm{BL}(\mathcal{L}(G_{n,P}), \mathcal{L}(G_P)) = 0,$$

where G_P is a tight Gaussian process for every $P \in \mathcal{P}$. We will also write

$$G_{n,P} \overset{w}{\to} G_P \quad \text{uniformly over } P \in \mathcal{P}.$$

The main tool for establishing that a function class is uniformly Donsker is Theorem 2.8.4 in Van der Vaart and Wellner (1996), which we state as follows.

Theorem 3.B.10. *Let $\mathcal{H}$ be a class of measurable functions with envelope function F and let $\mathcal{P}$ be a class of probability measures such that*

$$\lim_{M\to\infty} \sup_{P\in\mathcal{P}} PF^2\left\{F > M\right\} = 0$$

$$\int_0^\infty \sup_{P\in\mathcal{P}} \sqrt{\log \mathcal{N}_{[]}\left(\epsilon \|F\|_{P,2}, \mathcal{H}, \mathrm{L}_2(P)\right)}\, \mathrm{d}\epsilon < \infty,$$

where $\|F\|_{P,r} = \left(\int_{\mathbb{Y}} |F|^r \,\mathrm{d}P\right)^{1/r}$, and $\mathcal{N}_{[]}$ denotes the bracketing number.

Then, $\mathcal{H}$ is Donsker and pre-Gaussian uniformly in $P \in \mathcal{P}$.

Definition 3.B.11. The function class $\mathcal{H}$ is pre-Gaussian uniformly in $P \in \mathcal{P}$ if the associated Brownian bridge processes satisfy the two conditions:

$$\sup_{P\in\mathcal{P}} \mathbb{E}\|G_P\|_{\mathcal{H}} < \infty, \qquad \lim_{\delta\to 0} \sup_{P\in\mathcal{P}} \mathbb{E} \sup_{\rho_P(f,g)<\delta} |G_P(f) - G_P(g)| = 0,$$

where

$$\rho_P(f, f') = \left[\int_{\mathbb{Y}} (f(y) - f'(y))^2 \,\mathrm{d}P(y) - \left(\int_{\mathbb{Y}} (f(y) - f'(y))\,\mathrm{d}P(y)\right)^2\right]^{\frac{1}{2}}.$$

Remark 3.B.12. There is a relation between the semi-metric $\rho_P(\cdot,\cdot)$, the $\mathrm{L}_2(P)$ norm, and the supnorm. In particular, since

$$\rho_P(f, f')^2 = \|f - f'\|^2_{\mathrm{L}_2(P)} - \left(\int_{\mathbb{Y}} (f(y) - f'(y))\,\mathrm{d}P(y)\right)^2,$$

it follows that

$$\|f - f'\|_\infty \geq \|f - f'\|_{\mathrm{L}_2(P)} \geq \rho_P(f, f').$$

Appendix 3.C Proofs for Section 3.5

In this section, we define the empirical process arising in our setting and prove the main Theorem 3.5.1.

We consider the space $(\ell^\infty(\mathcal{D}^*), \|\cdot\|_{\mathcal{D}^*})$ (see Appendix 3.A for the definition of $\mathcal{D}^*$) with $\|G\|_{\mathcal{D}^*} = \sup_{h \in \mathcal{D}^*} |G(h)|$. We equip this space with the Borel sigma-algebra generated by the distance function induced by the norm $\|\cdot\|_{\mathcal{D}^*}$, and work with random variables taking values in $\ell^\infty(\mathcal{D}^*)$. Finally, we set $K = \mathrm{diam}(\mathbb{X})$.

Given $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$ and the hierarchical empirical estimators $\widehat{Q}_{n,m}$ (see Definition 3.2.3), we define a sequence of empirical processes $G_{n,m,Q}$, for each $m \geq 1$, by

$$G_{n,m,Q}(h) = \sqrt{n}\left(\widehat{Q}_{n,m}(h) - \mathcal{T}_m[Q](h)\right), \quad h \in \mathcal{D}^*.$$

For brevity, we denote $G_{n,m,Q}$ by $G_{n,m}$. To show that it takes values in $\ell^\infty(\mathcal{D}^*)$, note that

$$\begin{aligned}
\sup_{h \in \mathcal{D}^*} |G_{n,m}(h)| &= \sqrt{n} \sup_{h \in \mathcal{D}^*} \left|\widehat{Q}_{n,m}(h) - \mathcal{T}_m[Q](h)\right| \\
&\leq \sqrt{n} \sup_{h \in \mathcal{D}^*} \widehat{Q}_{n,m}(|h|) + \sqrt{n} \sup_{h \in \mathcal{D}^*} \mathcal{T}_m[Q](|h|) \\
&\leq 2K\sqrt{n},
\end{aligned}$$

since $\mathcal{D}^*$ is uniformly bounded by K.

Note also that, as $\mathcal{D}^*$ is uniformly bounded, we have

$$\sup_{h \in \mathcal{D}^*} |h(P) - \mathcal{T}_m[Q](h)| < \infty$$

for all $P \in \mathcal{P}(\mathbb{X})$. Before proving the main Theorem 3.5.1, we study the asymptotic distribution of the test statistic and the threshold via the permutation approach.

Theorem 3.C.1 (Asymptotic distribution of test statistic). *Let $\mathbb{X} = [a, b]$ and $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Let $T_{n,m} = \sqrt{\frac{n}{2}} d_{\mathrm{Lip}}\left(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}\right)$ for $n, m \geq 1$. Then the following statements hold:*

- *If $Q^1 = Q^2$, then $T_{n,m} \xrightarrow{w} \|G_{Q^1}\|_{\mathcal{D}^*}$ as $n, m \to \infty$, where G_{Q^1} is a tight Q^1-Brownian bridge process indexed by $\mathcal{D}^*$.*
- *If $Q^1 \neq Q^2$, then $T_{n,m} \xrightarrow{p} \infty$ as $n, m \to \infty$.*

Proof. We first consider the case $Q^1 = Q^2$.

Recall our setting (3.1). Under the null hypothesis, we have

$$\mathcal{T}_m[Q^1] = \mathcal{T}_m[Q^2].$$

We write the test statistic $T_{n,m}$ as

$$\begin{aligned}
T_{n,m} &= \sqrt{\frac{n}{2}} d_{\mathrm{Lip}}(\widehat{Q}^1_{n,m}, \widehat{Q}^2_{n,m}) \\
&= \sup_{h\in\mathcal{D}^*} \left| \sqrt{\frac{n}{2}} \left(\widehat{Q}^1_{n,m}(h) - \widehat{Q}^2_{n,m}(h) \right) \right| \\
&= \sup_{h\in\mathcal{D}^*} \left| \frac{1}{\sqrt{2}} \sqrt{n} \left(\widehat{Q}^1_{n,m}(h) - \mathcal{T}_m[Q^1](h) \right) - \frac{1}{\sqrt{2}} \sqrt{n} \left(\widehat{Q}^2_{n,m}(h) - \mathcal{T}_m[Q^2](h) \right) \right|.
\end{aligned}$$

Since n and m diverge to infinity independently, let $(n_q)_{q\geq 1}$ and $(m_q)_{q\geq 1}$ be sequences such that

$$\lim_{q\to\infty} n_q = \infty \quad \text{and} \quad \lim_{q\to\infty} m_q = \infty.$$

Our goal is to show that

$$T_{n_q,m_q} \xrightarrow{w} \|G_{Q^1}\|_{\mathcal{D}^*}$$

as $q \to \infty$.

Recall that

$$T_{n_q,m_q} = \sup_{h\in\mathcal{D}^*} \left| \frac{1}{\sqrt{2}} \sqrt{n_q} \left(\widehat{Q}^1_{n_q,m_q}(h) - \mathcal{T}_{m_q}[Q^1](h) \right) - \frac{1}{\sqrt{2}} \sqrt{n_q} \left(\widehat{Q}^2_{n_q,m_q}(h) - \mathcal{T}_{m_q}[Q^2](h) \right) \right|.$$

Let $Q = Q^1$. We decompose the proof into four steps:

1. Empirical processes converge uniformly:

$$\lim_{n\to\infty} \sqrt{n} \left(\widehat{Q}_{n,m}(h) - \mathcal{T}_m[Q](h) \right) = G_{\mathcal{T}_m[Q]} \quad \text{uniformly over } m \geq 1.$$

2. $\lim_{m\to\infty} G_{\mathcal{T}_m[Q]} = G_Q$
3. $\lim_{q\to\infty} \sqrt{n_q} \left(\widehat{Q}_{n_q,m_q}(h) - \mathcal{T}_{m_q}[Q](h) \right) = G_Q$
4. $\lim_{q\to\infty} T_{n_q,m_q} = \|G_{Q^1}\|_{\mathcal{D}^*}$

Step 1.

Let $G_{n,m} = \sqrt{n}(\widehat{Q}_{n,m} - \mathcal{T}_m[Q])$ for $n, m \geq 1$. To show that the function class $\mathcal{D}^*$ is uniformly Donsker over the laws $\mathcal{T}_m[Q]$ for all $m = 1, 2, \ldots,$ we use Theorem 3.B.10.

The envelope function of the class $\mathcal{D}^*$ is the constant function equal to K. Hence, the first assumption is trivially satisfied. Moreover,

$$\|F\|_{\mathcal{T}_m[Q],2} = K.$$

By Lemma 3.C.9, we obtain

$$\begin{aligned}\int_0^\infty \sup_{m\geq 1} \sqrt{\log \mathcal{N}_{[]}(\epsilon K, \mathcal{D}^*, \mathrm{L}_2(\mathcal{T}_m[Q]))}\,\mathrm{d}\epsilon &\leq \int_0^\infty \sup_{m\geq 1} \sqrt{\log \mathcal{N}\big(\tfrac{\epsilon}{2}K, \mathcal{D}^*, \|\cdot\|_\infty\big)}\,\mathrm{d}\epsilon \\ &= \int_0^\infty \sqrt{\log \mathcal{N}\big(\tfrac{\epsilon}{2}K, \mathcal{D}^*, \|\cdot\|_\infty\big)}\,\mathrm{d}\epsilon.\end{aligned}$$

Since $\|h - h'\|_\infty \leq 2K$ for all $h, h' \in \mathcal{D}^*$, it follows that for all $\epsilon \geq 4$,

$$\mathcal{N}\big(\tfrac{\epsilon K}{2}, \mathcal{D}^*, \|\cdot\|_\infty\big) = 1.$$

Consequently, it suffices to consider

$$\int_0^4 \sqrt{\log \mathcal{N}\big(\tfrac{\epsilon}{2}K, \mathcal{D}^*, \|\cdot\|_\infty\big)}\,\mathrm{d}\epsilon,$$

since the remainder of the integral vanishes. By Lemma 3.C.10, there exists a constant $C > 0$ such that

$$\int_0^4 \sqrt{\log \mathcal{N}\big(\tfrac{\epsilon}{2}K, \mathcal{D}^*, \|\cdot\|_\infty\big)}\,\mathrm{d}\epsilon \leq \int_0^4 C\epsilon^{-1/2}\,\mathrm{d}\epsilon < \infty.$$

Therefore, by Theorem 2.8.4 in Van der Vaart and Wellner (1996),

$$G_{n,m} \xrightarrow{w} G_{\mathcal{T}_m[Q]} \quad \text{uniformly in } m,$$

where $\{G_{\mathcal{T}_m[Q]}\}_{m\geq 1}$ is a family of tight Gaussian processes and the class $\mathcal{D}^*$ is uniformly pre-Gaussian.

Step 2.

To show the convergence of Gaussian processes, we apply Theorem 3.B.4. We equip $\mathcal{D}^*$ with the supremum norm $\|\cdot\|_\infty$. By Lemma 3.C.10, $(\mathcal{D}^*, \|\cdot\|_\infty)$ is totally bounded. It therefore suffices to verify convergence of finite-dimensional distributions and asymptotic equicontinuity.

Fix $h_1, \ldots, h_N \in \mathcal{D}^*$ for some integer $N \geq 1$. For each m,

$$\big(G_{\mathcal{T}_m[Q]}(h_1), \ldots, G_{\mathcal{T}_m[Q]}(h_N)\big) \sim \mathcal{N}(0, \Sigma^m),$$

where

$$\Sigma_{ij}^m = \mathcal{T}_m[Q](h_i h_j) - \mathcal{T}_m[Q](h_i)\mathcal{T}_m[Q](h_j).$$

Let

$$\Sigma_{ij} = Q^1(h_i h_j) - Q^1(h_i)Q^1(h_j)$$

denote the covariance matrix of $\big(G_{Q^1}(h_1), \dots, G_{Q^1}(h_N)\big)$. By Lemma 3.C.5,

$$\lim_{m\to\infty} d_{\mathrm{Lip}}(\mathcal{T}_m[Q], Q) = 0.$$

This implies that $\lim_{m\to\infty} \mathcal{T}_m[Q](h_i) = Q(h_i)$ for all $i = 1, \dots, N$. Hence, the second terms in Σ_{ij}^m converge to the corresponding terms in Σ_{ij}.

Since d_{Lip} metrizes weak convergence on $\mathcal{P}(\mathcal{P}(\mathbb{X}))$, we have

$$\mathcal{T}_m[Q](h) \to Q(h)$$

for every function h that is bounded and continuous with respect to weak convergence on $\mathcal{P}(\mathbb{X})$. Therefore, to show the convergence of the first terms of the covariance matrices, it suffices to show that the product $h_i h_j$ is continuous and bounded.

Each h_i is bounded by K, hence $\|h_i h_j\|_\infty \le K^2$. Moreover, if $P_n \xrightarrow{w} P$ in $\mathcal{P}(\mathbb{X})$, then

$$\lim_{n\to\infty} h_i(P_n) = \lim_{n\to\infty} g_i(P_n(f_i)) = g_i\left(\lim_{n\to\infty} P_n(f_i)\right) = g_i\left(P(f_i)\right) = h_i(P),$$

since both f_i and g_i are continuous and f_i is bounded. Thus $h_i h_j$ is bounded and continuous, implying

$$\Sigma_{ij}^m \to \Sigma_{ij}.$$

Therefore, finite-dimensional distributions converge.

Uniform pre-Gaussianity yields asymptotic equicontinuity. Indeed, by Remark 3.B.12

$$\rho_{\mathcal{T}_m[Q]}(h, h') \le \|h - h'\|_\infty,$$

we have

$$\sup_{m\ge 1} \mathbb{E} \sup_{\|h-h'\|_\infty<\delta} |G_{\mathcal{T}_m[Q]}(h) - G_{\mathcal{T}_m[Q]}(h')| \le \sup_{m\ge 1} \mathbb{E} \sup_{\rho_{\mathcal{T}_m[Q]}(h,h')<\delta} |G_{\mathcal{T}_m[Q]}(h) - G_{\mathcal{T}_m[Q]}(h')|.$$

The latter converges to 0 as $\delta \to 0$ and asymptotic equicontinuity follows from Markov's inequality. Hence, by Theorem 3.B.4,

$$\mathcal{L}(G_{\mathcal{T}_m[Q]}) \xrightarrow{w} \mathcal{L}(G_Q).$$

Step 3. Since $\ell^\infty(\mathcal{D}^*)$ is complete, the law of the tight Gaussian process G_Q is separable (see Lemma 1.3.2 in Van der Vaart and Wellner (1996)); therefore, we can use the metrization of weak convergence on $\ell^\infty(\mathcal{D}^*)$ (see (3.14)).

Consider the sequences $(n_q)_{q\geq 1}$ and $(m_q)_{q\geq 1}$ defined above. Then

$$\begin{aligned}
\mathrm{BL}\left(\mathcal{L}\left(G_{n_q,m_q}\right),\mathcal{L}\left(G_Q\right)\right) &\leq \mathrm{BL}\left(\mathcal{L}\left(G_{n_q,m_q}\right),\mathcal{L}\left(G_{\mathcal{T}_{m_q}[Q]}\right)\right) + \mathrm{BL}\left(\mathcal{L}\left(G_{\mathcal{T}_{m_q}[Q]}\right),\mathcal{L}\left(G_{Q^1}\right)\right) \\
&\leq \sup_{m\geq 1} \mathrm{BL}\left(\mathcal{L}\left(G_{n_q,m}\right),\mathcal{L}\left(G_{\mathcal{T}_m[Q]}\right)\right) \\
&\quad + \mathrm{BL}\left(\mathcal{L}\left(G_{\mathcal{T}_{m_q}[Q]}\right),\mathcal{L}\left(G_{Q^1}\right)\right),
\end{aligned}$$

which converges to 0 by Steps 1 and 2 as $q \to \infty$.

Step 4. Recall that

$$T_{n_q,m_q} = \sup_{h\in\mathcal{D}^*}\left|\frac{1}{\sqrt{2}}\sqrt{n_q}\left(\widehat{Q}^1_{n_q,m_q}(h) - \mathcal{T}_{m_q}[Q^1](h)\right) - \frac{1}{\sqrt{2}}\sqrt{n_q}\left(\widehat{Q}^2_{n_q,m_q}(h) - \mathcal{T}_{m_q}[Q^2](h)\right)\right|.$$

The term inside the absolute value converges weakly in the space $\ell^\infty(\mathcal{D}^*)$ to the difference of Gaussian processes

$$\frac{1}{\sqrt{2}}G_{Q^1} - \frac{1}{\sqrt{2}}G_{Q^2} \overset{d}{=} G_{Q^1}.$$

Since $\|\cdot\|_{\mathcal{D}^*}$ is a continuous function from $\ell^\infty(\mathcal{D}^*)$ to $\mathbb{R}$, the Continuous Mapping Theorem yields

$$T_{n_q,m_q} \xrightarrow{w} \|G_{Q^1}\|_{\mathcal{D}^*} \quad \text{as } q \to \infty.$$

We now address the case $Q^1 \neq Q^2$. We decompose the test statistic as:

$$\begin{aligned}
T_{n_q,m_q} &= \sup_{h\in\mathcal{D}^*}\left|\sqrt{\frac{n_q}{2}}\left(\widehat{Q}^1_{n_q,m_q}(h) - \widehat{Q}^2_{n_q,m_q}(h)\right)\right| \\
&= \sup_{h\in\mathcal{D}^*}\left|\frac{1}{\sqrt{2}}\sqrt{n_q}\left(\widehat{Q}^1_{n_q,m_q}(h) - \mathcal{T}_{m_q}[Q^1](h)\right) - \frac{1}{\sqrt{2}}\sqrt{n_q}\left(\widehat{Q}^2_{n_q,m_q}(h) - \mathcal{T}_{m_q}[Q^2](h)\right)\right. \\
&\qquad\left. + \frac{1}{\sqrt{2}}\sqrt{n_q}\left(\mathcal{T}_{m_q}[Q^1](h) - \mathcal{T}_{m_q}[Q^2](h)\right)\right|.
\end{aligned}$$

Let $T_{n_q,m_q} = \|a - b + c\|_{\mathcal{D}^*}$, where for $h \in \mathcal{D}^*$,

$$\begin{aligned}
a(h) &= \frac{1}{\sqrt{2}}\sqrt{n_q}\left(\widehat{Q}^1_{n_q,m_q}(h) - \mathcal{T}_{m_q}[Q^1](h)\right), \\
b(h) &= \frac{1}{\sqrt{2}}\sqrt{n_q}\left(\widehat{Q}^2_{n_q,m_q}(h) - \mathcal{T}_{m_q}[Q^2](h)\right), \\
c(h) &= \frac{1}{\sqrt{2}}\sqrt{n_q}\left(\mathcal{T}_{m_q}[Q^1](h) - \mathcal{T}_{m_q}[Q^2](h)\right).
\end{aligned}$$

Since

$$\|c\|_{\mathcal{D}^*} = \|(a-b+c)-(a-b)\|_{\mathcal{D}^*} \leq \|a-b+c\|_{\mathcal{D}^*} + \|a-b\|_{\mathcal{D}^*},$$

we obtain

$$\|a-b+c\|_{\mathcal{D}^*} \geq \|c\|_{\mathcal{D}^*} - \|a-b\|_{\mathcal{D}^*} \geq \|c\|_{\mathcal{D}^*} - \|a\|_{\mathcal{D}^*} - \|b\|_{\mathcal{D}^*}.$$

Thus,

$$\begin{aligned} T_{n_q,m_q} \geq &\sup_{h\in\mathcal{D}^*}\left|\frac{1}{\sqrt{2}}\sqrt{n_q}\left(\mathcal{T}_{m_q}[Q^1](h) - \mathcal{T}_{m_q}[Q^2](h)\right)\right| \\ &- \sup_{h\in\mathcal{D}^*}\left|\frac{1}{\sqrt{2}}\sqrt{n_q}\left(\widehat{Q}^1_{n_q,m_q}(h) - \mathcal{T}_{m_q}[Q^1](h)\right)\right| \\ &- \sup_{h\in\mathcal{D}^*}\left|\frac{1}{\sqrt{2}}\sqrt{n_q}\left(\widehat{Q}^2_{n_q,m_q}(h) - \mathcal{T}_{m_q}[Q^2](h)\right)\right|. \end{aligned}$$

In the first part of the proof, we showed that

$$\begin{aligned} &\sup_{h\in\mathcal{D}^*}\left|\sqrt{n_q}\left(\widehat{Q}^1_{n_q,m_q}(h) - \mathcal{T}_{m_q}[Q^1](h)\right)\right| \xrightarrow{w} \|G_{Q^1}\|_{\mathcal{D}^*}, \\ &\sup_{h\in\mathcal{D}^*}\left|\sqrt{n_q}\left(\widehat{Q}^2_{n_q,m_q}(h) - \mathcal{T}_{m_q}[Q^2](h)\right)\right| \xrightarrow{w} \|G_{Q^2}\|_{\mathcal{D}^*}, \end{aligned}$$

where G_{Q^1} and G_{Q^2} are tight Gaussian processes. By Lemma 3.C.6, the random variables $\|G_{Q^1}\|_{\mathcal{D}^*}$ and $\|G_{Q^2}\|_{\mathcal{D}^*}$ are almost surely bounded.

For brevity, define

$$\begin{aligned} A_q &= \sup_{h\in\mathcal{D}^*}\left|\sqrt{n_q}\left(\widehat{Q}^1_{n_q,m_q}(h) - \mathcal{T}_{m_q}[Q^1](h)\right)\right| \\ B_q &= \sup_{h\in\mathcal{D}^*}\left|\sqrt{n_q}\left(\widehat{Q}^2_{n_q,m_q}(h) - \mathcal{T}_{m_q}[Q^2](h)\right)\right| \\ C_q &= \sup_{h\in\mathcal{D}^*}\left|\sqrt{n_q}\left(\mathcal{T}_{m_q}[Q^1](h) - \mathcal{T}_{m_q}[Q^2](h)\right)\right|. \end{aligned}$$

Note that $C_q = \sqrt{n_q} d_{\text{Lip}}\left(\mathcal{T}_{m_q}[Q^1], \mathcal{T}_{m_q}[Q^2]\right) \to \infty$. Indeed, by the triangle inequality,

$$\left|d_{\text{Lip}}(\mathcal{T}_{m_q}[Q^1], \mathcal{T}_{m_q}[Q^2]) - d_{\text{Lip}}(Q^1, Q^2)\right| \leq d_{\text{Lip}}(Q^1, \mathcal{T}_{m_q}[Q^1]) + d_{\text{Lip}}(Q^2, \mathcal{T}_{m_q}[Q^2]),$$

which converges to 0 by Lemma 3.C.5. As $d_{\text{Lip}}(Q^1, Q^2) > 0$, it follows that $C_q \to \infty$ as $q \to \infty$. Recall that our goal is to show that for every $M > 0$ and $\delta \in (0,1)$, there exists $q_0 \geq 1$ such that for all $q \geq q_0$,

$$\mathbb{P}\left(T_{n_q,m_q} > M\right) > \delta.$$

Now fix $M > 0$ and $\delta \in (0,1)$. Also fix $\delta' > \frac{\delta+1}{2}$. As $\|G_{Q^1}\|_{\mathcal{D}^*}$ and $\|G_{Q^2}\|_{\mathcal{D}^*}$ are almost surely bounded, there exists a constant $c' \in \mathbb{R}$ such that

$$\mathbb{P}\left(\|G_{Q^1}\|_{\mathcal{D}^*} < c'\right) > \delta' \quad \text{and} \quad \mathbb{P}\left(\|G_{Q^2}\|_{\mathcal{D}^*} < c'\right) > \delta'.$$

As $A_q \xrightarrow{w} \|G_{Q^1}\|_{\mathcal{D}^*}$ and $B_q \xrightarrow{w} \|G_{Q^2}\|_{\mathcal{D}^*}$ as $q \to \infty$, by the Portmanteau Theorem it follows that there exists $q' \geq 1$ such that for all $q \geq q'$,

$$\mathbb{P}\left(A_q < c'\right) > \delta' \quad \text{and} \quad \mathbb{P}\left(B_q < c'\right) > \delta'.$$

Note that

$$\begin{aligned}\mathbb{P}\left(A_q < c', B_q < c'\right) &= 1 - \mathbb{P}\left(A_q > c' \text{ or } B_q > c'\right) \geq 1 - \mathbb{P}\left(A_q > c'\right) - \mathbb{P}\left(B_q > c'\right)\\ &\geq 1 - (1-\delta') - (1-\delta') = 2\delta' - 1.\end{aligned}$$

Recall that $C_q \to \infty$. This implies that there exists q'' such that

$$C_q > \sqrt{2}M + c' + c' \quad \text{for all } q \geq q''.$$

Now, let $S_1 = \{A_q < c'\}$ and $S_2 = \{B_q < c'\}$ and take $q \geq \max\{q', q''\}$. Then

$$\begin{aligned}\mathbb{P}\left(T_{n_q,m_q} > M\right) &\geq \mathbb{P}\left(\frac{1}{\sqrt{2}}C_q - \frac{1}{\sqrt{2}}A_q - \frac{1}{\sqrt{2}}B_q > M\right)\\ &\geq \mathbb{P}\left(\left\{C_q > \sqrt{2}M + A_q + B_q\right\} \cap S_1 \cap S_2\right)\\ &\geq \mathbb{P}\left(\left\{C_q > \sqrt{2}M + c' + c'\right\} \cap S_1 \cap S_2\right)\\ &= \mathbb{P}\left(S_1 \cap S_2\right) \geq 2\delta' - 1 > \delta.\end{aligned}$$

□

We now study the asymptotic behavior of the threshold under the permutation approach. As in the case of the test statistic, to derive the asymptotic distribution of the threshold, we first analyze the asymptotic distribution of the underlying empirical process. Before stating the result, let us set up some notation.

Recall the definition of the permuted hierarchical empirical estimators $\widetilde{Q}^1_{n,m}$ and $\widetilde{Q}^2_{n,m}$ (3.10). We also define

$$H_{n,m} = \frac{1}{2}\widehat{Q}^1_{n,m} + \frac{1}{2}\widehat{Q}^2_{n,m}, \quad H = \frac{1}{2}Q^1 + \frac{1}{2}Q^2.$$

Note that given the hierarchical samples $\mathbf{X}_{n,m}, \mathbf{Y}_{n,m}$,

$$\frac{1}{2}\widetilde{Q}^1_{n,m} + \frac{1}{2}\widetilde{Q}^2_{n,m} = \frac{1}{2}\widehat{Q}^1_{n,m} + \frac{1}{2}\widehat{Q}^2_{n,m}.$$

Theorem 3.C.2. *Let $\mathbb{X} = [a,b]$ and $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Given hierarchical samples $(X_{n,m})_{n,m\geq 1}$,$(Y_{n,m})_{n,m\geq 1}$, define*

$$\widetilde{G}_{n,m} = \sqrt{n}(\widetilde{Q}^1_{n,m} - H_{n,m})$$

for $n, m \geq 1$.

Then, for almost every infinite size hierarchical samples $(X_{n,m})_{n,m\geq 1}$,$(Y_{n,m})_{n,m\geq 1}$ from Q^1 and Q^2,

$$\widetilde{G}_{n,m} \overset{w}{\to} \sqrt{\frac{1}{2}}G_H \quad \textit{as } n, m \to \infty,$$

where G_H is the H-Brownian bridge process indexed by $\mathcal{D}^*$.

Proof. Let us consider the sequences $(n_q)_{q\geq 1}$ and $(m_q)_{q\geq 1}$ such that

$$\lim_{q\to\infty} n_q = \lim_{q\to\infty} m_q = \infty.$$

We adapt the notation accordingly.

Let $(X_{n,m})_{n,m\geq 1}$,$(Y_{n,m})_{n,m\geq 1}$ be hierarchical samples of infinite size from Q^1 and Q^2 respectively. For $q \geq 1$, denote by Z_q the random vector of length $2n_q$, defined by

$$Z_q = \left(Z_{1,q}, \ldots, Z_{2n_q,q}\right) = \left(\widehat{P}^1_{1,m_q}, \ldots, \widehat{P}^1_{n_q,m_q}, \widehat{P}^2_{1,m_q}, \ldots, \widehat{P}^2_{n_q,m_q}\right),$$

and by $R_q = \left(R_{1,q}, \ldots, R_{2n_q,q}\right)$, a random vector taking values uniformly over the set of all permutations of $\{1, \ldots, 2n_q\}$. We also denote

$$\widehat{Q}^1_q = \widehat{Q}^1_{n_q,m_q}, \qquad \widehat{Q}^2_q = \widehat{Q}^2_{n_q,m_q}, \qquad \widetilde{Q}^1_q = \widetilde{Q}^1_{n_q,m_q}, \qquad \widetilde{Q}^2_q = \widetilde{Q}^2_{n_q,m_q}.$$

and

$$H_q = \frac{1}{2}\widetilde{Q}^1_q + \frac{1}{2}\widetilde{Q}^2_q, \quad H = \frac{1}{2}Q^1 + \frac{1}{2}Q^2.$$

We divide the proof of the convergence of $\widetilde{G}_q = \widetilde{G}_{n_q,m_q}$ into two main steps: finite-dimensional convergence and asymptotic equicontinuity. The result then follows by Theorem 3.B.4.

Step 1. Finite-dimensional convergence

This part is further divided into two steps: first, we show convergence for a single function in $\mathcal{D}^*$, and then, using the Cramér–Wold theorem, we extend the result to a finite number of functions.

- Step 1.1. Marginal convergence.

Fix a measurable function $h : \mathcal{P}(\mathbb{X}) \to \mathbb{R}$ such that $\|h\|_\infty < \infty$. Note that every function in $\mathcal{D}^*$ has a bounded sup norm. We want to show that almost surely on $(X_{n,m})_{n,m\geq 1}$,$(Y_{n,m})_{n,m\geq 1}$ from Q^1 and Q^2,

$$\widetilde{G}_q(h) \xrightarrow{w} \mathcal{N}\left(0, \frac{1}{2}\mathrm{Var}_H(h)\right),$$

where $\mathrm{Var}_H(h) = H\left(h^2\right) - \left(H(h)\right)^2$.

If $h = 0$, then the result follows trivially. Assume that $h \neq 0$. We have

$$\begin{aligned}
\widetilde{G}_q(h) &= \sqrt{n_q}\left(\sum_{i=1}^{n_q} \frac{1}{n_q} h(Z_{R_{i,q},q}) - H_q(h)\right) \\
&= \frac{1}{\sqrt{n_q}} \sum_{i=1}^{n_q} \left(h\left(Z_{R_{i,q},q}\right) - H_q(h)\right) \\
&= \frac{1}{\sqrt{n_q}} \left(\sum_{i=1}^{n_q} h\left(Z_{R_{i,q},q}\right) - n_q H_q(h)\right).
\end{aligned}$$

Define, for $q \geq 1$,

$$S_q = \sum_{i=1}^{2n_q} a_{R_{i,q},q} b_{i,q},$$

where

$$\begin{aligned}
a_q &= \left(h\left(Z_{1,q}\right), \ldots, h\left(Z_{2n_q,q}\right)\right) \\
&= \left(h\left(\widehat{P}^1_{1,m_q}\right), \ldots, h\left(\widehat{P}^1_{n_q,m_q}\right), h\left(\widehat{P}^2_{1,m_q}\right), \ldots, h\left(\widehat{P}^2_{n_q,m_q}\right)\right),
\end{aligned}$$

and $b_q = \left(b_{1,q}, \ldots, b_{2n_q,q}\right)$, where

$$b_{i,q} = \begin{cases} 1, & \text{if } i \leq n_q, \\ 0, & \text{if } i > n_q. \end{cases}$$

We compute

$$\mathbb{E}_{R_q} S_q = \sum_{i=1}^{2n_q} \mathbb{E}_{R_q} a_{R_{i,q},q} b_{i,q} = \sum_{i=1}^{n_q} \mathbb{E}_{R_q} a_{R_{i,q},q} = \sum_{i=1}^{n_q} H_q(h) = n_q H_q(h).$$

Thus,

$$\widetilde{G}_q(h) = \frac{1}{\sqrt{n_q}}\left(S_q - \mathbb{E}_{R_q} S_q\right) = \frac{\sqrt{\operatorname{Var}(S_q)}}{\sqrt{n_q}} \frac{\left(S_q - \mathbb{E}_{R_q} S_q\right)}{\sqrt{\operatorname{Var}(S_q)}}.$$

First, we show that

$$\frac{\left(S_q - \mathbb{E}_{R_q} S_q\right)}{\sqrt{\operatorname{Var}(S_q)}} \xrightarrow{w} \mathcal{N}(0,1),$$

using Theorem 4.1 in Hajek (1961). For that, define

$$A_q^2 = \sum_{i=1}^{2n_q} \left(a_{i,q} - \bar{a}_q\right)^2, \quad B_q^2 = \sum_{i=1}^{2n_q} \left(b_{i,q} - \bar{b}_q\right)^2,$$

where $\bar{a}_q$ and $\bar{b}_q$ denote averages. We verify the three conditions of Theorem 4.1 in Hajek (1961).

i) $\lim\limits_{q\to\infty} \dfrac{\max_{1\le i\le 2n_q}\left(a_{i,q} - \bar{a}_q\right)^2}{A_q^2} = 0.$

For fixed $q \ge 1$,

$$\frac{\max_{1\le i\le 2n_q}\left(a_{i,q} - \bar{a}_q\right)^2}{A_q^2} = \frac{\frac{\max_{1\le i\le 2n_q}\left(a_{i,q} - \bar{a}_q\right)^2}{2n_q}}{\frac{\sum_{i=1}^{2n_q}\left(a_{i,q} - \bar{a}_q\right)^2}{2n_q}}.$$

Consider the denominator:

$$\frac{A_q^2}{2n_q} = \frac{\sum_{i=1}^{2n_q}\left(a_{i,q} - \bar{a}_q\right)^2}{2n_q} = \frac{1}{2n_q}\sum_{i=1}^{n_q} h^2\left(\widehat{P}^1_{i,m_q}\right) + \frac{1}{2n_q}\sum_{i=1}^{n_q} h^2\left(\widehat{P}^2_{i,m_q}\right) - \left(H_q(h)\right)^2.$$

Since h is bounded, by Lemma 3.C.4

$$\frac{A_q^2}{2n_q} \xrightarrow{a.s.} \frac{1}{2}Q^1(h^2) + \frac{1}{2}Q^2(h^2) - (H(h))^2 = H(h^2) - (H(h))^2 = \operatorname{Var}_H(h).$$

For the numerator, note that

$$\frac{\max_{1\le i\le 2n_q}\left(a_{i,q} - \bar{a}_q\right)^2}{2n_q} \le \max_{1\le i\le 2n_q} \frac{a_{i,q}^2 + \bar{a}_q^2 + 2|a_{i,q}||\bar{a}_q|}{2n_q}.$$

We have already shown that $\bar{a}_q \xrightarrow{a.s.} H(h)$. Hence, it suffices to show that

$$\max_{1\le i\le 2n_q} \frac{a_{i,q}^2}{2n_q} \xrightarrow{a.s.} 0 \quad \text{and} \quad \max_{1\le i\le 2n_q} \frac{|a_{i,q}|}{2n_q} \xrightarrow{a.s.} 0.$$

Since $a_{i,q} = h\left(Z_{i,q}\right)$ and the function h is bounded, the desired conclusion follows.

ii) $\lim\limits_{q\to\infty} \dfrac{\max_{1\le i\le 2n_q}\left(b_{i,q} - \bar{b}_q\right)^2}{B_q^2} = 0.$

Note that

$$B_q^2 = \sum_{i=1}^{2n_q} (b_{i,q} - \bar{b}_q)^2 = \sum_{i=1}^{n_q} \left(1 - \frac{1}{2}\right)^2 + \sum_{i=n_q+1}^{2n_q} \left(-\frac{1}{2}\right)^2 = \frac{n_q}{2}.$$

Hence,

$$\lim_{q\to\infty} \frac{\max_{1\le i\le 2n_q} \left(b_{i,q} - \bar{b}_q\right)^2}{B_q^2} = \frac{\frac{1}{4}}{\frac{n_q}{2}} = 0.$$

iii) For all $\tau > 0$,

$$\lim_{q\to\infty} \frac{1}{2n_q} \sum_{i,j:|\delta_{q,i,j}|>\tau} \delta_{q,i,j}^2 = 0,$$

where

$$\delta_{q,i,j} = \frac{\left(b_{i,q} - \bar{b}_q\right)\left(a_{q,j} - \bar{a}_q\right)}{\left(\frac{1}{2n_q} A_q^2 B_q^2\right)^{1/2}}, \quad 1 \le i, j \le 2n_q,\ q \ge 1.$$

We show that, almost surely, for every $\tau > 0$ there exists $q' \ge 1$ such that for all $q \ge q'$, no pair of indices $i, j \in \{1, \ldots, 2n_q\}$ satisfies $|\delta_{q,i,j}| > \tau$. Indeed, note that

$$\delta_{q,i,j} = \frac{\left(\left(b_{i,q} - \bar{b}_q\right)^2 \left(a_{q,j} - \bar{a}_q\right)^2\right)^{1/2}}{\left(\frac{1}{2n_q} A_q^2 B_q^2\right)^{1/2}} \le \left(\frac{\frac{1}{4} \max_{1\le j\le 2n_q} \left(a_{q,j} - \bar{a}_q\right)^2}{\frac{1}{2n_q} A_q^2 \frac{n_q}{2}}\right)^{1/2} \overset{a.s.}{\to} 0,$$

by point i). This implies that

$$\max_{1\le i,j\le 2n_q} |\delta_{q,i,j}| \overset{a.s.}{\to} 0.$$

Therefore, for almost every $(X_{n,m})_{n,m\ge1}$ and $(Y_{n,m})_{n,m\ge1}$,

$$\frac{\left(S_q - \mathbb{E}_{R_q} S_q\right)}{\sqrt{\operatorname{Var}\left(S_q\right)}} \overset{w}{\to} \mathcal{N}\left(0, 1\right).$$

Next, we show that, almost surely,

$$\frac{\sqrt{\operatorname{Var}\left(S_q\right)}}{\sqrt{n_q}} \to \sqrt{\frac{1}{2} \operatorname{Var}_H(h)}.$$

By Lemma 3.C.11,

$$\operatorname{Var}(S_q) = \frac{n_q^2 A_q^2}{2n_q(2n_q - 1)} = \frac{n_q}{2(2n_q - 1)} A_q^2.$$

From point i), we have shown that

$$\frac{A_q^2}{2n_q} \overset{a.s.}{\to} \operatorname{Var}_H(h).$$

Therefore,

$$\frac{\sqrt{\operatorname{Var}\left(S_q\right)}}{\sqrt{n_q}} = \sqrt{\frac{\frac{n_q}{2(2n_q-1)}A_q^2}{n_q}} \xrightarrow{a.s.} \sqrt{\frac{1}{2}\operatorname{Var}_H(h)}.$$

Consequently,

$$\widetilde{G}_q(h) \xrightarrow{w} \sqrt{\frac{1}{2}}\, G_H(h),$$

for almost every choice of infinite hierarchical samples from Q^1 and Q^2.

Next, we use this result to show finite-dimensional convergence.

- Step 1.2. Finite-dimensional convergence.

Let $k \geq 1$ and let $h_1, \ldots, h_k \in \mathcal{D}^*$. We want to show that

$$\left(\widetilde{G}_q\left(h_1\right), \ldots, \widetilde{G}_q\left(h_k\right)\right) \xrightarrow{w} \sqrt{\frac{1}{2}}\left(G_H\left(h_1\right), \ldots, G_H\left(h_k\right)\right)$$

for almost every $(X_{n,m})_{n,m\geq 1}, (Y_{n,m})_{n,m\geq 1}$ from Q^1 and Q^2 respectively.

Note that, by Theorem 29.4 in Billingsley (1995), the above convergence is equivalent to

$$\sum_{i=1}^k t_i \widetilde{G}_q\left(h_i\right) \xrightarrow{w} \sum_{i=1}^k \sqrt{\frac{1}{2}}\, t_i G_H\left(h_i\right),$$

for $t_1, \ldots, t_k \in \mathbb{R}$. Therefore, it suffices to prove the latter.

Observe that

$$\sum_{i=1}^k t_i \widetilde{G}_q(h_i) = \sum_{i=1}^k t_i \sqrt{n_q}\left(\widetilde{Q}_q^1 - H_q\right)(h_i) = \sqrt{n_q}\left(\widetilde{Q}_q^1 - H_q\right)\left(\sum_{i=1}^k t_i h_i\right) = \widetilde{G}_q\left(\sum_{i=1}^k t_i h_i\right).$$

Moreover, $\left\|\sum_{i=1}^k t_i h_i\right\|_\infty < \infty$. By Step 1,

$$\sum_{i=1}^k t_i \widetilde{G}_q(h_i) = \widetilde{G}_q\left(\sum_{i=1}^k t_i h_i\right) \xrightarrow{w} \mathcal{N}\left(0, \frac{1}{2}\operatorname{Var}_H\left(\sum_{i=1}^k t_i h_i\right)\right),$$

for almost every hierarchical sample from Q^1 and Q^2.

It remains to show that

$$\sum_{i=1}^k \sqrt{\frac{1}{2}}\, t_i G_H\left(h_i\right) \sim \mathcal{N}\left(0, \frac{1}{2}\operatorname{Var}_H\left(\sum_{i=1}^k t_i h_i\right)\right).$$

Note that

$$\begin{aligned}
\mathrm{Var}_H\left(\sum_{i=1}^k t_i h_i\right) &= H\left(\left(\sum_{i=1}^k t_i h_i\right)^2\right) - \left(H\left(\sum_{i=1}^k t_i h_i\right)\right)^2 \\
&= H\left(\sum_{i=1}^k \sum_{j=1}^k t_i t_j h_i h_j\right) - \left(\sum_{i=1}^k t_i H(h_i)\right)^2 \\
&= \sum_{i=1}^k \sum_{j=1}^k t_i t_j H\left(h_i h_j\right) - \sum_{i=1}^k \sum_{j=1}^k t_i t_j H(h_i) H(h_j) \\
&= \sum_{i=1}^k \sum_{j=1}^k t_i t_j \left(H\left(h_i h_j\right) - H(h_i) H(h_j)\right) \\
&= t^\top \Sigma^k t,
\end{aligned}$$

where Σ^k is the covariance matrix of the random vector $(G_H\left(h_1\right), \ldots, G_H\left(h_k\right))$.

Consequently,

$$\frac{1}{2}\mathrm{Var}_H\left(\sum_{i=1}^k t_i h_i\right) = \frac{1}{2} t^\top \Sigma^k t = \mathrm{Var}\left(\sum_{i=1}^k \sqrt{\frac{1}{2}}\, t_i G_H(h_i)\right).$$

Since this is a sum of centered Gaussian random variables, we obtain

$$\sum_{i=1}^k \sqrt{\frac{1}{2}}\, t_i G_H(h_i) \sim \mathcal{N}\left(0, \frac{1}{2}\mathrm{Var}_H\left(\sum_{i=1}^k t_i h_i\right)\right).$$

Step 2. Asymptotic equicontinuity.

We equip $\mathcal{D}^*$ with the supremum norm $\|\cdot\|_\infty$; by Lemma 3.C.10, $(\mathcal{D}^*, \|\cdot\|_\infty)$ is totally bounded.

Now fix $\epsilon > 0$ and any hierarchical samples $(X_{n,m})_{n,m\geq 1}, (Y_{n,m})_{n,m\geq 1}$ from Q^1 and Q^2 respectively. Note that, since we fix the hierarchical samples, the only randomness of the empirical process comes from the permutation. We want to show that

$$\lim_{\delta\to 0} \limsup_{q\to\infty} \mathbb{P}_{R_q}\left(\sup_{\substack{h,h'\in\mathcal{D}^*\\ \|h-h'\|_\infty<\delta}} \left|\widetilde{G}_q(h) - \widetilde{G}_q(h')\right| > \epsilon\right) = 0.$$

Now fix $\delta > 0$. By Markov's inequality,

$$\mathbb{P}_{R_q}\left(\sup_{\substack{h,h'\in\mathcal{D}^*\\ \|h-h'\|_\infty<\delta}} \left|\widetilde{G}_q(h) - \widetilde{G}_q(h')\right| > \epsilon\right) \leq \frac{1}{\epsilon}\mathbb{E}_{R_q} \sup_{\substack{h,h'\in\mathcal{D}^*\\ \|h-h'\|_\infty<\delta}} \left|\widetilde{G}_q(h-h')\right| = \frac{1}{\epsilon}\mathbb{E}_{R_q}\left\|\widetilde{G}_q\right\|_{\mathcal{D}^*_\delta},$$

where $\mathcal{D}_\delta^* = \{h - h' : h, h' \in \mathcal{D}^*,\ \|h - h'\|_\infty < \delta\}$.

To bound the expectation above, we would like to use a maximal inequality (Lemma 3.B.8). However, since the empirical process $\widetilde{G}_q$ does not sample i.i.d. random variables from H_q, we first bound it by an empirical process based on i.i.d. random variables. For this, we use Hoeffding's inequality (Proposition A.1.9 in Van der Vaart and Wellner (1996)).

Note that the function

$$\ell^\infty(\mathcal{D}^*) \ni G \mapsto \|G\|_{\mathcal{D}_\delta^*}$$

is convex.

Consider the set

$$C_q := \left\{ \frac{1}{\sqrt{n_q}} \left(\delta_{Z_{1,q}} - H_q\right), \dots, \frac{1}{\sqrt{n_q}} \left(\delta_{Z_{2n_q,q}} - H_q\right) \right\}.$$

Let $U_1, \dots, U_{n_q}$ be drawn from C_q without replacement, and let $V_1, \dots, V_{n_q}$ be drawn from C_q with replacement. Then, by Proposition A.1.9 in Van der Vaart and Wellner (1996), we have

$$\mathbb{E} \left\| \sum_{i=1}^{n_q} U_i \right\|_{\mathcal{D}_\delta^*} \le \mathbb{E} \left\| \sum_{i=1}^{n_q} V_i \right\|_{\mathcal{D}_\delta^*}.$$

Note that

$$\sum_{i=1}^{n_q} U_i = \widetilde{G}_q.$$

Therefore,

$$\mathbb{E}_{R_q} \left\| \widetilde{G}_q \right\|_{\mathcal{D}_\delta^*} \le \mathbb{E}_{\widehat{Z}_q} \left\| \frac{1}{\sqrt{n_q}} \sum_{i=1}^{n_q} \left(\delta_{\widehat{Z}_{i,q}} - H_q\right) \right\|_{\mathcal{D}_\delta^*},$$

where $\widehat{Z}_{1,q}, \dots, \widehat{Z}_{n_q,q} \overset{\text{i.i.d}}{\sim} H_q$.

Now we apply the maximal inequality.

Note that $\|h - h'\|_\infty < \delta$ implies $H_q\left((h - h')^2\right) < \delta^2$. Hence, by Lemma 3.B.8,

$$\mathbb{E}_{\widehat{Z}_q} \left\| \frac{1}{\sqrt{n_q}} \sum_{i=1}^{n_q} \left(\delta_{\widehat{Z}_{i,q}} - H_q\right) \right\|_{\mathcal{D}_\delta^*} \lesssim J_{[]}(\delta, \mathcal{D}_\delta^*, \mathrm{L}_2(H_q)) + \sqrt{n_q}\, H_q \left\{K \mathbf{1}\big(K > \sqrt{n_q}\alpha_q(\delta)\big)\right\},$$

where

$$\alpha_q(\delta) = \frac{\delta}{\sqrt{\log \mathcal{N}_{[]}\left(\delta, \mathcal{D}_\delta^*, \mathrm{L}_2(H_q)\right)}},$$

and K is an envelope function of $\mathcal{D}^*$, which is a constant function. The constant in the inequality is universal; it does not depend on the probability measures or the function class, so we ignore it.

First, we bound the second term. By Lemma 3.C.9,

$$\mathcal{N}_{[]}\left(\delta, \mathcal{D}^*_\delta, \mathrm{L}_2(H_q)\right) \leq \mathcal{N}\left(\frac{\delta}{2}, \mathcal{D}^*_\delta, \|\cdot\|_\infty\right).$$

This implies that

$$\alpha_q(\delta) \geq \frac{\delta}{\sqrt{\log \mathcal{N}\left(\frac{\delta}{2}, \mathcal{D}^*_\delta, \|\cdot\|_\infty\right)}} =: \alpha(\delta).$$

Therefore,

$$H_q\left\{K\mathbf{1}\big(K > \sqrt{n_q}\alpha_q(\delta)\big)\right\} \leq H_q\left\{K\mathbf{1}\big(K > \sqrt{n_q}\alpha(\delta)\big)\right\}.$$

Since K is a constant function, the set over which we integrate is empty for sufficiently large q. Hence,

$$\lim_{\delta\to 0}\limsup_{q\to\infty} \sqrt{n_q}\, H_q\left\{K\mathbf{1}\big(K > \sqrt{n_q}\alpha_q(\delta)\big)\right\} = 0.$$

Now we focus on the first term in the maximal inequality. By Lemmas 3.C.9 and 3.C.10,

$$\begin{aligned} J_{[]}(\delta, \mathcal{D}^*_\delta, \mathrm{L}_2(H_q)) &= \int_0^\delta \sqrt{\log \mathcal{N}_{[]}\left(\epsilon, \mathcal{D}^*_\delta, \mathrm{L}_2(H_q)\right)}\, d\epsilon \\ &\leq \int_0^\delta \sqrt{\log \mathcal{N}\left(\frac{\epsilon}{2}, \mathcal{D}^*, \|\cdot\|_\infty\right)}\, d\epsilon = C'\sqrt{\delta} \overset{\delta\to 0}{\to} 0, \end{aligned}$$

for some constant C' that does not depend on q or on δ. Therefore,

$$\lim_{\delta\to 0}\limsup_{q\to\infty} J_{[]}(\delta, \mathcal{D}^*_\delta, \mathrm{L}_2(H_q)) = 0.$$

□

We now establish the convergence of the threshold under the permutation approach.

Theorem 3.C.3. *Given $\mathbb{X} = [a, b]$ and $Q^1, Q^2 \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$, let $r(n, m, \theta)$ be the permutation threshold (3.11). Then,*

$$r(n, m, \theta) \overset{a.s.}{\to} q_{1-\theta}(\|G_H\|_{\mathcal{D}^*}) \quad \text{as } n, m \to \infty,$$

where $q_{1-\theta}$ denotes the $(1-\theta)$-quantile.

Proof. Note that, as $H_{n,m} = \frac{1}{2}\widetilde{Q}^1_{n,m} + \frac{1}{2}\widetilde{Q}^2_{n,m}$, we have that

$$\sqrt{\frac{n}{2}}\left(\widetilde{Q}^1_{n,m} - \widetilde{Q}^2_{n,m}\right) = \sqrt{2}\sqrt{n}\left(\widetilde{Q}^1_{n,m} - H_{n,m}\right).$$

By Theorem 3.C.2,

$$\sqrt{n}\left(\widetilde{Q}^1_{n,m} - H_{n,m}\right) \xrightarrow{w} \sqrt{\frac{1}{2}}G_H$$

for almost every infinite size hierarchical samples $(X_{n,m})_{n,m\geq 1}, (Y_{n,m})_{n,m\geq 1}$. This implies that

$$\sqrt{\frac{n}{2}}\left(\widetilde{Q}^1_{n,m} - \widetilde{Q}^2_{n,m}\right) \xrightarrow{w} G_H.$$

Since $\|\cdot\|_{\mathcal{D}^*}$ is a continuous function from $\ell^\infty(\mathcal{D}^*)$ to $\mathbb{R}$, the Continuous Mapping Theorem yields

$$\sqrt{\frac{n}{2}}\, d_{\text{Lip}}\left(\widetilde{Q}^1_{n,m}, \widetilde{Q}^2_{n,m}\right) \xrightarrow{w} \|G_H\|_{\mathcal{D}^*}.$$

Finally, as the random variable $\|G_H\|_{\mathcal{D}^*}$ has an absolutely continuous density (see Proposition 2 of Beran and Millar (1986)), its associated quantiles converge for every θ by Lemma 21.2 in Van der Vaart (1998). □

Proof of Theorem 3.5.1

Proof. Suppose first that $Q^1 = Q^2$. Let $(n_q)_{q\geq 1}$ and $(m_q)_{q\geq 1}$ be sequences such that

$$\lim_{q\to\infty} n_q = \lim_{q\to\infty} m_q = \infty.$$

By the first part of Theorem 3.C.1, the sequence of random variables T_{n_q,m_q} converges in distribution to $\|G_{Q^1}\|_{\mathcal{D}^*}$. The latter is almost surely bounded by Lemma 3.C.6 and has an absolutely continuous density (Proposition 2 of Beran and Millar (1986)).

In addition, by Theorem 3.C.3, $r(n_q, m_q, \theta)$ converges almost surely to the $(1-\theta)$-quantile of $\|G_{Q^1}\|_{\mathcal{D}^*}$ for all $\theta \in (0,1)$.

Then, by Lemma 3.C.7, for all $\theta \in (0,1)$

$$\lim_{q\to\infty} \mathbb{P}\big(T_{n_q,m_q} > r(n_q, m_q, \theta)\big) = \theta.$$

Now suppose that $Q^1 \neq Q^2$. By the second part of Theorem 3.C.1, the sequence of random variables T_{n_q,m_q} converges in probability to ∞.

Moreover, by Theorem 3.C.3, $r(n_q, m_q, \theta)$ converges almost surely to the $(1-\theta)$-quantile of $\left\|G_{\frac{1}{2}Q^1+\frac{1}{2}Q^2}\right\|_{\mathcal{D}^*}$ for all $\theta \in (0,1)$. This random variable is almost surely bounded by Lemma 3.C.6, and hence its $(1-\theta)$-quantile is finite.

Therefore, by Lemma 3.C.8, for all $\theta \in (0,1)$

$$\lim_{q\to\infty} \mathbb{P}\big(T_{n_q,m_q} > r(n_q, m_q, \theta)\big) = 1.$$

□

Now we prove the rest of the auxiliary lemmas.

Lemma 3.C.4 (Triangular SLLN)**.** *Let $(X_{n,m})_{n,m\geq 1}$ be a sequence of random variables in a bounded interval of $\mathbb{R}$. Let also $X, X_1, X_2, \ldots \overset{\text{i.i.d}}{\sim} P$. Assume that $X_{n,m} \overset{a.s.}{\to} X_n$ for all $n \geq 1$ and $(X_n, X_{n,1}, X_{n,2}, \ldots)$ are i.i.d. over n. Then,*

$$\left| \frac{1}{n} \sum_{i=1}^{n} X_{i,m} - \mathbb{E}X \right| \overset{a.s.}{\to} 0 \quad \textit{as } n, m \to \infty.$$

Proof. Let $(n_q)_{q\geq 1}, (m_q)_{q\geq 1}$ be any sequences such that $n_q \to \infty$ and $m_q \to \infty$. We upper bound the quantity of interest as

$$\left| \frac{1}{n_q} \sum_{i=1}^{n_q} X_{i,m_q} - \mathbb{E}X \right| \leq \left| \frac{1}{n_q} \sum_{i=1}^{n_q} X_{i,m_q} - \frac{1}{n_q} \sum_{i=1}^{n_q} X_i \right| + \left| \frac{1}{n_q} \sum_{i=1}^{n_q} X_i - \mathbb{E}X \right|.$$

The second term converges to 0 almost surely by the strong law of large numbers. It therefore remains to show that the first term converges to zero almost surely.

$$\limsup_{q\to\infty} \left| \frac{1}{n_q} \sum_{i=1}^{n_q} X_{i,m_q} - \frac{1}{n_q} \sum_{i=1}^{n_q} X_i \right| \leq \limsup_{q\to\infty} \frac{1}{n_q} \sum_{i=1}^{n_q} \left| X_{i,m_q} - X_i \right|.$$

Note that for all $M \geq 1$, we have

$$\limsup_{q\to\infty} \frac{1}{n_q} \sum_{i=1}^{n_q} \left| X_{i,m_q} - X_i \right| \leq \limsup_{q\to\infty} \frac{1}{n_q} \sum_{i=1}^{n_q} \sup_{m\geq M} \left| X_{i,m} - X_i \right|.$$

We define $Y_{i,M} = \sup_{m\geq M} |X_{i,m} - X_i|$ for $i, M \geq 1$. Then, for fixed M, the $Y_{i,M}$ are i.i.d. random variables over i. Therefore, by the SLLN,

$$\frac{1}{n_q} \sum_{i=1}^{n_q} Y_{i,M} \overset{a.s.}{\to} \mathbb{E}Y_{1,M} \quad \text{as } q \to \infty.$$

Note that this almost sure event depends on the integer M. However, since we have a countable number of such sets, by taking the countable intersection, we obtain almost sure convergence for all $M \geq 1$. Moreover, by our assumption on $X_{i,m}$, we have $Y_{i,M} \overset{a.s.}{\to} 0$ as $M \to \infty$. Therefore, by the dominated convergence theorem,

$$\lim_{M\to\infty} \mathbb{E}Y_{1,M} = 0.$$

Consequently,

$$\lim_{M\to\infty} \limsup_{q\to\infty} \frac{1}{n_q} \sum_{i=1}^{n_q} Y_{i,M} = \lim_{M\to\infty} \mathbb{E}Y_{1,M} = 0 \quad \text{a.s.}$$

□

Lemma 3.C.5. *Let $\mathbb{X}$ be a compact Polish space and $Q \in \mathcal{P}(\mathcal{P}(\mathbb{X}))$. Then*

$$\lim_{m\to\infty} d_{\mathrm{Lip}}(\mathcal{T}_m[Q], Q) = 0.$$

Proof. By de Finetti's theorem,

$$\frac{1}{m}\sum_{j=1}^{m} \delta_{X_j} \xrightarrow{w} \widetilde{P} \quad \text{a.s.},$$

where $\widetilde{P} \sim Q$. Since almost sure convergence implies weak convergence, we obtain

$$\mathcal{L}\left(\frac{1}{m}\sum_{j=1}^{m} \delta_{X_j}\right) \xrightarrow{w} \mathcal{L}(\widetilde{P}).$$

Since d_{Lip} metrizes weak convergence (see Theorem 3.1 and Lemma 3.2 in Catalano and Lavenant (2024)), the desired result follows. □

Lemma 3.C.6. *Let G be a tight random variable taking values in $\ell^\infty(\mathcal{D}^*)$, and let f be a continuous function from $\ell^\infty(\mathcal{D}^*)$ to $\mathbb{R}$. Then $f(G)$ is a tight random variable taking values in $\mathbb{R}$ and it is almost surely bounded.*

Proof. Fix $\epsilon \in (0,1)$. Since G is tight, there exists a compact subset $S \subset \ell^\infty(\mathcal{D}^*)$ such that

$$\mathbb{P}(G \in S) > \epsilon.$$

Since f is continuous, we also have that $f(S)$ is compact. Then,

$$\mathbb{P}\{f(G) \in f(S)\} = \mathbb{P}\left\{G \in f^{-1}(f(S))\right\} \geq \mathbb{P}\{G \in S\} > \epsilon,$$

since $S \subset f^{-1}(f(S))$. This shows that $f(G)$ is tight.

Now we show that $f(G)$ is almost surely bounded, that is,

$$\lim_{M\to\infty} \mathbb{P}\{f(G) \in [-M, M]\} = 1.$$

Suppose that

$$\lim_{M\to\infty} \mathbb{P}\{f(G) \in [-M, M]\} = \delta$$

for some $\delta < 1$ (note that the limit always exists, since the sets are increasing). Take $\epsilon > \delta$. We know that there exists a compact subset $B \subset \mathbb{R}$ such that

$$\mathbb{P}(f(G) \in B) > \epsilon.$$

Since B is bounded, there exists $M' > 0$ such that $B \subset [-M', M']$, and hence

$$\mathbb{P}\left(f(G) \in [-M', M']\right) \geq \mathbb{P}\left(f(G) \in B\right) > \epsilon > \delta,$$

which leads to a contradiction. Therefore, $f(G)$ is almost surely bounded. $\square$

Lemma 3.C.7. *Let T be an almost surely bounded and absolutely continuous random variable, and let $\{T_n\}_{n\geq 1}$ be a sequence of random variables such that $T_n \xrightarrow{w} T$. Let $q(\theta)$ denote the $(1-\theta)$-quantile of T, and let $r(n,\theta)$ be a sequence of random variables such that*

$$r(n,\theta) \xrightarrow{p} q(\theta) \quad \text{as } n \to \infty$$

for every $\theta \in (0,1)$. Then,

$$\lim_{n\to\infty} \mathbb{P}(T_n > r(n,\theta)) = \theta \quad \text{for all } \theta \in (0,1).$$

Proof. Fix $\theta \in (0,1)$. By Slutsky's theorem,

$$T_n - r(n,\theta) \xrightarrow{w} T - q(\theta).$$

As T is absolutely continuous, $\mathbb{P}(T - q(\theta) = 0) = 0$. Therefore, by the Portmanteau theorem,

$$\lim_{n\to\infty} \mathbb{P}(T_n - r(n,\theta) > 0) = \mathbb{P}(T - q(\theta) > 0) = \theta.$$

$\square$

Lemma 3.C.8. *Let $\{T_n\}_{n\geq 1}$ be a sequence of random variables such that $T_n \xrightarrow{p} \infty$. Let $r(n,\theta)$ be a sequence of random variables such that*

$$r(n,\theta) \xrightarrow{p} r(\theta) < \infty \quad \text{as } n \to \infty$$

for every $\theta \in (0,1)$, where $r(\theta) \in \mathbb{R}$. Then,

$$\lim_{n\to\infty} \mathbb{P}(T_n > r(n,\theta)) = 1,$$

for all $\theta \in (0,1)$.

Proof. Fix $\epsilon > 0$ and $\delta \in (0,1)$. Since $T_n \xrightarrow{p} \infty$ and $r(n,\theta) \xrightarrow{p} r(\theta)$, there exists $N \geq 1$ such that for all $n \geq N$,

$$\begin{aligned}\mathbb{P}(T_n > r(\theta) + \epsilon) &> \delta,\\ \mathbb{P}(r(n,\theta) < r(\theta) + \epsilon) &> \delta.\end{aligned}$$

Define the events

$$A_n = \{T_n > r(\theta) + \epsilon\}, \qquad B_n = \{r(n, \theta) < r(\theta) + \epsilon\}.$$

Then, for every $n \geq N$,

$$\mathbb{P}(A_n \cap B_n) = 1 - \mathbb{P}(A_n^c \cup B_n^c) \geq 1 - \mathbb{P}(A_n^c) - \mathbb{P}(B_n^c) > 2\delta - 1.$$

On the event $A_n \cap B_n$, we have $T_n > r(\theta) + \epsilon > r(n, \theta)$. Therefore, for all $n \geq N$,

$$\mathbb{P}(T_n > r(n, \theta)) \geq \mathbb{P}(\{T_n > r(n, \theta)\} \cap A_n \cap B_n) = \mathbb{P}(A_n \cap B_n) > 2\delta - 1.$$

Since $\delta \in (0, 1)$ was arbitrary, the claim follows. □

Lemma 3.C.9. *Let $\mathcal{H}$ be a class of functions from $\mathbb{Y}$ to $\mathbb{R}$, and let P be a probability measure on $\mathbb{Y}$. Then*

$$\mathcal{N}_{[]}(\epsilon, \mathcal{H}, \mathrm{L}_2(P)) \leq \mathcal{N}\left(\frac{\epsilon}{2}, \mathcal{H}, \|\cdot\|_\infty\right).$$

Proof. Let $\{h_1, \ldots, h_N\}$ be an $\epsilon/2$-cover of $\mathcal{H}$ with respect to the $\|\cdot\|_\infty$ norm. Define

$$l_i = h_i - \frac{\epsilon}{2}, \qquad u_i = h_i + \frac{\epsilon}{2},$$

for $i = 1, \ldots, N$. Then $\|u_i - l_i\|_{\mathrm{L}_2(P)} = \epsilon$ for each i. Moreover, for every $h \in \mathcal{H}$, there exists an index i such that $\|h - h_i\|_\infty \leq \epsilon/2$, which implies

$$h_i(y) - \frac{\epsilon}{2} \leq h(y) \leq h_i(y) + \frac{\epsilon}{2},$$

for all $y \in \mathbb{Y}$. Thus, $h \in [l_i, u_i]$, and the collection $\{[l_i, u_i]\}_{i=1}^N$ forms a family of ϵ-brackets that covers $\mathcal{H}$. □

Lemma 3.C.10. *Let $\mathbb{X} \subset \mathbb{R}^d$ be a set with bounded diameter. Then there exists a constant $C_d \in \mathbb{R}$ such that*

$$\log \mathcal{N}(\epsilon, \mathcal{D}^*, \|\cdot\|_\infty) \leq C_d \frac{1}{\epsilon^d},$$

for all $\epsilon > 0$.

Proof. Let $K = \mathrm{diam}(\mathbb{X})$. As shown on page 17 of Catalano and Lavenant (2024), we have

$$\log \mathcal{N}(\epsilon, \mathcal{D}^*, \|\cdot\|_\infty) \leq \log 2 \left\lceil \frac{4K}{\epsilon} \right\rceil + \log \mathcal{N}\left(\frac{\epsilon}{2}, \mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R}), \|\cdot\|_\infty\right).$$

By Theorem 2.7.1 in Van der Vaart and Wellner (1996), there exists a constant C_d depending on $\mathrm{diam}(\mathbb{X})$ such that $\log \mathcal{N}(\epsilon, \mathrm{Lip}_1^*(\mathbb{X}, \mathbb{R}), \|\cdot\|_\infty) \leq C_d \epsilon^{-d}$, which yields the desired result. □

Lemma 3.C.11. *Let $n \geq 1$ and let $\{a_1, \ldots, a_{2n}\} \subset \mathbb{R}$. Consider a random permutation $R = (R_1, \ldots, R_{2n})$ of $\{1, \ldots, 2n\}$, and define*

$$S_n = \sum_{i=1}^{n} a_{R_i}.$$

Then

$$\operatorname{Var}(S_n) = \frac{n^2 \operatorname{Var}(a_{R_1})}{2n-1},$$

where $\bar{a} = \frac{1}{2n} \sum_{i=1}^{2n} a_i$.

Proof. We have

$$\operatorname{Var}(S_n) = \sum_{i=1}^{n} \operatorname{Var}(a_{R_i}) + 2 \sum_{i<j} \operatorname{Cov}(a_{R_i}, a_{R_j}) = n \operatorname{Var}(a_{R_1}) + n(n-1) \operatorname{Cov}(a_{R_1}, a_{R_2}).$$

We now compute $\operatorname{Cov}(a_{R_1}, a_{R_2})$:

$$\operatorname{Cov}(a_{R_1}, a_{R_2}) = \mathbb{E}[a_{R_1} a_{R_2}] - \bar{a}^{\,2}.$$

For the first term, we have

$$\begin{aligned}
\mathbb{E}[a_{R_1} a_{R_2}] &= \sum_{i=1}^{2n} \sum_{j \neq i} \frac{1}{2n} \frac{1}{2n-1} a_i a_j \\
&= \sum_{i=1}^{2n} \frac{1}{2n-1} a_i \left(\sum_{j=1}^{2n} \frac{a_j}{2n} - \frac{a_i}{2n} \right) \\
&= \sum_{i=1}^{2n} \frac{1}{2n-1} a_i \bar{a} - \frac{1}{2n-1} \frac{1}{2n} \sum_{i=1}^{2n} a_i^2 \\
&= \frac{2n}{2n-1} \bar{a}^{\,2} - \frac{1}{2n-1} \left(\operatorname{Var}(a_{R_1}) + \bar{a}^{\,2} \right) \\
&= \bar{a}^{\,2} - \frac{1}{2n-1} \operatorname{Var}(a_{R_1}).
\end{aligned}$$

Therefore,

$$\operatorname{Cov}(a_{R_1}, a_{R_2}) = \bar{a}^{\,2} - \frac{1}{2n-1} \operatorname{Var}(a_{R_1}) - \bar{a}^{\,2} = -\frac{1}{2n-1} \operatorname{Var}(a_{R_1}).$$

Substituting this into the expression for $\mathrm{Var}(S_n)$ yields

$$\begin{aligned}\mathrm{Var}(S_n) &= n\,\mathrm{Var}(a_{R_1}) + n(n-1)\left(-\frac{1}{2n-1}\mathrm{Var}(a_{R_1})\right)\\ &= n\,\mathrm{Var}(a_{R_1})\left(1-\frac{n-1}{2n-1}\right) = \frac{n^2\,\mathrm{Var}(a_{R_1})}{2n-1}.\end{aligned}$$

□

Appendix 3.D Fréchet mean and variance

In this section, we compute Fréchet means and variances for a specific class of probability measures, namely normal distributions. Let $\mathcal{W}_2$ denote the Wasserstein-2 distance on $\mathcal{P}_2(\mathbb{X})$, which is the space of probability measures with finite second moments.

Let $\mathbb{X} = \mathbb{R}$ and consider the class of probability measures $\mathcal{P}_{\mathcal{N}}$ defined as

$$\mathcal{P}_{\mathcal{N}} = \left\{\mathcal{N}(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 \in (0, \infty)\right\}. \tag{3.16}$$

Note that for $P^1, P^2 \in \mathcal{P}_{\mathcal{N}}$,

$$\mathcal{W}_2^2(P^1, P^2) = (\mu_1 - \mu_2)^2 + (\sigma_1 - \sigma_2)^2,$$

where μ_1, μ_2 and σ_1^2, σ_2^2 are the means and variances of P^1 and P^2, respectively. In this setting, the Fréchet mean and variance for the law of a random probability measure $Q \in \mathcal{P}(\mathcal{P}_{\mathcal{N}})$ can be computed from the following lemma:

Lemma 3.D.1. *Let $Q \in \mathcal{P}(\mathcal{P}_{\mathcal{N}})$ where $\mathcal{P}_{\mathcal{N}}$ is as in (3.16). Let $\widetilde{P} \sim Q$, and define the random mean and variance as*

$$\widetilde{\mu} = \int_{\mathbb{R}} x\widetilde{P}(dx), \quad \widetilde{\sigma}^2 = \int_{\mathbb{R}} x^2\widetilde{P}(dx) - \left(\int_{\mathbb{R}} x\widetilde{P}(dx)\right)^2.$$

Then,

$$\mu_F = \mathcal{N}(\mathbb{E}[\widetilde{\mu}], (\mathbb{E}[\widetilde{\sigma}])^2), \quad V_F = \mathrm{Var}(\widetilde{\mu}) + \mathrm{Var}(\widetilde{\sigma})$$

are the Fréchet mean and variance, respectively.

Proof. It is a well-known fact that if $P, P' \in \mathcal{P}_2(\mathbb{R})$, then

$$\mathcal{W}_2^2(P, P') \geq \mathcal{W}_2^2(\mathcal{N}(\mu_X, \sigma_X^2), \mathcal{N}(\mu_Y, \sigma_Y^2)),$$

where μ_X, σ_X^2 and μ_Y, σ_Y^2 are the means and variances of P and P', respectively. Indeed, let $X \sim P$ and $Y \sim P'$, and consider an optimal coupling $(X^*, Y^*) \sim \pi^*$. We have:

$$
\begin{aligned}
\mathcal{W}_2^2(P, P') &= \mathbb{E}[(X^* - Y^*)^2] \\
&= \mathbb{E}[(X^*)^2] + \mathbb{E}[(Y^*)^2] - 2\mathbb{E}[X^* Y^*] \\
&= \mathrm{Var}(X^*) + \mathrm{Var}(Y^*) + (\mathbb{E}[X^*])^2 + (\mathbb{E}[Y^*])^2 - 2\mathbb{E}[X^* Y^*] \\
&= \mathrm{Var}(X^*) + \mathrm{Var}(Y^*) + (\mathbb{E}[X^*])^2 + (\mathbb{E}[Y^*])^2 - 2\mathrm{Cov}(X^*, Y^*) - 2\mathbb{E}[X^*]\mathbb{E}[Y^*] \\
&\geq \mathrm{Var}(X^*) + \mathrm{Var}(Y^*) + (\mathbb{E}[X^*])^2 + (\mathbb{E}[Y^*])^2 \\
&- 2\sqrt{\mathrm{Var}(X^*)}\sqrt{\mathrm{Var}(Y^*)} - 2\mathbb{E}[X^*]\mathbb{E}[Y^*] \\
&= (\mathbb{E}[X^*] - \mathbb{E}[Y^*])^2 + (\sqrt{\mathrm{Var}(X^*)} - \sqrt{\mathrm{Var}(Y^*)})^2 \\
&= (\mathbb{E}[X] - \mathbb{E}[Y])^2 + (\sqrt{\mathrm{Var}(X)} - \sqrt{\mathrm{Var}(Y)})^2 \\
&= \mathcal{W}_2^2(\mathcal{N}(\mu_X, \sigma_X^2), \mathcal{N}(\mu_Y, \sigma_Y^2)).
\end{aligned}
$$

Thus, we obtain

$$
\mathbb{E}[\mathcal{W}_2^2(P, \widetilde{P})] \geq \mathbb{E}[\mathcal{W}_2^2(\mathcal{N}(\mu_P, \sigma_P^2), \mathcal{N}(\widetilde{\mu}, \widetilde{\sigma}^2))],
$$

where μ_P and σ_P^2 denote the mean and variance of $X \sim P$. We now lower bound the Fréchet variance:

$$
\begin{aligned}
\min_{P \in \mathcal{P}_2(\mathbb{X})} \mathbb{E}[\mathcal{W}_2^2(P, \widetilde{P})] &\geq \min_{P \in \mathcal{P}_2(\mathbb{X})} \mathbb{E}[\mathcal{W}_2^2(\mathcal{N}(\mu_P, \sigma_P^2), \mathcal{N}(\widetilde{\mu}, \widetilde{\sigma}^2))] \\
&= \min_{\mu \in \mathbb{R}, \sigma \in (0, \infty)} \mathbb{E}[(\mu - \widetilde{\mu})^2 + (\sigma - \widetilde{\sigma})^2] \\
&\overset{(*)}{=} \mathbb{E}[(\mathbb{E}[\widetilde{\mu}] - \widetilde{\mu})^2 + (\mathbb{E}[\widetilde{\sigma}] - \widetilde{\sigma})^2] \\
&= \mathrm{Var}(\widetilde{\mu}) + \mathrm{Var}(\widetilde{\sigma}),
\end{aligned}
$$

where $(*)$ follows from the fact that the expectation minimizes the mean squared error. By defining the candidate for the Fréchet mean as

$$
\mu_F = \mathcal{N}(\mathbb{E}[\widetilde{\mu}], (\mathbb{E}[\widetilde{\sigma}])^2),
$$

we observe that $\mu_F \in \mathcal{P}_2(\mathbb{X})$, and furthermore,

$$
\mathbb{E}[\mathcal{W}_2^2(\mu_F, \widetilde{P})] = \mathbb{E}[(\mathbb{E}[\widetilde{\mu}] - \widetilde{\mu})^2 + (\mathbb{E}[\widetilde{\sigma}] - \widetilde{\sigma})^2] = \mathrm{Var}(\widetilde{\mu}) + \mathrm{Var}(\widetilde{\sigma}).
$$

□

Remark 3.D.2. A special case of the above occurs when $Q \in \mathcal{P}(\mathcal{P}_\mathcal{N})$ is concentrated on the class $\mathcal{P}'_\mathcal{N}$ of normal distributions with unit variance, i.e.,

$$
\mathcal{P}'_\mathcal{N} = \{\mathcal{N}(\mu, 1) : \mu \in \mathbb{R}\}.
$$

In this case, if $\widetilde{P} \sim Q$, then $\widetilde{\sigma}^2 \overset{a.s.}{=} 1$. Consequently,

$$\mathcal{N}(\mathbb{E}[\widetilde{\mu}], 1) \quad \text{and} \quad \mathrm{Var}(\widetilde{\mu})$$

are the Fréchet mean and variance, respectively. Furthermore, since $\mathcal{W}_1$ and $\mathcal{W}_2$ coincide on this space, their associated Fréchet means and variances also coincide.

Remark 3.D.3. Using Lemma 3.D.1, we can also compute the Fréchet mean and variance for a mixture of the laws of RPMs. Specifically, let Q^1 and Q^2 be in $\mathcal{P}(\mathcal{P}_{\mathcal{N}})$ with the same Fréchet means and variances. Then, the Fréchet mean and variance for the mixture $\lambda Q^1 + (1-\lambda)Q^2$ coincide with those of Q^1 and Q^2. This follows from the following fact:

Let $P^1, P^2 \in \mathcal{P}(\mathbb{R})$ and $Y_1 \sim P^1$, $Y_2 \sim P^2$ be independent random variables. Then for every $\lambda \in (0,1)$ and $Z \sim \lambda P^1 + (1-\lambda)P^2$

$$\begin{aligned}
&\mathbb{E}[Z] = \lambda \mathbb{E} Y_1 + (1-\lambda)\mathbb{E}[Y_2] \\
&\mathrm{Var}\,(Z) = \lambda \mathrm{Var}(Y_1) + (1-\lambda)\mathrm{Var}(Y_2) + \lambda(1-\lambda)\left(\mathbb{E}[Y_1] - \mathbb{E}[Y_2]\right)^2.
\end{aligned}$$

Chapter 4

Merging Rate of Opinions

4.1 Introduction to Merging

In Bayesian statistics, learning about the data-generating mechanism begins with the specification of a prior distribution over a class of probability distributions that could have generated the data. This prior may be uninformative (e.g., uniform over a suitable class), reflecting limited prior information, or more concentrated when substantive prior knowledge is available. Upon observing data, the statistician updates these beliefs via Bayes' rule, obtaining a posterior distribution over the candidate data-generating laws.

The goal of this section is to compare posterior beliefs arising from different prior distributions. This problem is commonly referred to as the merging of opinions. To study it rigorously, several modeling choices must be clarified. In particular, what assumptions are made about the sequence of observations? Are restrictions imposed on the class of prior distributions?

This problem was studied in Blackwell and Dubins (1962), where the convergence of predictive distributions is analyzed. Their framework considers probability measures defined on the space of infinite sequences of observations. After observing the first n observations, one forms a predictive distribution over future observations via conditional probabilities. They show that if one probability measure Q is absolutely continuous with respect to another probability measure P, then their corresponding predictive distributions merge almost surely with respect to Q. This result applies to Bayesian settings in which priors are defined over infinite-dimensional spaces and implies that the corresponding predictive distributions, and hence posterior beliefs, merge. However, it relies on a key restriction: one of the probability measures on the infinite sequence of observations, induced by a prior, must be absolutely continuous with respect to the other. A sufficient condition guaranteeing this is to impose absolute continuity at the level of prior distributions. However, this condition can fail even in standard Bayesian nonparametric models; for example, two Dirichlet process priors with mutually singular base measures are themselves mutually singular. Moreover, the result is formulated under the assumption that the observations are generated according to one of the probability measures under consideration. If the objective is instead to compare posterior updating procedures themselves, it is natural to avoid strong assumptions about the true data-generating mechanism. This perspective is adopted in Catalano and Lavenant (2026), and the goal of this chapter is to extend the study of merging to another example of a Bayesian nonparametric model.

The rest of this chapter is organized as follows. In Section 4.2, we review known results on merging rates from Catalano and Lavenant (2026), covering the merging between two Dirichlet processes and between the Dirichlet process and the normalized generalized Gamma process. In Section 4.3, we derive a new result on the merging rate between the Pitman–Yor and Dirichlet processes.

4.2 Known Results on Merging Rates

In this section, we summarize known results on merging between two posterior distributions arising from the Bayesian model 2.1 when different priors are used. As discussed earlier, we investigate merging independently of the data-generating mechanism. That is, for a fixed sequence of observations, we compare how the corresponding posterior distributions differ under different prior specifications.

Since we work with distances between random measures, the key step is to quantify the distance between posterior distributions. To this end, we consider two priors given by completely random measures, derive the corresponding posterior distributions, and then compare them. Due to the identifiability issue discussed in Subsection 2.2.3, we compute distances between scaled versions of these posterior distributions.

We start with the case of merging between two Dirichlet processes. Recall that the Dirichlet process is obtained by normalizing a Gamma CRM. By Theorem 15 in Catalano and Lavenant (2026), for two Gamma CRMs $\tilde{\mu}^1$ and $\tilde{\mu}^2$ with parameters $(\alpha_i, 1, P_0^i)$ for $i = 1, 2$ (see Definition 2.2.7), and for any sequence of observations $(x_n)_{n\geq 1}$ in $\mathbb{X}$, it holds that

$$\mathrm{AD}(\nu_S^{1,*}, \nu_S^{2,*}) \asymp \frac{1}{n}, \qquad \mathcal{W}_{\mathrm{BL}}\big(\mathcal{L}(\tilde{\mu}_S^{1,*}), \mathcal{L}(\tilde{\mu}_S^{2,*})\big) \lesssim \frac{1}{n},$$

where $\tilde{\mu}_S^{k,*}$ denotes the scaled version of the posterior CRM associated with $\tilde{\mu}^k$ and $\nu_S^{k,*}$ denotes the associated Lévy intensity for $k = 1, 2$. It is important to note that the Lévy intensities associated with scaled posterior CRMs are not random, since the posterior CRMs are no longer Cox CRMs after scaling.

The result says that the posterior distributions always merge at rate $1/n$, regardless of the observed data and independently of the data-generating mechanism.

This behavior admits an intuitive explanation. The posterior distribution induced by a Dirichlet process converges weakly to the corresponding predictive distribution, which in turn converges to the empirical measure. Equivalently, one can say that the posterior distribution concentrates around the empirical measure as the sample size increases. Since this limiting object does not depend on the prior, different posterior distributions merge.

We now discuss another known result on merging between the Dirichlet process and the generalized Gamma CRM.

Definition 4.2.1. A completely random measure is said to be a generalized Gamma CRM if its Lévy intensity is given by

$$\mathrm{d}\nu(s,x) = \frac{\alpha}{\Gamma(1-\sigma)} \frac{e^{-s}}{s^{1+\sigma}} \,\mathrm{d}s\,\mathrm{d}P_0(x),$$

where $\alpha > 0$ is the total mass parameter, $\sigma \in (0,1)$ is the discount parameter, and $P_0 \in \mathcal{P}(\mathbb{X})$ is the base probability measure.

By Theorem 19 in Catalano and Lavenant (2026), for a generalized Gamma CRM $\tilde{\mu}^1$ with parameters (α, σ, P_0) and a Gamma CRM $\tilde{\mu}^2$ with parameters $(\alpha, 1, P_0)$, and for any sequence of observations $(x_n)_{n\geq 1}$ in $\mathbb{X}$, it holds that

$$\mathcal{W}_{\mathrm{AD}}\big(\mathcal{L}(\tilde{\nu}_S^{1,*}), \mathcal{L}(\tilde{\nu}_S^{2,*})\big) \asymp \max\left\{ n^{-\frac{1}{1+\sigma}}, \frac{k_n}{n} \right\},$$

and

$$\mathcal{W}_{\mathrm{BL}}\big(\mathcal{L}(\tilde{\mu}_S^{1,*}), \mathcal{L}(\tilde{\mu}_S^{2,*})\big) \lesssim \max\left\{ n^{-\frac{1}{1+\sigma}}, \frac{k_n}{n} \right\},$$

where k_n denotes the number of distinct observations among $x_1, \dots, x_n$.

This result shows that merging occurs provided that the number of distinct observations does not grow at the same rate as the sample size. In such cases, the rate of merging depends on the parameter σ. In contrast to the Dirichlet process versus Dirichlet process case, merging may fail when $k_n \sim n$. This situation can arise, for instance, when the data are generated i.i.d. from a continuous distribution.

4.3 Merging between the Pitman–Yor and Dirichlet Processes

We now present a result on merging between the Pitman–Yor and Dirichlet processes. We first recall the posterior distribution derived from the Pitman–Yor process. The strategy for establishing the merging rate is to first obtain the scaled version of the posterior distributions from each completely random measure, and then study the adapted Wasserstein discrepancy.

Let $\tilde{\mu}$ be a Cox CRM with random Lévy intensity $\tilde{\nu}$, as in Definition 2.2.11. The random Lévy intensity has the form

$$\mathrm{d}\tilde{\nu}_\xi(s,x) = \frac{\xi\sigma}{\Gamma(1-\sigma)}\frac{e^{-s}}{s^{1+\sigma}}\mathrm{d}s\,\mathrm{d}P_0(x),$$

where $\xi \sim \Gamma\left(\frac{\theta}{\sigma}, 1\right)$.

The posterior distribution of $\tilde{\mu}$ under the Bayesian model 2.1 has the following form (Hjort et al., 2010, p. 114):

Given observations $x_1, \dots, x_n$ from model 2.1, the posterior $\widetilde{\mu}^*$ of $\tilde{\mu}$ is a Cox CRM characterized by

$$\widetilde{\mu}^* \mid U \overset{d}{=} \tilde{\mu}_U + \sum_{i=1}^{k} J_i^U \delta_{x_i^*},$$

where

- $\tilde{\mu}_U$ is a generalized Gamma CRM with Lévy intensity

$$\mathrm{d}\tilde{\nu}(s,x) = \frac{\sigma}{\Gamma(1-\sigma)}\frac{e^{-Us}}{s^{1+\sigma}}\mathrm{d}s\,\mathrm{d}P_0(x).$$

  Note that we have omitted the subscript U for brevity.

- $J_i^U \sim \mathrm{Gamma}(n_i - \sigma, U)$

- $f_U(u) = \frac{\sigma}{\Gamma\left(k+\frac{\theta}{\sigma}\right)} u^{\theta+k\sigma-1} e^{-u^\sigma} \mathbf{1}_{(0,\infty)}(u)$

and $(x_1^*, \dots, x_k^*)$ are the distinct atoms with multiplicities $n_1, \dots, n_k$.

Note that both k and U implicitly depend on n and are thus sequences; however, since we do not need to track this dependence explicitly before deriving the merging rate, we drop the subscript n for brevity.

Proposition 4.3.1. *Let $\tilde{\mu}$ be a Cox CRM associated with the Pitman–Yor process with parameters (σ, θ, P_0), and let $\widetilde{\mu}^*$ denote its posterior. Then the scaled version of $\widetilde{\mu}^*$ has random Lévy intensity*

$$\mathrm{d}\tilde{\nu}_S^*(s,x) = \mathrm{d}\rho_x^*(s)\,\mathrm{d}P_0^*(x),$$

where

$$P_0^* = \frac{\sigma U^\sigma}{C}P_0 + \frac{1}{C}\sum_{i=1}^{k}(n_i - \sigma)\delta_{x_i^*},$$

$$\mathrm{d}\rho_x^*(s) = \frac{C^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-Cs}}{s^{1+\sigma}}\mathbf{1}_{\mathbb{X}\setminus\{x_1^*,\dots,x_k^*\}}(x)\,\mathrm{d}s + C\frac{e^{-Cs}}{s}\mathbf{1}_{\{x_1^*,\dots,x_k^*\}}(x)\,\mathrm{d}s,$$

$$C = C(\sigma, n, k, U) = \sigma U^\sigma + n - k\sigma,$$

and U is as defined above.

Proof. Since $\widetilde{\mu}^* \mid U$ is a sum of CRMs, we compute its Lévy intensity.

By iteratively applying Lemma SM4 in the Supplementary Material Catalano and Lavenant (2026), for a general CRM $\tilde{\mu}$ with Lévy intensity ν, independent and infinitely divisible random variables $J_1, \ldots, J_k$ with Lévy measures π_i, and fixed atoms $y_1, \ldots, y_k$, the Lévy intensity of $\tilde{\mu} + \sum_{i=1}^k J_i \delta_{y_i}$ is

$$\mathrm{d}\nu(s,x) + \sum_{i=1}^{k} \mathrm{d}\left(\pi_i \otimes \delta_{y_i}\right)(s,x),$$

where ν is the Lévy intensity of $\tilde{\mu}$.

The Lévy measure of $J_i^U \sim \mathrm{Gamma}(n_i - \sigma, U)$ is

$$\mathrm{d}\pi_i(s) = \frac{n_i - \sigma}{s} e^{-Us} \mathbf{1}_{(0,\infty)}(s)\,\mathrm{d}s.$$

Then, for $(s,x) \in (0,\infty) \times \mathbb{X}$,

$$\begin{aligned}
\mathrm{d}\nu(s,x) + \sum_{i=1}^{k} \mathrm{d}\left(\pi_i \otimes \delta_{x_i^*}\right)(s,x) \\
&= \frac{\sigma}{\Gamma(1-\sigma)} \frac{e^{-Us}}{s^{1+\sigma}} \mathrm{d}s\,\mathrm{d}P_0(x) + \sum_{i=1}^{k} \frac{n_i - \sigma}{s} e^{-Us}\,\mathrm{d}s\,\mathrm{d}\delta_{x_i^*}(x) \\
&= \frac{\sigma}{\Gamma(1-\sigma)} \frac{e^{-Us}}{s^{1+\sigma}} \mathrm{d}s\,\mathrm{d}P_0(x) + \frac{e^{-Us}}{s}\,\mathrm{d}s \sum_{i=1}^{k} (n_i - \sigma)\,\mathrm{d}\delta_{x_i^*}(x) =: \mathrm{d}\tilde{\nu}^*(s,x).
\end{aligned}$$

So $\widetilde{\mu}^*$ is a Cox CRM with random Lévy intensity $\tilde{\nu}^*$. Denote

$$\begin{aligned}
\mathrm{d}\rho^1(s) &= \frac{\sigma}{\Gamma(1-\sigma)} \frac{e^{-Us}}{s^{1+\sigma}}\,\mathrm{d}s, \\
\mathrm{d}\rho^2(s) &= \frac{e^{-Us}}{s}\,\mathrm{d}s.
\end{aligned}$$

Then,

$$\mathrm{d}\tilde{\nu}^*(s,x) = \mathrm{d}\rho^1(s)\,\mathrm{d}P_0(x) + \mathrm{d}\rho^2(s) \sum_{i=1}^{k} (n_i - \sigma)\,\mathrm{d}\delta_{x_i^*}(x).$$

The next step is to obtain the scaled version of $\widetilde{\mu}^*$. Definition 2.2.6, we obtain the scaled version of the latent CRM for fixed U.

As discussed in Subsection 2.2.3, if ν is the Lévy intensity of a CRM $\tilde{\mu}$, then the Lévy intensity of its scaled version is

$$\mathrm{d}\nu_S(s,x) = \mathbb{E}[\tilde{\mu}(\mathbb{X})]\,\rho_x(\mathbb{E}[\tilde{\mu}(\mathbb{X})]\,s)\,\mathrm{d}P_0(x),$$

by Lemma SM3 in the Supplementary Material Catalano and Lavenant (2026). By applying the same argument to

$$\mathrm{d}\tilde{\nu}^*(s,x) = \mathrm{d}\rho^1(s)\,\mathrm{d}P_0^1(x) + \mathrm{d}\rho^2(s)\,\mathrm{d}P_0^2(x),$$

we get

$$\begin{aligned}\mathrm{d}\tilde{\nu}_S^*(s,x) = \mathbb{E}[\widetilde{\mu}^*(\mathbb{X}) \mid U]\, \rho_x^1(\mathbb{E}[\widetilde{\mu}^*(\mathbb{X}) \mid U]\, s)\, \mathrm{d}P_0^1(x) \\ + \mathbb{E}[\widetilde{\mu}^*(\mathbb{X}) \mid U]\, \rho_x^2(\mathbb{E}[\widetilde{\mu}^*(\mathbb{X}) \mid U]\, s)\, \mathrm{d}P_0^2(x).\end{aligned}$$

We therefore first compute $\mathbb{E}[\widetilde{\mu}^*(\mathbb{X}) \mid U]$. By Campbell's theorem,

$$\mathbb{E}[\widetilde{\mu}^*(\mathbb{X}) \mid U] = \int_{(0,\infty)\times\mathbb{X}} s\,\mathrm{d}\tilde{\nu}^*(s,x).$$

So,

$$\begin{aligned}\mathbb{E}[\widetilde{\mu}^*(\mathbb{X}) \mid U] &= \int_{(0,\infty)\times\mathbb{X}} s\frac{\sigma}{\Gamma(1-\sigma)}\frac{e^{-Us}}{s^{1+\sigma}}\mathrm{d}s\,\mathrm{d}P_0(x) + \int_{(0,\infty)\times\mathbb{X}} s\frac{e^{-Us}}{s}\,\mathrm{d}s\sum_{i=1}^{k}(n_i-\sigma)\,\mathrm{d}\delta_{x_i^*}(x) \\ &= \frac{\sigma}{\Gamma(1-\sigma)}\int_0^\infty s^{-\sigma}e^{-Us}\,\mathrm{d}s + (n-k\sigma)\int_0^\infty e^{-Us}\,\mathrm{d}s \\ &= \frac{\sigma}{\Gamma(1-\sigma)}\frac{\Gamma(1-\sigma)}{U^{1-\sigma}} + \frac{n-k\sigma}{U} \\ &= \frac{\sigma U^\sigma}{U} + \frac{n-k\sigma}{U} = \frac{C}{U},\end{aligned}$$

where $C = C(\sigma,n,k,U) = \sigma U^\sigma + n - k\sigma$.

Plugging this into the scaling formula, we obtain the scaled Lévy intensity of the latent CRM:

$$\begin{aligned}\mathrm{d}\tilde{\nu}_S^*(s,x) &= \frac{C}{U}\,\mathrm{d}\rho^1\left(\frac{C}{U}s\right)\mathrm{d}P_0(x) + \frac{C}{U}\,\mathrm{d}\rho^2\left(\frac{C}{U}s\right)\sum_{i=1}^{k}(n_i-\sigma)\,\mathrm{d}\delta_{x_i^*}(x) \\ &= \frac{C}{U}\frac{\sigma}{\Gamma(1-\sigma)}\frac{e^{-Cs}}{C^{1+\sigma}s^{1+\sigma}}U^{1+\sigma}\,\mathrm{d}s\,\mathrm{d}P_0(x) + \frac{e^{-Cs}}{s}\,\mathrm{d}s\sum_{i=1}^{k}(n_i-\sigma)\,\mathrm{d}\delta_{x_i^*}(x) \\ &= \frac{\sigma}{\Gamma(1-\sigma)}\frac{e^{-Cs}}{s^{1+\sigma}}\left(\frac{U}{C}\right)^\sigma\,\mathrm{d}s\,\mathrm{d}P_0(x) + \frac{e^{-Cs}}{s}\,\mathrm{d}s\sum_{i=1}^{k}(n_i-\sigma)\,\mathrm{d}\delta_{x_i^*}(x).\end{aligned}$$

The above expression is not yet in canonical form; we wish to express it as a product of a transition kernel for jumps and a density for atoms.

Note that

$$\mathcal{M}_1(\tilde{\nu}_S^*) = \int_{(0,\infty)\times\mathbb{X}} s\,\mathrm{d}\tilde{\nu}_S^*(s,x) = 1,$$

since it is the Lévy intensity of a scaled CRM. The measure $s\,\mathrm{d}\tilde{\nu}_S^*$ can therefore be disintegrated as

$$s\,\mathrm{d}\tilde{\nu}_S^*(s,x) = \mathrm{d}\varrho_x(s)\,\mathrm{d}Q(x),$$

where Q is the second marginal of $s\,\mathrm{d}\tilde{\nu}_S^*$ and ϱ_x is the probability transition kernel.

Since

$$Q(\mathbb{X}) = \int_{(0,\infty)\times\mathbb{X}} s\,\mathrm{d}\tilde{\nu}_S^*(s,x) = 1,$$

Q is a probability measure. It follows that

$$\mathrm{d}\tilde{\nu}_S^*(s,x) = \frac{1}{s}\,\mathrm{d}\varrho_x(s)\,\mathrm{d}Q(x).$$

To find the terms in this decomposition, we first determine Q. For $A \in \mathcal{B}(\mathbb{X})$,

$$\begin{aligned}
Q(A) &= \int_{(0,\infty)\times A} s\,\mathrm{d}\tilde{\nu}_S^*(s,x) \\
&= \frac{\sigma}{\Gamma(1-\sigma)}\left(\frac{U}{C}\right)^{\sigma} P_0(A)\int_0^\infty s^{-\sigma}e^{-Cs}\,\mathrm{d}s + \sum_{i=1}^k (n_i-\sigma)\delta_{x_i^*}(A)\int_0^\infty e^{-Cs}\,\mathrm{d}s \\
&= \frac{\sigma}{\Gamma(1-\sigma)}\left(\frac{U}{C}\right)^{\sigma} P_0(A)\frac{\Gamma(1-\sigma)}{C^{1-\sigma}} + \frac{1}{C}\sum_{i=1}^k (n_i-\sigma)\delta_{x_i^*}(A) \\
&= \frac{\sigma U^\sigma}{C}P_0(A) + \frac{1}{C}\sum_{i=1}^k (n_i-\sigma)\delta_{x_i^*}(A).
\end{aligned}$$

We denote

$$P_0^*(\cdot) = \frac{\sigma U^\sigma}{C}P_0(\cdot) + \frac{1}{C}\sum_{i=1}^k (n_i-\sigma)\delta_{x_i^*}(\cdot).$$

It remains to determine ϱ_x, which must satisfy

$$\frac{\sigma}{\Gamma(1-\sigma)}\frac{e^{-Cs}}{s^{\sigma+1}}\left(\frac{U}{C}\right)^{\sigma}\mathrm{d}s\,\mathrm{d}P_0(x) + \frac{e^{-Cs}}{s}\,\mathrm{d}s\sum_{i=1}^k (n_i-\sigma)\,\mathrm{d}\delta_{x_i^*}(x) = \frac{1}{s}\,\mathrm{d}\varrho_x(s)\,\mathrm{d}P_0^*(x).$$

Since P_0 is non-atomic, we obtain

$$\mathrm{d}\varrho_x(s) = \frac{C^{1-\sigma}}{\Gamma(1-\sigma)}s^{-\sigma}e^{-Cs}\mathbf{1}_{\mathbb{X}\setminus\{x_1^*,\dots,x_k^*\}}(x)\,\mathrm{d}s + Ce^{-Cs}\mathbf{1}_{\{x_1^*,\dots,x_k^*\}}(x)\,\mathrm{d}s.$$

The conclusion follows by setting

$$\mathrm{d}\rho_x^*(s) = \frac{1}{s}\mathrm{d}\varrho_x(s) = \frac{C^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-Cs}}{s^{1+\sigma}}\mathbf{1}_{\mathbb{X}\setminus\{x_1^*,\dots,x_k^*\}}(x)\,\mathrm{d}s + C\frac{e^{-Cs}}{s}\mathbf{1}_{\{x_1^*,\dots,x_k^*\}}(x)\,\mathrm{d}s.$$

$\square$

We are now ready to study the merging between the Pitman–Yor and Dirichlet processes.

Proposition 4.3.2. *Let $\tilde\mu^1$ be a Pitman–Yor CRM with parameters (σ, θ, P_0) and let $\tilde\mu^2$ be a Gamma CRM with parameters $(\alpha, 1, P_0)$. Then*

$$\mathcal{W}_{\mathrm{AD}}\left(\mathcal{L}\left(\tilde\nu_S^{*,1}\right), \mathcal{L}\left(\tilde\nu_S^{*,2}\right)\right) = \mathbb{E}_U\left[\mathcal{A}^U + J_1^U + J_2^U\right],$$

where

$$\begin{aligned}
\mathcal{A}^U &= \mathcal{W}_{d_{\mathbb{X},t}}\left(\frac{\sigma U^\sigma}{C} P_0 + \frac{1}{C}\sum_{i=1}^k (n_i - \sigma)\delta_{x_i^*}, \frac{\alpha}{\alpha+n} P_0 + \frac{1}{\alpha+n}\sum_{i=1}^n \delta_{x_i}\right),\\
J_1^U &= \frac{\sigma U^\sigma}{C} \mathcal{W}_*\left(\frac{C^{1-\sigma}}{\Gamma(1-\sigma)} \frac{e^{-Cs}}{s^{1+\sigma}}\,\mathrm{d}s, (\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\,\mathrm{d}s\right),\\
J_2^U &= \frac{n - k\sigma}{C} \mathcal{W}_*\left(\frac{Ce^{-Cs}}{s}\,\mathrm{d}s, (\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\,\mathrm{d}s\right),
\end{aligned}$$

where U and C are as defined in Proposition 4.3.1.

Proof. Denote by $\tilde\nu_S^{*,2}$ the Lévy intensity of the scaled Gamma posterior; its derivation can be found in the proof of Lemma 12 in Catalano and Lavenant (2026), and it has the following form:

$$\mathrm{d}\tilde\nu_S^{*,2}(s,x) = (\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\,\mathrm{d}s\left(\frac{\alpha}{\alpha+n}\,\mathrm{d}P_0(x) + \frac{1}{\alpha+n}\sum_{i=1}^n \mathrm{d}\delta_{x_i}(x)\right).$$

Note that the scaled Gamma posterior is no longer a Cox CRM. Since it is homogeneous, Remark 2.3.9 gives

$$\begin{aligned}
\mathcal{W}_{\mathrm{AD}}\left(\mathcal{L}\left(\tilde\nu_S^{*,1}\right), \mathcal{L}\left(\tilde\nu_S^{*,2}\right)\right) &= \mathbb{E}_{U\sim f_U}\mathrm{AD}\left(\tilde\nu_S^{*,1}, \tilde\nu_S^{*,2}\right)\\
&= \mathbb{E}_{U\sim f_U}\mathcal{W}_{d_{\mathbb{X},t}}\left(P_0^{*,1}, P_0^{*,2}\right) + \mathbb{E}_{U\sim f_U}\mathbb{E}_{X\sim P_0^{*,1}}\mathcal{W}_*\left(\rho_X^{*,1}, \rho^{*,2}\right),
\end{aligned}$$

where

$$\begin{aligned}
\mathrm{d}\rho_x^{*,1}(s) &= \frac{C^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-Cs}}{s^{1+\sigma}}\mathbf{1}_{\mathbb{X}\setminus\{x_1^*,\dots,x_k^*\}}(x)\,\mathrm{d}s + C\frac{e^{-Cs}}{s}\mathbf{1}_{\{x_1^*,\dots,x_k^*\}}(x)\,\mathrm{d}s,\\
\mathrm{d}\rho^{*,2}(s) &= (\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\,\mathrm{d}s,\\
P_0^{*,1}(\cdot) &= \frac{\sigma U^\sigma}{C}P_0(\cdot) + \frac{1}{C}\sum_{i=1}^k (n_i-\sigma)\delta_{x_i^*}(\cdot),\\
P_0^{*,2}(\cdot) &= \frac{\alpha}{\alpha+n}P_0(\cdot) + \frac{1}{\alpha+n}\sum_{i=1}^n \delta_{x_i}(\cdot).
\end{aligned}$$

The first term is

$$\mathbb{E}_{U\sim f_U}\mathcal{W}_{d_{\mathbb{X},t}}\left(\frac{\sigma U^\sigma}{C}P_0+\frac{1}{C}\sum_{i=1}^{k}(n_i-\sigma)\delta_{x_i^*},\frac{\alpha}{\alpha+n}P_0+\frac{1}{\alpha+n}\sum_{i=1}^{n}\delta_{x_i}\right)=\mathbb{E}_{U\sim f_U}\mathcal{A}^U.$$

The second term is

$$\begin{aligned}\mathbb{E}_{U\sim f_U}\mathbb{E}_{X\sim P_0^{*,1}}\mathcal{W}_*\big(\rho_X^{*,1},\rho^{*,2}\big)&=\mathbb{E}_{U\sim f_U}\frac{\sigma U^\sigma}{C}\mathbb{E}_{X\sim P_0}\mathcal{W}_*\big(\rho_X^{*,1},\rho^{*,2}\big)\\&\quad+\mathbb{E}_{U\sim f_U}\frac{1}{C}\sum_{i=1}^{k}(n_i-\sigma)\mathcal{W}_*\Big(\rho_{x_i^*}^{*,1},\rho^{*,2}\Big)\\&=\mathbb{E}_{U\sim f_U}\frac{\sigma U^\sigma}{C}\mathcal{W}_*\left(\frac{C^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-Cs}}{s^{1+\sigma}}\,\mathrm{d}s,(\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\,\mathrm{d}s\right)\\&\quad+\mathbb{E}_{U\sim f_U}\frac{n-k\sigma}{C}\mathcal{W}_*\left(\frac{Ce^{-Cs}}{s}\,\mathrm{d}s,(\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\,\mathrm{d}s\right)\\&=\mathbb{E}_{U\sim f_U}J_1^U+\mathbb{E}_{U\sim f_U}J_2^U.\end{aligned}$$

□

Theorem 4.3.3. *Let $\tilde{\mu}^1$ and $\tilde{\mu}^2$ be as in Proposition 4.3.2 and let $(x_n)_{n\geq 1}$ be any sequence in $\mathbb{X}$. Then, as $n\to\infty$,*

$$\mathcal{W}_{\mathrm{AD}}\big(\mathcal{L}\left(\tilde{\nu}_S^{*,1}\right),\mathcal{L}\left(\tilde{\nu}_S^{*,2}\right)\big)\asymp\frac{k_n}{n}.$$

In particular, $\mathcal{W}_{\mathrm{BL}}\big(\mathcal{L}\left(\tilde{\mu}_S^{*,1}\right),\mathcal{L}\left(\tilde{\mu}_S^{*,2}\right)\big)\lesssim\frac{k_n}{n}$.

Proof. First, note that the $\mathcal{W}_{\mathrm{BL}}$ bound follows from Theorem 6 of Catalano and Lavenant (2026).

From now on, we attach the subscript n to k, U, and C, writing k_n, U_n, and C_n. Recall that U_n has density function $f_U(u)=\frac{\sigma}{\Gamma\left(k_n+\frac{\theta}{\sigma}\right)}u^{\theta+k_n\sigma-1}e^{-u^\sigma}\mathbf{1}_{(0,\infty)}(u)$. By the change of variables $y=u^\sigma$, the random variable $U_n^\sigma\sim\mathrm{Gamma}\left(k_n+\frac{\theta}{\sigma},1\right)$.

In view of Proposition 4.3.2, we study the behaviour of each term in

$$\mathbb{E}_U\left(J_1^{U_n}+J_2^{U_n}+\mathcal{A}^{U_n}\right).$$

- First term: $\mathbb{E}_{U_n}J_1^{U_n}$.

Recall that

$$J_1^{U_n}=\frac{\sigma U_n^\sigma}{C_n}\mathcal{W}_*\left(\frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-C_ns}}{s^{1+\sigma}}\mathrm{d}s,(\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\mathrm{d}s\right)=\frac{\sigma U_n^\sigma}{C_n}W_n,$$

where $C_n = \sigma U_n^\sigma + n - k_n\sigma$ and

$$W_n = \mathcal{W}_*\left(\frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-C_n s}}{s^{1+\sigma}}\mathrm{d}s, (\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\mathrm{d}s\right).$$

We first establish an upper bound for the expected value.

The extended Wasserstein distance satisfies

$$\begin{aligned}
W_n &= \int_0^\infty \left|\int_s^\infty \frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-C_n t}}{t^{1+\sigma}}\mathrm{d}t - \int_s^\infty (\alpha+n)\frac{e^{-(\alpha+n)t}}{t}\mathrm{d}t\right|\mathrm{d}s \\
&\leq \int_0^\infty\int_s^\infty \frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-C_n t}}{t^{1+\sigma}}\mathrm{d}t\,\mathrm{d}s + \int_0^\infty\int_s^\infty (\alpha+n)\frac{e^{-(\alpha+n)t}}{t}\mathrm{d}t\,\mathrm{d}s \\
&\leq \int_0^\infty s\frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-C_n s}}{s^{1+\sigma}}\mathrm{d}s + \int_0^\infty s(\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\mathrm{d}s,
\end{aligned}$$

where the last inequality follows from Fubini's theorem. Since both integrals equal 1, we conclude that $W_n \leq 2$.

Note also that

$$\frac{\sigma U_n^\sigma}{C_n} = \frac{\sigma U_n^\sigma}{\sigma U_n^\sigma + n - k_n\sigma} \leq \frac{\sigma}{1-\sigma}\frac{U_n^\sigma}{n}.$$

Taking expectations, we obtain

$$\mathbb{E}\frac{\sigma U_n^\sigma}{C_n} \leq \frac{\sigma}{1-\sigma}\frac{k_n + \frac{\theta}{\sigma}}{n}.$$

Therefore,

$$\mathbb{E}J_1^{U_n} \leq 2\frac{\sigma}{1-\sigma}\frac{k_n + \frac{\theta}{\sigma}}{n},$$

implying that $\mathbb{E}J_1^{U_n} \lesssim \frac{k_n}{n}$.

We now turn to the lower bound. To this end, we distinguish two cases:

$$\sup k_n < \infty \quad \text{and} \quad \sup k_n = \infty.$$

- Case: $\sup k_n < \infty$

Since k_n is a nondecreasing sequence of integers, there exists $N \geq 1$ such that $k_n = k$ for all $n \geq N$, where $k = \sup k_n$. This implies that for all $n \geq N$,

$$U_n^\sigma \sim \mathrm{Gamma}\left(k + \frac{\theta}{\sigma}, 1\right).$$

Let $p \in (0,1)$ and $t_1 < t_2$ be such that for all $n \geq N$,

$$\mathbb{P}\left(t_1 < U_n^\sigma < t_2\right) = p.$$

Denote $A_n = \{t_1 < U_n^\sigma < t_2\}$. Then on A_n,

$$\frac{\sigma U_n^\sigma}{C_n} = \frac{\sigma U_n^\sigma}{\sigma U_n^\sigma + n - k_n \sigma} \geq \frac{t_1 \sigma}{t_2 \sigma + n - k_n \sigma}.$$

We now obtain a positive lower bound for W_n on the set A_n.

$$\begin{aligned}
&\mathcal{W}_*\left(\frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-C_n s}}{s^{1+\sigma}}\mathrm{d}s, (\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\mathrm{d}s\right) \\
&\qquad = \int_0^\infty \left| \int_s^\infty \frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-C_n t}}{t^{1+\sigma}}\mathrm{d}t - \int_s^\infty (\alpha+n)\frac{e^{-(\alpha+n)t}}{t}\mathrm{d}t \right| \mathrm{d}s \\
&\qquad = \int_0^\infty \left| \int_{C_n s}^\infty \frac{C_n^{1-\sigma}}{\Gamma(1-\sigma)}\frac{e^{-y}}{y^{1+\sigma}} C_n^{1+\sigma}\frac{1}{C_n}\mathrm{d}y - \int_{(\alpha+n)s}^\infty (\alpha+n)\frac{e^{-y}}{y}\mathrm{d}y \right| \mathrm{d}s \\
&\qquad = \int_0^\infty \left| \frac{C_n}{\Gamma(1-\sigma)}\Gamma\left(-\sigma, C_n s\right) - (\alpha+n)\Gamma(0,(\alpha+n)s) \right| \mathrm{d}s,
\end{aligned}$$

where $\Gamma(z,t) = \int_t^\infty x^{z-1}e^{-x}\,\mathrm{d}x$ denotes the incomplete gamma function.

By the change of variables $t = C_n s$, the last expression reduces to

$$\int_0^\infty \left| \frac{\Gamma(-\sigma,t)}{\Gamma(1-\sigma)} - \frac{\alpha+n}{C_n}\Gamma\left(0, \frac{\alpha+n}{C_n}t\right) \right| \mathrm{d}t.$$

Note that on A_n,

$$\frac{n}{t_2\sigma + n} \leq \frac{\alpha+n}{\sigma U_n^\sigma + n - k_n\sigma} \leq \frac{\alpha+n}{t_1\sigma + n(1-\sigma)},$$

implying that there exists $N' \geq N$ such that for all $n \geq N'$, on A_n, we have $\frac{\alpha+n}{C_n} \in \left[\frac{1}{2}, c^*\right]$ for some $c^* \in \left(1, \frac{1}{1-\sigma}\right)$. Define the function $F : (0,\infty) \to \mathbb{R}$,

$$F(r) = \int_0^\infty \left| \frac{\Gamma(-\sigma,t)}{\Gamma(1-\sigma)} - r\Gamma\left(0, rt\right) \right| \mathrm{d}t.$$

The function F is continuous. We wish to show that F is bounded below by a positive constant on $\left[\frac{1}{2}, c^*\right]$, which will allow us to deduce that W_n is bounded below on A_n.

Suppose that $F(r) = 0$ for some $r \in \left(\frac{1}{2}, c^*\right)$. Since the integrand is continuous, it follows that for every $t \in (0,\infty)$,

$$\left| \frac{\Gamma(-\sigma,t)}{\Gamma(1-\sigma)} - r\Gamma\left(0, rt\right) \right| = 0.$$

Hence,

$$\frac{\Gamma(-\sigma,t)}{\Gamma(1-\sigma)} = r\Gamma(0, rt).$$

Differentiating with respect to t, we obtain, for all $t > 0$,

$$-\frac{t^{-(1+\sigma)}e^{-t}}{\Gamma(1-\sigma)} = -r^2(rt)^{-1}e^{-rt},$$

that is,

$$t^{-\sigma}e^{t(r-1)} = r\Gamma(1-\sigma).$$

Since the left-hand side is not constant while the right-hand side is, we obtain a contradiction. Therefore, $F(r) > 0$ for all $r \in \left[\frac{1}{2}, c^*\right]$. Since F is continuous, it attains its minimum on this compact interval; denote this minimum by $w > 0$. Consequently, since $\frac{\alpha+n}{C_n} \in \left[\frac{1}{2}, c^*\right]$ on A_n, we have $W_n \geq w$ on A_n.

Combining the above,

$$\mathbb{E}J_1^{U_n} = \mathbb{E}\frac{\sigma U_n^\sigma W_n}{C_n} \geq \mathbb{E}\frac{\sigma U_n^\sigma W_n}{C_n}\mathbf{1}_{A_n} \geq \frac{t_1\sigma}{t_2\sigma + n - k_n\sigma}\, w\, p$$

for $n \geq N'$. Together with the upper bound, we deduce that $\mathbb{E}J_1^{U_n} \asymp \frac{1}{n}$.

- Case: $\sup k_n = \infty$

Our strategy is to show that $\frac{n}{k_n}J_1^{U_n}$ converges in probability to a positive constant and is uniformly integrable. This will yield L^1 convergence and allow us to bound the expectation asymptotically.

We begin by establishing the L^1 convergence of $\frac{U_n^\sigma}{k_n}$ to 1.

$$\mathbb{E}\left|\frac{U_n^\sigma}{k_n} - 1\right|^2 = \mathbb{E}\left|\frac{U_n^\sigma - k_n}{k_n}\right|^2 = \mathbb{E}\left|\frac{U_n^\sigma - \mathbb{E}U_n^\sigma + \frac{\theta}{\sigma}}{k_n}\right|^2 = \frac{\mathrm{Var}(U_n^\sigma)}{k_n^2} + \left(\frac{\frac{\theta}{\sigma}}{k_n}\right)^2.$$

Since $\mathrm{Var}(U_n^\sigma) = k_n + \frac{\theta}{\sigma}$, we get

$$\lim_{n\to\infty}\mathbb{E}\left|\frac{U_n^\sigma}{k_n} - 1\right|^2 = 0.$$

Since L^2 convergence implies L^1 convergence, the desired result follows.

We now show that $\frac{n}{C_n}$ converges in probability to a positive value.

$$\frac{n}{C_n} = \frac{n}{\sigma U_n^\sigma + n - k_n\sigma} = \frac{1}{\frac{\sigma U_n^\sigma}{n} + 1 - \frac{k_n\sigma}{n}}.$$

Note that if $k_n \asymp n$, then by the above, $\frac{U_n^\sigma}{k_n} \xrightarrow{P} 1$, so $\frac{\sigma U_n^\sigma}{n} - \frac{k_n\sigma}{n} \xrightarrow{P} 0$; otherwise both $\frac{U_n^\sigma}{n}$

and $\frac{k_n\sigma}{n}$ converge to 0. In either case, the denominator converges in probability to 1, so

$$\frac{n}{C_n} \xrightarrow{P} 1.$$

We now establish the convergence in probability of W_n. As shown above,

$$W_n = \int_0^\infty \left| \frac{\Gamma(-\sigma, t)}{\Gamma(1-\sigma)} - \frac{\alpha+n}{C_n}\Gamma\left(0, \frac{\alpha+n}{C_n}t\right)\right| \mathrm{d}t.$$

Since $\frac{n}{C_n}$ converges in probability, so does $\frac{\alpha+n}{C_n}$. By the continuous mapping theorem, the last expression converges in probability to

$$\int_0^\infty \left| \frac{\Gamma(-\sigma, t)}{\Gamma(1-\sigma)} - \Gamma(0,t)\right| \mathrm{d}t > 0.$$

We have thus shown that $\frac{n}{k_n}J_1^{U_n}$ converges in probability to some positive constant c. We now establish uniform integrability. Using $W_n \leq 2$ and $C_n \geq n - k_n\sigma$,

$$\frac{n}{k_n}\frac{\sigma U_n^\sigma}{C_n}W_n \leq 2\frac{n\sigma U_n^\sigma}{k_n(n-k_n\sigma)} \leq \frac{n\sigma U_n^\sigma}{k_n n(1-\sigma)} = \frac{\sigma}{1-\sigma}\frac{U_n^\sigma}{k_n}.$$

Since the last term converges in L^1, we obtain uniform integrability.

Therefore,

$$\mathbb{E}\frac{n}{k_n}J_1^{U_n} \to c,$$

from which we deduce that $\mathbb{E}J_1^{U_n} \asymp \frac{k_n}{n}$.

- Second term: $\mathbb{E}_{U_n}J_2^{U_n}$.

Note that

$$\frac{n-k_n\sigma}{C_n} = \frac{n-k_n\sigma}{\sigma U_n^\sigma + n - k_n\sigma} \leq \frac{n-k_n\sigma}{n-k_n\sigma} = 1.$$

It therefore suffices to bound the extended Wasserstein distance.

$$\begin{aligned}
\mathcal{W}_*&\left(C_n\frac{e^{-C_n s}}{s}\mathrm{d}s, (\alpha+n)\frac{e^{-(\alpha+n)s}}{s}\mathrm{d}s\right) \\
&= \int_0^\infty \left|\int_s^\infty C_n\frac{e^{-C_n t}}{t}\mathrm{d}t - \int_s^\infty (\alpha+n)\frac{e^{-(\alpha+n)t}}{t}\mathrm{d}t\right| \mathrm{d}s \\
&= \int_0^\infty \left|\int_{C_n s}^\infty C_n\frac{e^{-y}}{y}\mathrm{d}y - \int_{(\alpha+n)s}^\infty (\alpha+n)\frac{e^{-y}}{y}\mathrm{d}y\right| \mathrm{d}s \\
&= \int_0^\infty \left|C_n\Gamma\left(0, C_n s\right) - (\alpha+n)\Gamma(0,(\alpha+n)s)\right| \mathrm{d}s \\
&= J\left(C_n - n, \alpha, n\right),
\end{aligned}$$

where $J(\alpha_1, \alpha_2, n)$ is defined as

$$J(\alpha_1, \alpha_2, n) = \int_0^\infty |(\alpha_1 + n)\Gamma(0, (\alpha_1 + n)s) - (\alpha_2 + n)\Gamma(0, (\alpha_2 + n)s)| \, \mathrm{d}s.$$

By point 2 of Proposition 14 in Catalano and Lavenant (2026), we have that for every $n \geq 1$,

$$J(C_n - n, \alpha, n) \leq c \max\left\{\log\left(\frac{C_n}{\alpha + n}\right), \log\left(\frac{\alpha + n}{C_n}\right)\right\},$$

where $c = \int_0^\infty |\Gamma(0, t) - e^{-t}| \, \mathrm{d}t$.

Using the fact that $\log x \leq x - 1$, we have that

$$J(C_n - n, \alpha, n) \leq c \max\left\{\frac{C_n}{\alpha + n} - 1, \frac{\alpha + n}{C_n} - 1\right\}.$$

We bound each term in the maximum separately.

$$\frac{C_n}{\alpha + n} - 1 = \frac{\sigma U_n^\sigma - k_n \sigma - \alpha}{\alpha + n} \leq \sigma \frac{U_n^\sigma}{\alpha + n}.$$

Since

$$\mathbb{E} \frac{\sigma U_n^\sigma}{\alpha + n} = \frac{\sigma\left(k_n + \frac{\theta}{\sigma}\right)}{\alpha + n},$$

we have that

$$\mathbb{E}\left[\frac{C_n}{\alpha + n} - 1\right] \lesssim \frac{k_n}{n}.$$

We now bound the second term in the maximum.

$$\frac{\alpha + n}{C_n} - 1 = \frac{\alpha - \sigma U_n^\sigma + k_n \sigma}{n + \sigma U_n^\sigma - k_n \sigma} \leq \frac{\alpha + k_n \sigma}{n(1 - \sigma)}.$$

Therefore,

$$\mathbb{E}\left[\frac{\alpha + n}{C_n} - 1\right] \lesssim \frac{k_n}{n}.$$

- Third term: $\mathcal{A}^{U_n}$.

Finally, we bound

$$\mathcal{A}^{U_n} = \mathcal{W}_{\mathrm{d}_{\mathbb{X},t}}\left(\frac{\sigma U_n^\sigma}{C_n} P_0 + \frac{1}{C_n} \sum_{i=1}^{k_n} (n_i - \sigma)\delta_{x_i^*}, \frac{\alpha}{\alpha + n} P_0 + \frac{1}{\alpha + n} \sum_{i=1}^{n} \delta_{x_i}\right)$$

The proof uses two convexity properties of the Wasserstein distance. The first is that for all $\lambda \in (0, 1)$ and $P^1, P^2, Q^1, Q^2 \in \mathcal{P}(\mathbb{X})$,

$$\mathcal{W}_{d_{\mathbb{X},t}}\left(\lambda P^1 + (1 - \lambda)Q^1, \lambda P^2 + (1 - \lambda)Q^2\right) \leq \lambda \mathcal{W}_{d_{\mathbb{X},t}}(P^1, P^2) + (1 - \lambda)\mathcal{W}_{d_{\mathbb{X},t}}(Q^1, Q^2).$$

The second is that for all $\lambda \in (0,1)$ and $P, Q^1, Q^2 \in \mathcal{P}(\mathbb{X})$,

$$\mathcal{W}_{d_{\mathbb{X},t}}(\lambda P + (1-\lambda)Q^1, \lambda P + (1-\lambda)Q^2) = (1-\lambda)\mathcal{W}_{d_{\mathbb{X},t}}(Q^1, Q^2).$$

Since both measures share P_0, we aim to match the coefficient of P_0 in order to apply the second property. Since the relative magnitudes of these coefficients are not known a priori, we consider two cases:

$$\frac{\sigma U_n^\sigma}{C_n} > \frac{\alpha}{\alpha+n} \quad \text{and} \quad \frac{\sigma U_n^\sigma}{C_n} < \frac{\alpha}{\alpha+n}.$$

We decompose the expectation over the sets where these conditions hold. As shown below, on both sets the expectations are asymptotically bounded by

$$\frac{k_n}{n}.$$

Define the set

$$A_n = \left\{ \frac{\sigma U_n^\sigma}{C_n} > \frac{\alpha}{\alpha+n} \right\}.$$

It follows that

$$\mathbb{E}\mathcal{A}^{U_n} = \mathbb{E}\mathcal{A}^{U_n}\mathbf{1}_{A_n} + \mathbb{E}\mathcal{A}^{U_n}\mathbf{1}_{A_n^c}.$$

On A_n, we decompose

$$\frac{\sigma U_n^\sigma}{C_n} P_0 = \frac{\alpha}{\alpha+n} P_0 + \left(\frac{\sigma U_n^\sigma}{C_n} - \frac{\alpha}{\alpha+n} \right) P_0 = \frac{\alpha}{\alpha+n} P_0 + \lambda_1 P_0,$$

where

$$\lambda_1 = \frac{\sigma U_n^\sigma}{C_n} - \frac{\alpha}{\alpha+n}.$$

Note that $\lambda_1 \in (0,1)$. Then

$$\begin{aligned}
&\mathcal{W}_{d_{\mathbb{X},t}}\Big(\frac{\alpha}{\alpha+n}P_0 + \frac{n}{\alpha+n}\left(\frac{\alpha+n}{n}\lambda_1 P_0 + \frac{\alpha+n}{n}\frac{n-k_n\sigma}{C_n}\sum_{i=1}^{k_n}\frac{(n_i-\sigma)}{n-k_n\sigma}\delta_{x_i^*}\right),\\
&\qquad\qquad \frac{\alpha}{\alpha+n}P_0 + \frac{n}{\alpha+n}\sum_{i=1}^{n}\frac{1}{n}\delta_{x_i}\Big)\\
&\qquad = \frac{n}{\alpha+n}\mathcal{W}_{d_{\mathbb{X},t}}\left(\frac{\alpha+n}{n}\lambda_1 P_0 + \frac{\alpha+n}{n}\frac{n-k_n\sigma}{C_n}\sum_{i=1}^{k_n}\frac{(n_i-\sigma)}{n-k_n\sigma}\delta_{x_i^*}, \sum_{i=1}^{n}\frac{1}{n}\delta_{x_i}\right)\\
&\qquad = \frac{n}{\alpha+n}\mathcal{W}_{d_{\mathbb{X},t}}\Big(\frac{\alpha+n}{n}\lambda_1 P_0 + \frac{\alpha+n}{n}\frac{n-k_n\sigma}{C_n}\sum_{i=1}^{k_n}\frac{(n_i-\sigma)}{n-k_n\sigma}\delta_{x_i^*},\\
&\qquad\qquad \frac{\alpha+n}{n}\lambda_1\sum_{i=1}^{n}\frac{1}{n}\delta_{x_i} + \frac{\alpha+n}{n}\frac{n-k_n\sigma}{C_n}\sum_{i=1}^{n}\frac{1}{n}\delta_{x_i}\Big)\\
&\qquad \le \lambda_1\mathcal{W}_{d_{\mathbb{X},t}}\left(P_0, \sum_{i=1}^{n}\frac{1}{n}\delta_{x_i}\right) + \lambda_2\mathcal{W}_{d_{\mathbb{X},t}}\left(\sum_{i=1}^{k_n}\frac{(n_i-\sigma)}{n-k_n\sigma}\delta_{x_i^*}, \sum_{i=1}^{n}\frac{1}{n}\delta_{x_i}\right),
\end{aligned}$$

where

$$\lambda_2 = \frac{n-k_n\sigma}{C_n}.$$

We have shown that

$$\mathbb{E}\mathcal{A}^{U_n}\mathbf{1}_{A_n} \le \mathbb{E}\left[\mathbf{1}_{A_n}\lambda_1\mathcal{W}_{d_{\mathbb{X},t}}\left(P_0, \sum_{i=1}^{n}\frac{1}{n}\delta_{x_i}\right) + \mathbf{1}_{A_n}\lambda_2\mathcal{W}_{d_{\mathbb{X},t}}\left(\sum_{i=1}^{k_n}\frac{(n_i-\sigma)}{n-k_n\sigma}\delta_{x_i^*}, \sum_{i=1}^{n}\frac{1}{n}\delta_{x_i}\right)\right].$$

In the first term inside the expectation, since $d_{\mathbb{X},t} \le 2$ we can bound the Wasserstein distance by 2 and λ_1 by $\frac{\sigma U_n^\sigma}{C_n}$. Hence,

$$\mathbb{E}\left[\mathbf{1}_{A_n}\lambda_1\mathcal{W}_{d_{\mathbb{X},t}}\left(P_0, \sum_{i=1}^{n}\frac{1}{n}\delta_{x_i}\right)\right] \le \mathbb{E}\frac{\sigma U_n^\sigma}{C_n} \le \frac{\sigma}{1-\sigma}\frac{k_n+\frac{\theta}{\sigma}}{n},$$

so this term is asymptotically bounded by $\frac{k_n}{n}$.

We now bound

$$\mathbb{E}\left[\mathbf{1}_{A_n}\lambda_2\mathcal{W}_{d_{\mathbb{X},t}}\left(\sum_{i=1}^{k_n}\frac{(n_i-\sigma)}{n-k_n\sigma}\delta_{x_i^*}, \sum_{i=1}^{n}\frac{1}{n}\delta_{x_i}\right)\right].$$

Since $C_n \ge n-k_n\sigma$, we have $\lambda_2 \le 1$, and the Wasserstein distance does not depend on U_n. Note also that $\sum_{i=1}^{n}\frac{1}{n}\delta_{x_i} = \sum_{i=1}^{k_n}\frac{n_i}{n}\delta_{x_i^*}$. Since we can decompose

$$\sum_{i=1}^{k_n}\frac{n_i}{n}\delta_{x_i^*} = \sum_{i=1}^{k_n}\frac{n_i-\sigma}{n}\delta_{x_i^*} + \sum_{i=1}^{k_n}\frac{\sigma}{n}\delta_{x_i^*} = \frac{n-k_n\sigma}{n}\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*} + \frac{k_n\sigma}{n}\sum_{i=1}^{k_n}\frac{1}{k_n}\delta_{x_i^*},$$

applying the convexity of the Wasserstein distance, we obtain

$$
\begin{aligned}
&\mathcal{W}_{d_{\mathbb{X}},t}\left(\sum_{i=1}^{k_n}\frac{(n_i-\sigma)}{n-k_n\sigma}\delta_{x_i^*},\sum_{i=1}^{k_n}\frac{n_i}{n}\delta_{x_i^*}\right)=\\
&\qquad\qquad=\mathcal{W}_{d_{\mathbb{X}},t}\left(\frac{n-k_n\sigma}{n}\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*}+\frac{k_n\sigma}{n}\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*},\right.\\
&\qquad\qquad\quad\left.\frac{n-k_n\sigma}{n}\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*}+\frac{k_n\sigma}{n}\sum_{i=1}^{k_n}\frac{1}{k_n}\delta_{x_i^*}\right)\\
&\qquad\qquad=\frac{k_n\sigma}{n}\mathcal{W}_{d_{\mathbb{X}},t}\left(\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*},\sum_{i=1}^{k_n}\frac{1}{k_n}\delta_{x_i^*}\right)\\
&\qquad\qquad\leq 2\frac{k_n\sigma}{n}.
\end{aligned}
$$

We conclude that $\mathbb{E}\mathcal{A}^{U_n}\mathbf{1}_{A_n}\lesssim\frac{k_n}{n}$.

Now suppose we are on A_n^c. We decompose

$$
\frac{\alpha}{\alpha+n}P_0=\frac{\sigma U_n^\sigma}{C_n}P_0+\left(\frac{\alpha}{\alpha+n}-\frac{\sigma U_n^\sigma}{C_n}\right)P_0.
$$

Define

$$
\lambda_1'=\frac{\alpha}{\alpha+n}-\frac{\sigma U_n^\sigma}{C_n}.
$$

Note that, on A_n^c, $\lambda_1'\in(0,1)$. On A_n^c,

$$
\begin{aligned}
&\mathcal{W}_{d_{\mathbb{X}},t}\left(\frac{\sigma U_n^\sigma}{C_n}P_0+\frac{n-k_n\sigma}{C_n}\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*},\right.\\
&\qquad\qquad\left.\frac{\sigma U_n^\sigma}{C_n}P_0+\frac{n-k_n\sigma}{C_n}\left(\frac{C_n}{n-k_n\sigma}\lambda_1'P_0+\frac{C_n}{n-k_n\sigma}\frac{n}{\alpha+n}\sum_{i=1}^{k_n}\frac{n_i}{n}\delta_{x_i^*}\right)\right)\\
&\qquad\qquad=\frac{n-k_n\sigma}{C_n}\mathcal{W}_{d_{\mathbb{X}},t}\left(\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*},\right.\\
&\qquad\qquad\quad\left.\frac{C_n}{n-k_n\sigma}\lambda_1'P_0+\frac{C_n}{n-k_n\sigma}\frac{n}{\alpha+n}\sum_{i=1}^{k_n}\frac{n_i}{n}\delta_{x_i^*}\right)\\
&\qquad\qquad\leq\lambda_1'\mathcal{W}_{d_{\mathbb{X}},t}\left(\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*},P_0\right)\\
&\qquad\qquad\quad+\frac{n}{\alpha+n}\mathcal{W}_{d_{\mathbb{X}},t}\left(\sum_{i=1}^{k_n}\frac{n_i-\sigma}{n-k_n\sigma}\delta_{x_i^*},\sum_{i=1}^{k_n}\frac{n_i}{n}\delta_{x_i^*}\right).
\end{aligned}
$$

Since $\lambda_1'\leq\frac{\alpha}{\alpha+n}$, the first term is asymptotically bounded by $\frac{k_n}{n}$. As we have already bounded the second Wasserstein distance, we deduce that $\mathbb{E}\mathcal{A}^{U_n}\mathbf{1}_{A_n^c}\lesssim\frac{k_n}{n}$. $\square$

To summarize, unlike in the case of generalized Gamma CRM versus Gamma CRM, there is no dependence on the discount parameter for merging. What matters is the rate at which the number of distinct observations grows relative to the total number of observations. If they grow at the same rate, then the two posteriors do not merge; otherwise they merge at rate $\frac{k_n}{n}$.

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